Physics and Mathematics Reference

Generated 2026-03-19

1 Physical system overview

The audited execution graph models three coupled analysis layers.

1. A MATLAB heating model evaluates a discrete operating grid for a resistive nichrome coil inside a heater tube, followed by an aeration-tube network embedded in the vermicomposter bed `vermicomposter_air_heater_design_model.m:L1172-L1677`; `vermicomposter_air_heater_design_model.m:L1803-L2248`.
2. A Python summer branch replaces the electrical heater with a spot-cooled plenum and assist blower, then reuses the same reduced-order aeration, evaporation, and two-node bed closures with a blower/system operating-point correction when the requested flow exceeds the available assist-blower static `heat_transfer_study.py:L2760-L3378`; `heat_transfer_study.py:L4326-L4438`.
3. A Python annual layer reuses the selected heating and cooling reference points against a representative-year ambient driver `heat_transfer_study.py:L4547-L5012`; `heat_transfer_study.py:L5013-L5496`.

The physically important paths are electrical Joule heating of the wire, wire-to-air convection, heater-shell loss to ambient, sensible air-to-bed transfer through the aeration tube wall, direct discharge enthalpy through the perforations, latent heat consumption by evaporation, and bed-to-ambient loss through the vessel wall and top boundary condition.

This document is intentionally more explicit than `HeatTransferStudy/VermiHeatLoss.tex`. It restates the implemented equation families, identifies where they appear in the repository narrative, and records the exact code locations where each block is implemented or approximated.

2 Modeling boundaries and scope

- MATLAB is the authoritative source of the winter/heating operating-point grid and geometry-driven heater calculations `vermicomposter_air_heater_design_model.m:L1172-L1677`.
- Python is authoritative for the final reported heating recommendation, the full summer cooling workflow, the annual layer, and the current text summaries and plots `heat_transfer_study.py:L2220-L2499`; `heat_transfer_study.py:L2534-L3292`; `heat_transfer_study.py:L5013-L7020`.
- The model is reduced-order. Tubes are one-dimensional marching domains; the bed is a two-node lumped thermal network; plume and habitat calculations are post-processing estimates rather than CFD `vermicomposter_air_heater_design_model.m:L1803-L2248`; `heat_transfer_study.py:L1062-L1595`.

- The summer branch reuses the same bin wall, aeration-tube, evaporation, and two-node closures, but replaces electrical heating with spot-cooled inlet air and an assist-blower operating-point closure `heat_transfer_study.py:L2760-L3378`.
- This reference documents only equations actually found in the repository. Where `HeatTransferStudy/VermiHeatLoss.tex` contains broader narrative or optional transient discussion not used by the rooted execution graph, that is called out explicitly.

3 Modeling assumptions

- Air is treated as an ideal gas with Sutherland-law viscosity and fixed c_p and Pr in the property package `heat_transfer_study.py:L3607-L3675`; `vermicomposter_air_heater_design_model.m:L2333-L2359`.
- Heater and aeration tubes are discretized axially and solved by explicit marching rather than simultaneous PDE solution `vermicomposter_air_heater_design_model.m:L1803-L1999`; `heat_transfer_study.py:L2707-L2959`.
- Parallel aeration branches are identical at a given operating point. Total heat, water loss, and power are assembled from per-tube or per-string contributions `vermicomposter_air_heater_design_model.m:L1497-L1677`.
- The bed is represented by bottom and top thermal nodes with a single internal conductance between them `vermicomposter_air_heater_design_model.m:L971-L1075`; `heat_transfer_study.py:L4326-L4438`.
- Evaporation is represented as a local latent sink limited by diffusion, carrying capacity, and available sensible heat; there is no resolved liquid transport field in the compost `vermicomposter_air_heater_design_model.m:L2000-L2122`; `heat_transfer_study.py:L2567-L2603`.
- The vented-cover top is a repository-specific mixed boundary condition: insulated cover over most of the top plus free convection over a small vent area `vermicomposter_air_heater_design_model.m:L971-L1075`; `HeatTransferStudy/VermiHeatLoss.tex:L785-L847`.
- Summer spot-cooler behavior is a surrogate model based on configured rated quantities and simple psychrometric limiting, not a first-principles vapor-compression cycle model `heat_transfer_study.py:L2605-L2705`.

4 Energy balances

4.1 Air and psychrometric property package

Dry-air density is evaluated as

$$\rho = \frac{p}{RT}. \tag{1}$$

Dynamic viscosity uses the Sutherland form

$$\mu = \mu_0 \left(\frac{T}{T_0}\right)^{3/2} \frac{T_0 + S}{T + S}. \tag{2}$$

Thermal conductivity is then reconstructed from

$$k = \frac{\mu c_p}{Pr}. \tag{3}$$

For the summer psychrometric branch, humidity ratio is computed from vapor partial pressure p_v as

$$w = 0.62198 \frac{p_v}{p - p_v}, \tag{4}$$

and, for a given relative humidity ϕ,

$$p_v = \phi\, p_{\mathrm{sat}}(T). \tag{5}$$

Here $p_{\mathrm{sat}}(T)$ denotes the pure-water saturation pressure used by the implemented humid-air algebra. In the Python summer branch, that saturation-pressure call, the dew-point inversion built on it, and the latent-heat calls used later in the evaporation section are all routed through the imported `IAPWS_95.py` backend; the detailed Wagner–Pruss form is developed later in the moisture section.

Type. Equation (1) is exact for the ideal-gas approximation adopted here; (2) is a fitted transport-property relation; (3) is a derived closure using fixed c_p and Pr; (4)–(5) are exact algebraic psychrometric relations for the implemented moisture basis.
Repository-native narrative. `HeatTransferStudy/VermiHeatLoss.tex:L11-L37`; `HeatTransferStudy/VermiHeatLoss.tex:L1089-L1235`; `HeatTransferStudy/VermiHeatLoss.tex:L2084-L2245`.
Implementation. `vermicomposter_air_heater_design_model.m:L2333-L2359`; `IAPWS_95.py:L614-L674`; `heat_transfer_study.py:L3946-L4003`.
Deviation note. The property package uses constant c_p and Pr rather than temperature-dependent tabulated humid-air properties.

4.2 Tube-flow partitions, perforations, and electrical grouping

Let the total imposed inlet volumetric flow be $\dot{V}_{\mathrm{tot}}$ and the number of parallel aeration branches be N_{tubes}. The per-tube inlet volumetric flow is

$$\dot{V}_{\mathrm{tube,in}} = \frac{\dot{V}_{\mathrm{tot}}}{N_{\mathrm{tubes}}}. \tag{6}$$

Using the inlet density ρ_{in}, the corresponding per-tube inlet mass flow is

$$\dot{m}_{\mathrm{tube,in}} = \rho_{\mathrm{in}} \dot{V}_{\mathrm{tube,in}}. \tag{7}$$

For an aeration tube discretized into N_{aer} axial segments with N_h perforation holes of diameter d_h,

$$A_h = \frac{\pi d_h^2}{4}, \qquad A_{h,\mathrm{tube}} = N_h A_h, \qquad A_{h,\mathrm{seg}} = \frac{N_h}{N_{\mathrm{aer}}} A_h, \tag{8}$$

Following the general orifice discharge relation compiled by Idel'chik ([R12], pp. 31–32 of the accessible English translation), the closed-end aeration-tube model treats the segmentwise released mass flow as

$$\dot{m}_{\mathrm{rel},i} = C_d A_{h,\mathrm{seg}} \sqrt{2\rho_i \Delta p_i}, \qquad \Delta p_i = \max(p_{\mathrm{tube},i} - p_{\mathrm{bed},i}, 0)\,. \tag{9}$$

Here $p_{\mathrm{bed},i}$ is the local bed-side gauge pressure returned by the segmentwise porous-bed backpressure closure in Section 5. Because $p_{\mathrm{bed},i}$ itself depends on $\dot{m}_{\mathrm{rel},i}$ through that Ergun-based closure, (9) is an implicit local scalar relation rather than a one-shot explicit discharge evaluation. The corresponding segment release fraction is therefore an output,

$$\phi_{\mathrm{rel},i} = \frac{\dot{m}_{\mathrm{rel},i}}{\dot{m}_{\mathrm{tube,in}}}, \qquad \sum_{i=1}^{N_{\mathrm{aer}}} \phi_{\mathrm{rel},i} = 1, \tag{10}$$

where the unity sum follows from the same steady closed-end mass balance used later in the axial march: the segment update is $\dot{m}_{\mathrm{rem},i+1} = \dot{m}_{\mathrm{rem},i} - \dot{m}_{\mathrm{rel},i}$ with $\dot{m}_{\mathrm{rem},1} = \dot{m}_{\mathrm{tube,in}}$, and the inlet gauge pressure is chosen so that the terminal residual at the cap satisfies $\dot{m}_{\mathrm{rem},N_{\mathrm{aer}}+1} \approx 0$. Therefore

$$\sum_{i=1}^{N_{\mathrm{aer}}} \dot{m}_{\mathrm{rel},i} = \dot{m}_{\mathrm{tube,in}} - \dot{m}_{\mathrm{rem},N_{\mathrm{aer}}+1} \approx \dot{m}_{\mathrm{tube,in}},$$

so $\sum_i \phi_{\mathrm{rel},i} = 1$ holds to the mass-balance tolerance of the closed-end shooting solve rather than by an independent ad hoc normalization. Operationally, for a prescribed $\dot{m}_{\mathrm{tube,in}}$, the code solves these fractions by a scalar shooting method on inlet gauge pressure. It begins from a whole-hole-area estimate

$$p_{\mathrm{in,hi}} \sim \frac{1}{2\rho_{\mathrm{ref}}} \left(\frac{\dot{m}_{\mathrm{tube,in}}}{C_d A_{h,\mathrm{tot}}} \right)^2,$$

then marches the tube segment by segment at that trial pressure. In segment i, it solves the implicit local release equation (9) together with the porous-bed backpressure closure by bounded scalar bisection on $\dot{m}_{\mathrm{rel},i} \in [0, \dot{m}_{\mathrm{rem},i}]$, updates $\dot{m}_{\mathrm{rem},i+1}$, computes the local friction drop, and reduces the downstream tube pressure before moving to the next segment. After the last segment, the scalar residual for the trial inlet pressure is the remaining axial flow $\dot{m}_{\mathrm{rem},N_{\mathrm{aer}}+1}$. If that residual is still too large, the upper pressure bound is doubled until a bracket is found, and the code then bisects on inlet gauge pressure to find the smallest value for which

$$\dot{m}_{\mathrm{rem},N_{\mathrm{aer}}+1} \leq \max\left(10^{-9},\, 10^{-4}\dot{m}_{\mathrm{tube,in}}\right).$$

The converged segmentwise releases $\dot{m}_{\mathrm{rel},i}$ from that final marched state are then reported as $\phi_{\mathrm{rel},i} = \dot{m}_{\mathrm{rel},i}/\dot{m}_{\mathrm{tube,in}}$. and the local mean perforation-exit velocity is

$$u_{h,i} = \frac{\dot{V}_{\mathrm{rel},i}}{A_{h,\mathrm{seg}}}. \tag{11}$$

The coil helix geometry is built from the mean coil diameter D_{mean}, pitch p, and axial coil span L_{coil}. The per-turn length is

$$L_{\mathrm{turn}} = \sqrt{(\pi D_{\mathrm{mean}})^2 + p^2}, \tag{12}$$

and the wire length is

$$L_{\mathrm{wire}} = N_{\mathrm{turns}} L_{\mathrm{turn}}, \qquad N_{\mathrm{turns}} \approx \frac{L_{\mathrm{coil}}}{p}. \tag{13}$$

With wire resistivity $\rho_{\mathrm{wire}}(T)$ and cross-sectional area A_{wire},

$$R_{\mathrm{tube}} = \rho_{\mathrm{wire}}(T_{\mathrm{avg}}) \frac{L_{\mathrm{wire}}}{A_{\mathrm{wire}}}. \tag{14}$$

The current heating implementation does not assume that every heater tube is wired directly in parallel. Instead, each electrical *string* is one current path made by wiring one or more identical heater tubes in series, and the supply sees several such strings in parallel. A *group* is the subset of strings that all have the same number of series-connected tubes, so they share the same current and per-tube power calculation. For group g, let $n_{s,g}$ be the number of tubes in series in each string, and let $n_{\mathrm{str},g}$ be the number of parallel strings having that same series count. For example, 8 total

tubes default to four identical 2-tube strings, written $4 \times 2\mathrm{S}$; 7 total tubes default to one 3-tube string in parallel with two 2-tube strings, written $1 \times 3\mathrm{S} + 2 \times 2\mathrm{S}$. Then

$$I_{\mathrm{tube},g} = \frac{V}{n_{s,g} R_{\mathrm{tube}}}, \tag{15}$$

$$P_{\mathrm{tube},g} = I_{\mathrm{tube},g}^2 R_{\mathrm{tube}}, \tag{16}$$

and with $n_{\mathrm{tubes},g} = n_{s,g} n_{\mathrm{str},g}$,

$$I_{\mathrm{tot}} = \sum_g n_{\mathrm{str},g} I_{\mathrm{tube},g}, \qquad P_{\mathrm{tot}} = \sum_g n_{\mathrm{tubes},g} P_{\mathrm{tube},g}. \tag{17}$$

The heater-tube air inlet is the greenhouse ambient state,

$$T_{\mathrm{air,in}} = T_\infty, \tag{18}$$

and the aeration-tube inlet is the heater-tube outlet,

$$T_{\mathrm{air,in,aer}} = T_{\mathrm{air,out,heater}}. \tag{19}$$

For any tube of outer diameter D_o, the outside perimeter used to convert an axial length to an external heat-transfer area is

$$P_{\mathrm{out}} = \pi D_o, \qquad A_{\mathrm{seg,out}} = P_{\mathrm{out}} \Delta x. \tag{20}$$

Type. Equations (6), (7), (8), (10), and (20) are exact algebraic bookkeeping relations on the reduced-order model state. Equation (9) is the adopted closed-end perforation discharge closure based on [R12], pp. 31–32, coupled to the segmentwise Ergun bed-backpressure closure introduced later in Section 5.
Repository-native narrative. The same physical quantities appear in `HeatTransferStudy/VermiHeatLoss.tex:L961-L1034`, `HeatTransferStudy/VermiHeatLoss.tex:L1560-L1738`, and `HeatTransferStudy/VermiHeatLoss.tex:L2302-L2444`.
Implementation. `vermicomposter_air_heater_design_model.m:L763-L875`; `vermicomposter_air_heater_design_model.m:L1497-L1677`; `heat_transfer_study.py:L3895-L3924`; `vermicomposter_air_heater_design_model.m:L1684-L1727`.
Deviation note. The MATLAB implementation permits mixed string sizes for odd tube counts; the equations above are therefore written in grouped summation form rather than assuming a single equivalent series-parallel pattern.

4.3 Heater-tube energy balance

With heater length L_{heater} discretized into N_{heater} segments,

$$\Delta x_{\mathrm{heater}} = \frac{L_{\mathrm{heater}}}{N_{\mathrm{heater}}}, \tag{21}$$

and the local generated electrical heat assigned to segment i is

$$q_{\mathrm{gen},i} = P_{\mathrm{branch}} \frac{\Delta x_{\mathrm{heater}}}{L_{\mathrm{heater}}}. \tag{22}$$

For heater segment i, the air-energy update is

$$T_{\mathrm{air},i+1} = T_{\mathrm{air},i} + \frac{q_{\mathrm{gen},i} - q_{\mathrm{loss},i}}{\dot{m} c_p}. \tag{23}$$

Wire temperature is then recovered from the local wire-to-air temperature rise

$$\Delta T_{\mathrm{wire},i} = \frac{q_{\mathrm{gen},i}}{h_{\mathrm{wire},i} A'_{\mathrm{wire}} \Delta x}, \qquad T_{\mathrm{wire},i} = T_{\mathrm{air},i} + \Delta T_{\mathrm{wire},i}. \tag{24}$$

The segmentwise heater-shell loss is

$$q_{\mathrm{loss},i} = U_{\mathrm{heater}} P_{\mathrm{out}} \Delta x_{\mathrm{heater}} \left(T_{\mathrm{air},i} - T_\infty \right). \tag{25}$$

Type. Equation (22) is exact segment bookkeeping on the reduced model. Equation (23) is a discretized one-dimensional steady energy balance. Equations (24) and (25) are reduced-order algebraic temperature-drop and shell-loss closures.
Repository-native narrative. `HeatTransferStudy/VermiHeatLoss.tex:L38-L121`.
Implementation. `vermicomposter_air_heater_design_model.m:L1803-L1894`; `heat_transfer_study.py:L3775-L3895`.
Deviation note. The wire-to-air temperature difference is represented by a direct convective drop rather than a resolved wire conduction model.

4.4 Aeration-tube segment balance

With aeration length L_{aer} discretized into N_{aer} segments,

$$\Delta x_{\mathrm{aer}} = \frac{L_{\mathrm{aer}}}{N_{\mathrm{aer}}}, \tag{26}$$

with the remaining in-tube mass flow marched by

$$\dot{m}_{\mathrm{rem},i+1} = \dot{m}_{\mathrm{rem},i} - \dot{m}_{\mathrm{rel},i}, \qquad \dot{m}_{\mathrm{rem},1} = \dot{m}_{\mathrm{tube,in}}, \qquad \dot{m}_{\mathrm{rem},N_{\mathrm{aer}}+1} = 0. \tag{27}$$

The inlet gauge pressure is the value for which the coupled perforation-discharge march (9) and axial tube-friction march satisfy the terminal closed-end condition above. The mean throughflow carried by segment i is

$$\dot{m}_{\mathrm{mean},i} = \dot{m}_{\mathrm{rem},i} - \frac{1}{2} \dot{m}_{\mathrm{rel},i}. \tag{28}$$

This is the arithmetic average of the segment-inlet and segment-outlet axial throughflows,

$$\dot{m}_{\mathrm{mean},i} = \frac{\dot{m}_{\mathrm{rem},i} + \dot{m}_{\mathrm{rem},i+1}}{2},$$

obtained by substituting (27). The model uses that midpoint flow as the segment-average air heat-capacity rate for the wall-exchange step, which is why it appears in the exponential wall factor (36) and in $q_{\mathrm{wall},i}$. Using the segment average is a reduced-order approximation to continuously decreasing axial flow within a perforated segment: the inlet flow would overestimate the retained stream's thermal capacity over the whole segment, while the outlet flow would underestimate it. Thus the model does *not* force the last segment's axial throughflow to zero. Rather, the final segment may still carry a nonzero inlet and mean throughflow while releasing the remaining mass, and the zero-flow condition applies only to the post-segment residual $\dot{m}_{\mathrm{rem},N_{\mathrm{aer}}+1}$ at the closed cap. This discrete

closure is physically consistent provided the final modeled segment truly extends to the sealed end and contains the last active sidewall perforations near that sealed end. The end cap itself is not assumed to be perforated. Under that geometry, the final segment can legitimately carry nonzero axial flow while its released side flux drives the downstream residual to zero at the cap. If the real hardware instead includes an unperforated dead-end tail after the last perforation row, that tail should be modeled as an additional no-release terminal section; otherwise the present discretization lumps too much distal release directly against the cap and overcompresses the true near-tip pressure and release profile. The pressure-side closure for the same aeration march, including the local tube-to-bed driving pressure, the segment friction update, the reported branch pressure requirement, the distribution-network losses, and the full heating/cooling total pressure-drop assemblies, is consolidated in Section 5. For aeration segment i, the net bed-side energy contribution is

$$q_{\mathrm{bed},i} = q_{\mathrm{wall},i} + q_{\mathrm{jet},i} - q_{\mathrm{evap},i}. \tag{29}$$

Here $q_{\mathrm{wall},i}$ is the wall-mediated sensible exchange with the bed, $q_{\mathrm{jet},i}$ is the direct released-air enthalpy term, and $q_{\mathrm{evap},i}$ is the latent sink associated with the selected evaporation rate. The wall and release-temperature closures are introduced immediately below in this subsection; the evaporation closure is developed later in the moisture section. Summing over the segment set gives

$$Q_{\mathrm{bed,branch}} = \sum_i q_{\mathrm{bed},i}, \tag{30}$$

and the total bed heat rate across N_{tubes} identical branches is

$$Q_{\mathrm{bed,total}} = N_{\mathrm{tubes}} Q_{\mathrm{bed,branch}}. \tag{31}$$

Both current rooted branches apply the same wall exchange through an exponential segment factor,

$$\frac{d}{dx}[\dot{m}_{\mathrm{mean},i} c_p \left(T - T_{\mathrm{bed,ref}}\right)] = -U_{\mathrm{aer}} P_{\mathrm{out}} \left(T - T_{\mathrm{bed,ref}}\right), \tag{32}$$

which is the standard one-dimensional segment energy balance for a plug-flow air stream exchanging heat with a local bed-side temperature reservoir $T_{\mathrm{bed,ref}}$ through the overall conductance $U_{\mathrm{aer}} P_{\mathrm{out}}$. Over one discrete segment, the implementation holds $\dot{m}_{\mathrm{mean},i}$, c_p, U_{aer}, and $T_{\mathrm{bed,ref}}$ constant, so integrating (32) over Δx_{aer} gives

$$\frac{d(T - T_{\mathrm{bed,ref}})}{T - T_{\mathrm{bed,ref}}} = -\frac{U_{\mathrm{aer}} P_{\mathrm{out}}}{\dot{m}_{\mathrm{mean},i} c_p}\, dx, \tag{33}$$

and therefore, from the segment inlet $x = 0$ with $T = T_{\mathrm{air},i}$ to the segment outlet $x = \Delta x_{\mathrm{aer}}$ with $T = T_{\mathrm{air,after\,wall},i}$,

$$\int_{T_{\mathrm{air},i} - T_{\mathrm{bed,ref}}}^{T_{\mathrm{air,after\,wall},i} - T_{\mathrm{bed,ref}}} \frac{d\theta}{\theta} = -\int_0^{\Delta x_{\mathrm{aer}}} \frac{U_{\mathrm{aer}} P_{\mathrm{out}}}{\dot{m}_{\mathrm{mean},i} c_p}\, dx = -\frac{U_{\mathrm{aer}} P_{\mathrm{out}} \Delta x_{\mathrm{aer}}}{\dot{m}_{\mathrm{mean},i} c_p}, \tag{34}$$

so that

$$\ln\left(\frac{T_{\mathrm{air,after\,wall},i} - T_{\mathrm{bed,ref}}}{T_{\mathrm{air},i} - T_{\mathrm{bed,ref}}}\right) = -\frac{U_{\mathrm{aer}} P_{\mathrm{out}} \Delta x_{\mathrm{aer}}}{\dot{m}_{\mathrm{mean},i} c_p}. \tag{35}$$

Applying the exponential function to both sides of (35) then gives the segment decay factor. This is not a separate fitted relation; it is the direct algebraic solution of the standard first-order constant-property wall-exchange balance represented by (32), consistent with the generic heat-transfer treatment cited in [R5].

$$\Theta_i = \exp\left[-\frac{U_{\mathrm{aer}} P_{\mathrm{out}} \Delta x_{\mathrm{aer}}}{\dot{m}_{\mathrm{mean},i} c_p}\right], \tag{36}$$

$$\frac{T_{\text{air,after wall},i} - T_{\text{bed,ref}}}{T_{\text{air},i} - T_{\text{bed,ref}}} = \Theta_i, \tag{37}$$

so Θ_i is literally the fraction of the segment-inlet temperature excess above $T_{\text{bed,ref}}$ that remains after wall exchange across one aeration segment. Values near 1 indicate weak decay over that segment, whereas values near 0 indicate strong relaxation of the retained tube air toward the bed-side reference temperature.

$$T_{\text{air,after wall},i} = T_{\text{bed,ref}} + (T_{\text{air},i} - T_{\text{bed,ref}})\,\Theta_i, \tag{38}$$

This quantity is therefore a *derived segment-outlet retained-air temperature*, not a separate state that stores the inside-convection, wall-conduction, and outside-transfer contributions individually. Those mechanisms enter only through the upstream definition of U_{aer} and hence through Θ_i; once (38) is evaluated, $T_{\text{air,after wall},i}$ is just the resulting outlet temperature of the retained tube air. and the retained in-tube air state is marched to the next aeration segment by

$$T_{\text{air},i+1} = T_{\text{air,after wall},i}. \tag{39}$$

with

$$q_{\text{wall},i} = \dot{m}_{\text{mean},i} c_p \left(T_{\text{air},i} - T_{\text{air,after wall},i}\right). \tag{40}$$

The discharge-enthalpy term uses the segment-average release temperature

$$T_{\text{rel},i} = \frac{T_{\text{air},i} + T_{\text{air,after wall},i}}{2}. \tag{41}$$

Type. Derived reduced-order balance.
Repository-native narrative. `HeatTransferStudy/VermiHeatLoss.tex:L151-L205`; `HeatTransferStudy/VermiHeatLoss.tex:L1586-L1609`; `HeatTransferStudy/VermiHeatLoss.tex:L2168-L2238`.
Implementation. `vermicomposter_air_heater_design_model.m:L1895-L2076`; `heat_transfer_study.py:L2987-L3277`; `heat_transfer_study.py:L4182-L4323`.
Deviation note. The model treats released-air sensible exchange as a lumped enthalpy source rather than resolving jets in the pore space. The inlet gauge pressure is obtained by scalar shooting/bisection, so the terminal condition $\dot{m}_{\text{rem},N_{\text{aer}}+1} = 0$ is satisfied to numerical tolerance rather than by closed-form inversion.

4.5 Two-node bed balance

Here T_b and T_t are the bottom-node and top-node bed temperatures, respectively; T_∞ is the ambient air temperature; Q_b and Q_t are the net bed heat inputs assigned to the bottom and top nodes; $UA_{b,\infty}$ and $UA_{t,\infty}$ are the bottom-to-ambient and top-to-ambient conductances; and UA_{12} is the internal conductance coupling the two bed nodes. The split factor f_b assigns a fraction of the total bed heat rate $Q_{\text{bed,total}}$ to the bottom node. The bottom and top bed nodes are solved from the steady 2×2 linear system

$$\begin{bmatrix} UA_{b,\infty} + UA_{12} & -UA_{12} \\ -UA_{12} & UA_{t,\infty} + UA_{12} \end{bmatrix} \begin{bmatrix} T_b \\ T_t \end{bmatrix} = \begin{bmatrix} Q_b + UA_{b,\infty} T_\infty \\ Q_t + UA_{t,\infty} T_\infty \end{bmatrix}, \tag{42}$$

with

$$Q_b = f_b Q_{\text{bed,total}}, \qquad Q_t = (1 - f_b) Q_{\text{bed,total}}. \tag{43}$$

In variable-role terms, T_b and T_t are the dependent unknown node temperatures solved from the linear system. The ambient temperature T_∞, the total bed heat rate $Q_{\text{bed,total}}$, the split factor f_b, and the conductances $UA_{b,\infty}$, $UA_{t,\infty}$, and UA_{12} are treated as known inputs for the current operating point. The quantities Q_b and Q_t are therefore not additional unknowns; they are derived forcing terms assembled from $Q_{\text{bed,total}}$ and f_b.

Physically, (42) enforces steady energy balance on two lumped bed control volumes. The bottom node exchanges heat with ambient through $UA_{b,\infty}$, exchanges heat with the top node through UA_{12}, and receives the assigned bed load Q_b. The top node does the analogous thing with $UA_{t,\infty}$, UA_{12}, and Q_t. The diagonal terms in the matrix collect heat leakage out of each node, the off-diagonal terms represent internal node-to-node coupling, and the right-hand side collects the externally imposed heating/cooling plus the ambient forcing.

Numerically, this is solved as a direct dense 2×2 linear solve, not by iteration: MATLAB uses left division $A\backslash b$, and Python uses `np.linalg.solve`. In the rooted heating path and in the annual layer, that direct solve is the entire bed-temperature step once the conductances and $Q_{\text{bed,total}}$ are known. In the summer cooling branch, the same direct two-node solve is nested inside an outer scalar fixed-point/bisection closure for the bed reference temperature, but the inner two-node kernel itself remains this direct algebraic solve.

Type. Derived algebraic reduction of a two-node thermal network.
Repository-native narrative. `HeatTransferStudy/VermiHeatLoss.tex:L365-L424`; `HeatTransferStudy/VermiHeatLoss.tex:L2010-L2079`.
Implementation. `vermicomposter_air_heater_design_model.m:L2233-L2248`; `heat_transfer_study.py:L4432-L4438`.
Deviation note. The current rooted execution path does not use the optional transient MATLAB PID simulation; the authoritative audit path is the steady algebraic solve above.

5 Fluid mechanics, pressure-loss assumptions, and pressure-drop model

This section consolidates the fluid-mechanical assumptions and the exact pressure-drop quantities reported by the current code.

Fluid-mechanical assumptions used by the pressure-loss and aeration-discharge model.

- Internal passages are treated as smooth circular tubes. Tube friction is represented by Darcy–Weisbach pressure loss with $64/Re$ in laminar flow and Churchill's explicit smooth-tube relation otherwise, and abrupt area changes are represented by standard minor-loss coefficients rather than resolved separation or secondary-flow models.
- Flow is not treated with a full compressible momentum formulation. The code does update local density from the ideal-gas relation using the local temperature and pressure state, but the friction, minor-loss, and perforation-discharge closures are all used in their incompressible-form reduced-order versions.
- The optional upstream header loss and the unresolved internal splitter/manifold loss are represented by textbook fitting-loss surrogates from the Crowl–Louvar 2-K method.
- Parallel aeration branches are assumed identical at a given operating point, so one representative closed-end perforated branch is marched axially and then scaled up.
- Perforations are smeared uniformly over the configured segment count rather than resolved hole by hole. The end cap itself is not perforated.

- When an open top or top vent is present, the bed-side pressure opposing perforation discharge is derived segmentwise from a reduced-order Ergun porous-bed escape model leading to the open top or nearest vent. If no such ambient-side opening exists, the current code falls back to the configured bed-side boundary gauge pressure.
- The porous-bed closure is still reduced-order: it uses one representative tube-center height and one tributary top area per tube segment rather than a full three-dimensional porous-bed or headspace-flow network.
- The marched tube gauge pressure is not allowed to fall below the local bed-side gauge pressure used for that segment, so the reduced-order march does not permit reversal of the intended discharge direction.

All pressures in the aeration branch march are written on a gauge basis unless explicitly noted otherwise. Let $p_{\text{tube},i}$ denote the local tube gauge pressure at the inlet of aeration segment i, let $p_{\text{out,offset}}$ denote the configured outlet-boundary gauge-pressure offset supplied through `bedSideGaugePressure_Pa`, let $p_{\text{out,bnd}}$ denote the outlet-boundary gauge pressure at the end of the porous escape path, let $p_{\text{bed},i}$ denote the local bed-side gauge pressure predicted for segment i, and let $p_{\text{in,gauge}}$ denote the aeration-branch inlet gauge pressure returned by the closed-end shooting solve. Here $p_{\text{out,bnd}}$ is the pressure at the top opening or vent outlet reached by the porous escape path, not the pressure adjacent to the tube wall. For one branch assigned an equal share of N_{vent} identical circular top vents of diameter D_{vent}, the total branch-share vent area is

$$A_{\text{vent,branch}} = \frac{N_{\text{vent}} \pi D_{\text{vent}}^2}{4 N_{\text{tubes}}}. \tag{44}$$

By definition, the outlet-boundary gauge pressure $p_{\text{out,bnd}}$ is

$$p_{\text{out,bnd}} = \begin{cases} p_{\text{out,offset}}, & \text{open top,} \\ p_{\text{out,offset}} + \dfrac{1}{2\rho_{\text{out}}} \left(\dfrac{\dot{m}_{\text{tube,in}}}{C_{d,\text{vent}} A_{\text{vent,branch}}} \right)^2, & \text{covered top plus vents,} \\ p_{\text{out,offset}}, & \text{no explicit top opening modeled.} \end{cases} \tag{45}$$

Here $C_{d,\text{vent}} = 0.63$ is the simple-hole discharge coefficient implied by the Crowl–Louvar entrance-plus-exit loss treatment for a hole, for which they note $\sum K_f = 1.5$ and therefore $C_d \approx 0.63$ from equation (4-40) [R14, p. 126]. The boundary density ρ_{out} is evaluated with the same air-property package used elsewhere in the branch march, at the current bed-reference temperature and ambient absolute pressure p_∞. Thus an open top uses a direct atmospheric boundary up to the configured gauge offset, a covered vented top adds a Crowl–Louvar vent-hole discharge backpressure, and a fully closed-top case still falls back to the configured offset because no outlet geometry is available from which to derive a separate top-side discharge loss. Under the repository default $p_{\text{out,offset}} = 0\,\text{Pa}$ gauge, the open-top case therefore corresponds to an atmospheric boundary. The local absolute pressure used for property evaluation is

$$p_{\text{abs},i} = p_\infty + \max(p_{\text{tube},i}, p_{\text{out,bnd}}), \tag{46}$$

where p_∞ is the ambient absolute pressure. For the distribution network, u_{header} is the mean velocity in the upstream header, u_{conn} is the mean velocity in the branch connector, D_{header} is the header inside diameter, $D_{\text{split,in}}$ is the splitter-inlet inside diameter, D_{conn} is the branch-connector inside diameter, D_{down} is the downstream branch-tube diameter seen by the connector-to-branch

transition, and L_{conn} is the branch-connector length. Because the Crowl–Louvar 2-K fitting method is tabulated on an inch basis while the present document is written in SI units, use

$$1 \text{ in} = 0.0254 \text{ m}, \qquad D_{\text{in}} = \frac{D}{0.0254}, \qquad D_{\text{conn,in}} = \frac{D_{\text{conn}}}{0.0254},$$

when inserting SI diameters into the 2-K formulas below. The connector friction factor f_{conn} is the Churchill friction factor evaluated at the connector Reynolds number

$$Re_{\text{conn}} = \frac{\rho u_{\text{conn}} D_{\text{conn}}}{\mu}.$$

If a header is present, it is the shared upstream passage between the blower outlet and the splitter inlet; if no distinct common passage exists there, the model sets `headerEnabled=false` and omits the header term entirely.

The code constructs Reynolds number in the standard form

$$Re = \frac{\rho u D}{\mu}. \tag{47}$$

For laminar internal flow, the Darcy friction factor is

$$f = \frac{64}{Re}. \tag{48}$$

For turbulent internal flow, the implemented smooth-tube Churchill expression is

$$A = \left[2.457 \ln\left(\frac{1}{(7/Re)^{0.9}}\right)\right]^{16}, \qquad B = \left(\frac{37530}{Re}\right)^{16}, \tag{49}$$

$$f = 8\left[\left(\frac{8}{Re}\right)^{12} + \frac{1}{(A+B)^{3/2}}\right]^{1/12}. \tag{50}$$

Tube friction losses are accumulated with Darcy–Weisbach:

$$\Delta p_f = f \frac{L}{D} \frac{\rho u^2}{2}. \tag{51}$$

For the reduced-order porous escape model surrounding one representative aeration branch, let the filled bed height be

$$H_{\text{fill}} = H_{\text{bin}} f_{\text{fill}}, \tag{52}$$

where H_{bin} is the bin interior height and f_{fill} is the filled-volume fraction. In this porous-escape geometry, x denotes the axial coordinate measured from the upstream aeration-tube inlet toward the closed end, and y denotes the vertical coordinate measured upward from the bin floor. The index $i = 1, \ldots, N_{\text{aer}}$ enumerates the uniformly discretized aeration segments, so with $\Delta x_{\text{aer}} = L_{\text{aer}}/N_{\text{aer}}$ the segment-center location is derived directly from the midpoint of segment i. The quantity $y_{\text{tube,rep}}$ is not a solved axial profile; it is the one representative tube-center height used by the reduced-order porous-bed closure for all segments. If the user supplies `representativeTubeCenterHeight_m`, the code uses that value directly. Otherwise it defaults to a heuristic representative height of $0.10 H_{\text{fill}}$, clamped so the tube center remains at least one tube radius above the bin floor and one tube radius below the filled top surface.

Let the tributary top area assigned to one tube segment be

$$A_{\text{bed,trib}} = \frac{W_{\text{bin}}\Delta x_{\text{aer}}}{N_{\text{tubes}}}, \qquad x_i = \left(i - \frac{1}{2}\right)\Delta x_{\text{aer}}. \tag{53}$$

This tributary area is derived from an equal-strip allocation: each segment is assigned a bed strip of axial length Δx_{aer}, full bin width W_{bin}, and one-N_{tubes} share of that top area because the parallel aeration tubes are assumed hydraulically identical in the present reduced-order closure. If the top is open, the porous escape length for each segment is taken as the vertical distance to the filled top surface,

$$L_{\text{bed},i} = H_{\text{fill}} - y_{\text{tube,rep}}. \tag{54}$$

If the top is covered but vented with N_{vent} equal vents assumed evenly spaced over the aeration length L_{aer}, then the vent-center locations are

$$x_{\text{vent},j} = \left(j - \frac{1}{2}\right)\frac{L_{\text{aer}}}{N_{\text{vent}}}, \tag{55}$$

where $j = 1, \ldots, N_{\text{vent}}$ is simply the vent index. The present model does not solve vent positions from geometry data; it derives them from the simplifying assumption that N_{vent} identical vents are evenly spaced over the covered aeration length L_{aer}. and the porous escape length for segment i is approximated by the straight-line distance to the nearest vent center,

$$L_{\text{bed},i} = \sqrt{(H_{\text{fill}} - y_{\text{tube,rep}})^2 + \left(\min_j |x_i - x_{\text{vent},j}|\right)^2}. \tag{56}$$

The local superficial bed velocity associated with one segment release is then

$$u_{\text{bed},i} = \frac{\dot{m}_{\text{rel},i}}{\rho_i A_{\text{bed,trib}}}, \tag{57}$$

which is obtained directly from continuity on the porous-bed side using the released mass flow $\dot{m}_{\text{rel},i}$, the segment-i air density ρ_i, and the tributary bed area $A_{\text{bed,trib}}$. The present model uses this as a *superficial* velocity, meaning it is based on the total assigned cross-sectional bed area rather than on only the void area. It is therefore the standard Ergun velocity scale; the corresponding interstitial pore velocity would be $u_{\text{bed},i}/\varepsilon$, but that is not the quantity inserted into the current closure. and the segmentwise Ergun pressure gradient is

$$\frac{\Delta p_{\text{Ergun},i}}{L_{\text{bed},i}} = 150\frac{(1-\varepsilon)^2}{\varepsilon^3}\frac{\mu_i u_{\text{bed},i}}{d_p^2} + 1.75\frac{1-\varepsilon}{\varepsilon^3}\frac{\rho_i u_{\text{bed},i}^2}{d_p}, \tag{58}$$

where ε is the bed porosity (void fraction) used by the porous-flow closure. In the current repository, the default value is held constant at

$$\varepsilon = \frac{0.801 + 0.799}{2} = 0.800,$$

taken as the arithmetic mean of the Table 2 pre-composting total-porosity values at COM day 21 and day 63, which bracket the repository's nominal 30-day pre-composting window [R16, Table 2, p. 299]. The present reduced-order model keeps that porosity constant rather than evolving it in time because the same source shows only weak variation during the combined composting-plus-vermicomposting process: in the C + V data, total porosity changes from 79.9 to 81.0% v/v between day 63 and

day 203, a spread of only 1.1 percentage points over 140 days [R16, Table 2, p. 299]. A user override remains available through `bedPorosity` if later material-specific measurements are preferred. The local ρ_i and μ_i are the segment-i air density and viscosity returned by the same local air-property package used elsewhere in the branch march. The quantity d_p is the representative particle diameter used by the porous-flow closure. In the current repository, if no explicit `bedParticleDiameter_m` override is supplied, the Ergun model defaults to

$$d_p = d_{32},$$

reusing the same Table 3 Sauter mean diameter defined later by (124) for the particle-surface wetted-area estimate so that the porous-flow and surface-area submodels share the same default particle-size basis. A separate override remains available if later hydraulic calibration indicates that the effective porous-flow diameter should differ from the surface-area diameter. The code then takes the local bed-side gauge pressure as

$$p_{\mathrm{bed},i} = p_{\mathrm{out,bnd}} + \Delta p_{\mathrm{Ergun},i}. \tag{59}$$

If no open top or top vent is present, the present reduced-order implementation does not define a separate ambient-side porous escape path and therefore falls back to

$$p_{\mathrm{bed},i} = p_{\mathrm{out,offset}}.$$

For the closed-end aeration branch, the local pressure difference driving perforation discharge is

$$\Delta p_i = \max(p_{\mathrm{tube},i} - p_{\mathrm{bed},i}, 0)\,, \tag{60}$$

so the local release law (9) is evaluated on that tube-to-bed pressure difference. The explicit Darcy friction drop marched across aeration segment i is

$$\Delta p_{\mathrm{seg},i} = f_i \frac{\Delta x_{\mathrm{aer}}}{D_{\mathrm{aer}}} \frac{\rho_i u_i^2}{2}, \tag{61}$$

with axial pressure update

$$p_{\mathrm{tube},i+1} = \max(p_{\mathrm{tube},i} - \Delta p_{\mathrm{seg},i},\, p_{\mathrm{bed},i})\,, \qquad p_{\mathrm{tube},1} = p_{\mathrm{in,gauge}}. \tag{62}$$

The accumulated explicit tube-friction contribution is therefore

$$\Delta p_{\mathrm{tube,fric}} = \sum_{i=1}^{N_{\mathrm{aer}}} \Delta p_{\mathrm{seg},i}. \tag{63}$$

This quantity is the accumulated explicit friction *drop*, not a local tube-pressure state. The local tube gauge pressure at segment i is $p_{\mathrm{tube},i}$, whereas the cumulative friction drop up to that location may be written

$$\Delta p^{(i)}_{\mathrm{tube,fric}} = \sum_{k=1}^{i-1} \Delta p_{\mathrm{seg},k}. \tag{64}$$

If the bed-pressure floor in (62) never activates, then

$$p_{\mathrm{tube},i} = p_{\mathrm{in,gauge}} - \Delta p^{(i)}_{\mathrm{tube,fric}}. \tag{65}$$

More generally, $p_{\text{in,gauge}} - p_{\text{tube},i}$ is the realized cumulative drop to that location, while $\Delta p_{\text{tube,fric}}$ denotes only the explicit Darcy friction contribution summed over the aeration segments. The reported aeration-branch pressure requirement is the inlet gauge pressure returned by the closed-end discharge-plus-friction shooting solve,

$$\Delta p_{\text{aer,reported}} = p_{\text{in,gauge}}, \tag{66}$$

rather than $\Delta p_{\text{tube,fric}}$ alone. Under the default open-top and zero-offset configuration, $p_{\text{out,bnd}} = 0$ and this reported quantity is the branch pressure drop from the aeration inlet to the atmospheric outlet boundary seen by the porous escape path. Abrupt expansion is modeled as

$$\zeta_{\text{exp}} = \left(1 - \frac{A_1}{A_2}\right)^2, \qquad \Delta p_{\text{exp}} = \zeta_{\text{exp}} \frac{\rho u_1^2}{2}. \tag{67}$$

Here A_1 and A_2 are the upstream and downstream cross-sectional areas of the local area change, and u_1 is the upstream mean velocity. For a conical contraction from the larger area A_1 to the smaller area A_2, let

$$\epsilon = \frac{A_2}{A_1},$$

and let θ denote the included cone angle. The conical contraction surrogate uses an Idel'chik bellmouth coefficient $C(\epsilon, \theta)$ and applies

$$\zeta_{\text{contr}} = C(\epsilon, \theta)\left(\frac{1}{\epsilon} - 1\right), \qquad \Delta p_{\text{contr}} = \zeta_{\text{contr}} \frac{\rho u_2^2}{2}. \tag{68}$$

Here u_2 is the mean velocity in the smaller downstream passage. Whenever the text later writes a shorthand such as

$$\Delta p_{\text{contr}}(\dot{m}, D_1, D_2, \theta),$$

it means: evaluate (68) for a contraction carrying mass flow $\dot{m}$ from upstream diameter D_1 to downstream diameter D_2 at cone angle θ, with

$$A_1 = \frac{\pi D_1^2}{4}, \qquad A_2 = \frac{\pi D_2^2}{4}, \qquad \epsilon = \frac{A_2}{A_1}, \qquad u_2 = \frac{4\dot{m}}{\rho \pi D_2^2}.$$

Likewise, the shorthand

$$\Delta p_{\text{exp}}(\dot{m}, D_1, D_2)$$

means: evaluate (67) for an abrupt expansion carrying mass flow $\dot{m}$ from upstream diameter D_1 to downstream diameter D_2, with

$$A_1 = \frac{\pi D_1^2}{4}, \qquad A_2 = \frac{\pi D_2^2}{4}, \qquad u_1 = \frac{4\dot{m}}{\rho \pi D_1^2}.$$

These are only compact notational shorthands for repeated use of the already defined minor-loss relations; they are not separate empirical formulas.

Following the Crowl–Louvar presentation of Hooper's 2-K method [R14, pp. 124–126], the fitting-loss coefficient for a resolved fitting is

$$K_f = \frac{K_1}{Re} + K_\infty\left(1 + \frac{1}{D_{\text{in}}}\right), \tag{69}$$

where D_{in} is the flow-path inside diameter expressed in inches. For entrances and exits, Crowl and Louvar modify that relation to

$$K_{\text{entr/exit}} = \frac{K_1}{Re} + K_\infty. \tag{70}$$

In the present repository, the optional upstream-header term is idealized as a normal entrance into the separate header, so the code uses

$$K_{\text{header}} = \frac{160}{Re_{\text{header}}} + 0.50, \qquad Re_{\text{header}} = \frac{\rho u_{\text{header}} D_{\text{header}}}{\mu}. \tag{71}$$

Crowl and Louvar list these normal-entrance constants immediately below equation (4-39), stating $K_1 = 160$ and $K_\infty = 0.50$ for a normal entrance [R14, p. 126]. If `headerEnabled=false`, the model omits that header term entirely.

The commercial 4-way shutoff manifold does not publish an internal hydraulic-loss model, so one branch path through the splitter body is approximated as a standard threaded tee used as an elbow followed by one full-line open gate/ball/plug valve, both on the connector basis from Table 4-2 [R14, p. 125]:

$$K_{\text{tee,branch}} = \frac{500}{Re_{\text{conn}}} + 0.70\left(1 + \frac{1}{D_{\text{conn,in}}}\right), \tag{72}$$

$$K_{\text{valve,full}} = \frac{300}{Re_{\text{conn}}} + 0.10\left(1 + \frac{1}{D_{\text{conn,in}}}\right), \tag{73}$$

and therefore

$$K_{\text{splitter,body}} = K_{\text{tee,branch}} + K_{\text{valve,full}} \tag{74}$$

This remains a hardware surrogate because the manifold's actual internal geometry is unresolved, but it is now tied to explicit textbook fitting rows rather than to an arbitrary user-specified lumped K.

The individual distribution-network components reported by the code are all optional in the sense that they appear only when the corresponding geometric element exists in the modeled distribution path. Thus:

- Δp_{header} appears only when a separate upstream header is enabled.
- $\Delta p_{\text{header}\to\text{splitter}}$ appears only when that separate header exists and contracts into the splitter inlet.
- $\Delta p_{\text{splitter}}$ appears only when flow passes through the splitter/manifold body.
- $\Delta p_{\text{connector,fric}}$ appears only when $L_{\text{conn}} > 0$.
- $\Delta p_{\text{connector}\to\text{branch}}$ appears only when the downstream branch-tube diameter differs from the connector diameter.

Written individually, these contributions are

$$\Delta p_{\text{header}} = K_{\text{header}} \frac{\rho u_{\text{header}}^2}{2}, \tag{75}$$

$$\Delta p_{\text{header}\to\text{splitter}} = \Delta p_{\text{contr}}(\dot{m}_{\text{split}}, D_{\text{header}}, D_{\text{split,in}}, \theta_{\text{cone}}), \tag{76}$$

$$\Delta p_{\text{splitter}} = K_{\text{splitter,body}} \frac{\rho u_{\text{conn}}^2}{2}, \tag{77}$$

$$\Delta p_{\text{connector,fric}} = f_{\text{conn}} \frac{L_{\text{conn}}}{D_{\text{conn}}} \frac{\rho u_{\text{conn}}^2}{2}, \tag{78}$$

and

$$\Delta p_{\text{connector}\to\text{branch}} = \begin{cases} \Delta p_{\text{contr}}(\dot{m}_{\text{branch}}, D_{\text{conn}}, D_{\text{down}}, \theta_{\text{cone}}), & D_{\text{down}} < D_{\text{conn}}, \\ \Delta p_{\text{exp}}(\dot{m}_{\text{branch}}, D_{\text{conn}}, D_{\text{down}}), & D_{\text{down}} > D_{\text{conn}}, \\ 0, & D_{\text{down}} = D_{\text{conn}}. \end{cases} \tag{79}$$

Here $\dot{m}_{\text{split}}$ is the mass flow entering the splitter inlet, $\dot{m}_{\text{branch}}$ is the mass flow in one representative branch connector, and θ_{cone} is the configured included contraction-cone angle used in the Idel'chik contraction surrogate. The total distribution-network loss is the sum of whichever of these optional terms are active for the chosen geometry and configuration:

$$\Delta p_{\text{dist}} = \Delta p_{\text{header}} + \Delta p_{\text{header}\to\text{splitter}} + \Delta p_{\text{splitter}} + \Delta p_{\text{connector,fric}} + \Delta p_{\text{connector}\to\text{branch}}. \tag{80}$$

For the summer cooling branch, the exact reported total is

$$\Delta p_{\text{tot,cool}} = \Delta p_{\text{aer,reported}} + \Delta p_{\text{dist}}. \tag{81}$$

For heating electrical-topology group g, the heater-tube internal friction contribution is

$$\Delta p_{\text{heater},g} = \sum_j f_j \frac{\Delta x_{\text{heater}}}{D_{\text{heater}}} \frac{\rho_j u_j^2}{2}, \tag{82}$$

the branch expansion loss from the heater tube into the larger aeration tube is $\Delta p_{\text{exp},g}$, and the exact reported branch total is

$$\Delta p_{\text{branch},g} = \Delta p_{\text{heater},g} + \Delta p_{\text{exp},g} + \Delta p_{\text{aer,reported},g}. \tag{83}$$

Because multiple series-group topologies can coexist in parallel, the reported winter total takes the largest branch requirement across groups and then adds the shared distribution network:

$$\Delta p_{\text{tot,heat}} = \max_g(\Delta p_{\text{branch},g}) + \Delta p_{\text{dist}}. \tag{84}$$

For mixed series-group topologies, the current Python implementation further solves a redistribution closure instead of freezing equal branch flow across all groups. Define the group-private branch-path requirement as

$$\Delta p_{\text{path},g}(\dot{m}_g) = \Delta p_{\text{dist,private},g}(\dot{m}_g) + \Delta p_{\text{heater},g}(\dot{m}_g) + \Delta p_{\text{exp},g}(\dot{m}_g) + \Delta p_{\text{aer},g}(\dot{m}_g), \tag{85}$$

with

$$\Delta p_{\text{heater},g} = \Delta p_{\text{heater,fric},g} + \Delta p_{\text{coil},g}. \tag{86}$$

The coil-obstruction reduced-order model is evaluated segmentwise as

$$\Delta p_{\text{coil},g} = \sum_j \left[M_{\text{coil}}\, C_{D,\text{coil}} \frac{\rho_j u_{n,j}^2}{2} \left(\frac{a_{\text{proj}}}{A_{\text{tube}}} \right) \Delta x_{\text{heater}} \right], \qquad u_{n,j} = u_{\text{bulk},j}\, \phi_{\text{helix}}. \tag{87}$$

Here a_{proj} is the wire projected frontal area per unit heater length, A_{tube} is the heater internal flow area, ϕ_{helix} is the helix-angle factor, $C_{D,\text{coil}}$ is `heaterTube.coilDragCoefficient`, and M_{coil} is `heaterTube.coilLossMultiplier`. This term is a reduced-order surrogate informed by Idel'chik's wire-ribbing/spiral-ribbed resistance discussions [R17], not a direct one-chart transcription.

The redistribution closure introduces a common group-path pressure $\Delta p_{\text{branch,common}}$ such that

$$\Delta p_{\text{path},g}(\dot{m}_g) = \Delta p_{\text{branch,common}}, \qquad \forall g, \tag{88}$$

and enforces total-flow consistency,

$$\sum_g n_{\text{tubes},g}\, \dot{m}_g = \dot{m}_{\text{tot}}. \tag{89}$$

Equation (88) does *not* imply equal $\dot{m}_g$ across groups. It states that each active group connects the same splitter-outlet node to the same downstream outlet-boundary node, so each group is evaluated at its own $\dot{m}_g$ but must satisfy the same node-to-node pressure requirement. Writing common-node pressures explicitly,

$$\Delta p_{\text{dist,common}} = p_0 - p_s, \qquad \Delta p_{\text{path},g}(\dot{m}_g) = p_s - p_{\text{out}}, \tag{90}$$

and therefore

$$\Delta p_{\text{tot,heat}} = p_0 - p_{\text{out}} = \Delta p_{\text{dist,common}} + \Delta p_{\text{branch,common}}. \tag{91}$$

The group-specific branch drops are not summed across g: they are parallel alternatives between the same two nodes, not serial elements traversed by one fluid parcel.

Algorithmically, for a trial $\Delta p_{\text{branch,common}}$, each group inverts (88) by bounded bisection to obtain $\dot{m}_g$, then an outer bisection adjusts the trial pressure to satisfy (89). In residual form,

$$R(\Delta p_{\text{branch,common}}) = \sum_g n_{\text{tubes},g}\, \dot{m}_g(\Delta p_{\text{branch,common}}) - \dot{m}_{\text{tot}} = 0. \tag{92}$$

The explicit outer bracket and bisection used for (92) can be written as

$$\Delta p_{\text{lo}}^{(0)} = 0, \qquad \Delta p_{\text{hi}}^{(0)} = \max_g \left[\Delta p_{\text{path},g}\left(\frac{\dot{m}_{\text{tot}}}{n_{\text{tubes},g}} \right) \right], \tag{93}$$

where the equal-share flow $\dot{m}_{\text{tot}}/n_{\text{tubes},g}$ is used only to construct a conservative upper bracket and is not the final mixed-group operating point unless the groups are hydraulically identical.

$$\Delta p_{\text{mid}}^{(k)} = \frac{\Delta p_{\text{lo}}^{(k)} + \Delta p_{\text{hi}}^{(k)}}{2}, \tag{94}$$

with update

$$\begin{cases} \Delta p_{\text{hi}}^{(k+1)} = \Delta p_{\text{mid}}^{(k)}, \ \Delta p_{\text{lo}}^{(k+1)} = \Delta p_{\text{lo}}^{(k)}, & R\left(\Delta p_{\text{mid}}^{(k)}\right) \geq 0, \\ \Delta p_{\text{lo}}^{(k+1)} = \Delta p_{\text{mid}}^{(k)}, \ \Delta p_{\text{hi}}^{(k+1)} = \Delta p_{\text{hi}}^{(k)}, & R\left(\Delta p_{\text{mid}}^{(k)}\right) < 0, \end{cases} \tag{95}$$

and stop criterion

$$\Delta p_{\text{hi}}^{(k)} - \Delta p_{\text{lo}}^{(k)} \leq \max\left(10^{-6},\, 10^{-6} \max\left[\Delta p_{\text{hi}}^{(k)}, 1\right]\right). \tag{96}$$

For a given trial pressure $\Delta p_{\text{branch,common}}$, the inner group inversion in (88) uses

$$\dot{m}_{\text{lo},g}^{(0)} = 0, \qquad \dot{m}_{\text{hi},g}^{(0)} = \frac{\dot{m}_{\text{tot}}}{n_{\text{tubes},g}}, \tag{97}$$

$$\dot{m}_{\text{mid},g}^{(n)} = \frac{\dot{m}_{\text{lo},g}^{(n)} + \dot{m}_{\text{hi},g}^{(n)}}{2}, \tag{98}$$

then

$$\begin{cases} \dot{m}_{\mathrm{lo},g}^{(n+1)} = \dot{m}_{\mathrm{mid},g}^{(n)},\ \dot{m}_{\mathrm{hi},g}^{(n+1)} = \dot{m}_{\mathrm{hi},g}^{(n)}, & \Delta p_{\mathrm{path},g}(\dot{m}_{\mathrm{mid},g}^{(n)}) \le \Delta p_{\mathrm{branch,common}}, \\ \dot{m}_{\mathrm{hi},g}^{(n+1)} = \dot{m}_{\mathrm{mid},g}^{(n)},\ \dot{m}_{\mathrm{lo},g}^{(n+1)} = \dot{m}_{\mathrm{lo},g}^{(n)}, & \Delta p_{\mathrm{path},g}(\dot{m}_{\mathrm{mid},g}^{(n)}) > \Delta p_{\mathrm{branch,common}}, \end{cases} \tag{99}$$

with evaluated midpoint

$$\dot{m}_{\mathrm{eval},g}^{(n)} = \max\left(\dot{m}_{\mathrm{mid},g}^{(n)},\ \dot{m}_{\mathrm{min,branch}}\right), \tag{100}$$

where $\dot{m}_{\mathrm{min,branch}}$ is the solver-side small-flow floor used only to keep state evaluation well posed. The accepted value is $\dot{m}_g \approx \dot{m}_{\mathrm{lo},g}^{(n)}$ at convergence under

$$\dot{m}_{\mathrm{hi},g}^{(n)} - \dot{m}_{\mathrm{lo},g}^{(n)} \le \max\left(10^{-9},\ 10^{-6}\max\left[\dot{m}_{\mathrm{hi},g}^{(n)}, 1\right]\right). \tag{101}$$

These equations are the document-side statement of the implemented bracket/bisection closure.

Under this mixed-group closure, the reported winter total is

$$\Delta p_{\mathrm{tot,heat}} = \Delta p_{\mathrm{dist,common}} + \Delta p_{\mathrm{branch,common}}, \tag{102}$$

which reduces to (84) when only one group exists. If `headerEnabled=false`, then $\Delta p_{\mathrm{dist,common}} = 0$ and the modeled total is the solved branch-common pressure alone.

Type. Equations (47)–(102) are standard fluid-mechanics relations plus a repository-configured network assembly, textbook-derived fitting surrogates for unresolved hardware, the segmentwise Ergun porous-bed backpressure reduction, and the closed-end branch shooting closure.
Repository-native narrative. `HeatTransferStudy/VermiHeatLoss.tex:L862-L961`; `HeatTransferStudy/VermiHeatLoss.tex:L1899-L2009`.
Implementation. `vermicomposter_air_heater_design_model.m:L1542-L1668`; `vermicomposter_air_heater_design_model.m:L1948-L2034`; `vermicomposter_air_heater_design_model.m:L2203-L2341`; `heat_transfer_study.py:L3128-L3246`; `heat_transfer_study.py:L4515-L4618`; `heat_transfer_study.py:L4698-L4827`.
Primary-source citations. Equations (49)–(50) follow Churchill's explicit all-regime friction relation [R3, pp. 91–92]. Equation (58) uses the standard Ergun packed-bed pressure-drop relation [R15]. The constant default porosity used with that closure is taken from the total-porosity values reported in Fornes et al., Table 2 [R16, p. 299]. Equation (67) uses the Idel'chik sudden-expansion coefficient on the upstream-velocity basis [R6, p. 112]. Equation (68) uses Idel'chik's contraction relation, equation (3-3) [R12, pp. 83–84]. Equation (69) matches Crowl and Louvar's statement of Hooper's 2-K fitting method, equation (4-38) [R14, p. 124]. Equation (70) matches their entrance/exit modification, equation (4-39), with the normal-entrance constants used in (71) taken from the same page [R14, p. 126]. The tee and full-line valve constants used in (72) and (73) come from Table 4-2 [R14, p. 125]. The vented-top outlet-boundary closure (45) uses the same source's simple-hole discussion below equation (4-40), where Crowl and Louvar note $\sum K_f = 1.5$ and $C_d \approx 0.63$ for a hole [R14, p. 126]. Equation (87) is a repository-specific reduced-order projected-area drag surrogate motivated by Idel'chik's wire-ribbing and spiral-ribbed air-heater resistance discussions [R17].
Deviation note. The header and splitter-body terms are still surrogates for unresolved hardware, but they are now based on textbook 2-K fitting rows rather than on arbitrary standalone lumped K-values. The bed-side backpressure is likewise still reduced-order: the current implementation uses one representative tube-center height, one tributary top area per segment, the nearest-top-opening path length, and a branch-share vent-hole outlet closure when vents exist, but it does not solve a full three-dimensional porous-bed or vent-network flow field. The reported aeration-branch pressure requirement is the inlet gauge pressure needed by the coupled closed-end discharge-plus-friction march, not merely the sum of the explicit tube-friction terms.

6 Internal convection correlations

The internal convection coefficient is always formed from

$$h = \frac{Nu\,k}{D}. \tag{103}$$

For laminar developing flow, the implemented Hausen-type relation is

$$Gz = Re\,Pr\,\frac{D}{L}, \tag{104}$$

$$Nu = 3.66 + \frac{0.0668\,Gz}{1 + 0.04\,Gz^{2/3}}, \qquad Re < 2300. \tag{105}$$

For turbulent flow, the implemented Gnielinski relation is

$$Nu = \max\left[3.66,\ \frac{(f/8)(Re - 1000)Pr}{1 + 12.7(f/8)^{1/2}(Pr^{2/3} - 1)}\right], \qquad Re \geq 2300. \tag{106}$$

Type. Empirical correlations.
Repository-native narrative. `HeatTransferStudy/VermiHeatLoss.tex:L80-L103`; `HeatTransferStudy/VermiHeatLoss.tex:L862-L961`.
Implementation. `vermicomposter_air_heater_design_model.m:L2382-L2398`; `heat_transfer_study.py:L3764-L3774`.
Primary-source citations. Equation (105) is the Hausen entry-region correlation [R13, pp. 91–98]. Equation (106) follows Gnielinski's turbulent tube-flow relation [R2, pp. 359–368].
Deviation note. The same branch logic is used for both heater and aeration tubes. No separate roughness correction or transitional blending beyond the branch cutoff is implemented.

7 External convection correlations

For crossflow around the heater wire and perforation-scale characteristic diameter, the code uses the Churchill–Bernstein correlation

$$Nu = 0.3 + \frac{0.62Re^{1/2}Pr^{1/3}}{\left[1 + (0.4/Pr)^{2/3}\right]^{1/4}}\left[1 + \left(\frac{Re}{282000}\right)^{5/8}\right]^{4/5}. \tag{107}$$

For external vertical walls, the Rayleigh number is

$$Ra = \frac{g\beta\Delta T L^3}{\nu\alpha}, \tag{108}$$

and the Churchill–Chu plate relation is

$$Nu = \left[0.825 + \frac{0.387Ra^{1/6}}{\left(1 + (0.492/Pr)^{9/16}\right)^{8/27}}\right]^2, \qquad h = \frac{Nuk}{L}. \tag{109}$$

For horizontal top or vent surfaces, the implemented piecewise free-convection relation is

$$Nu = \begin{cases} 0.54Ra^{1/4}, & T_s \geq T_\infty,\ Ra \leq 10^7, \\ 0.15Ra^{1/3}, & T_s \geq T_\infty,\ Ra > 10^7, \\ 0.27Ra^{1/4}, & T_s < T_\infty. \end{cases} \tag{110}$$

For the Rayleigh evaluation in (108), the implemented property state is the film state

$$T_f = \frac{T_s + T_\infty}{2}, \qquad \beta = \frac{1}{T_{f,\mathrm{K}}}, \qquad \Delta T = |T_s - T_\infty|, \qquad \alpha = \frac{k}{\rho c_p}, \tag{111}$$

with $T_{f,\mathrm{K}} = T_f + 273.15$. Thus, conceptually: β is the thermal-expansion term driving buoyancy, ΔT is the wall-to-ambient buoyancy temperature difference, and α is thermal diffusivity.

For the elevated bottom, the same horizontal-plate family is used with downward-facing orientation logic (stable/unstable branch inverted relative to upward-facing surfaces):

$$Nu_{\mathrm{bottom}} = \begin{cases} 0.27 Ra_{\mathrm{bottom}}^{1/4}, & T_{s,\mathrm{bottom}} \geq T_\infty \text{ (stable; heated downward-facing)}, \\ 0.54 Ra_{\mathrm{bottom}}^{1/4}, & T_{s,\mathrm{bottom}} < T_\infty,\ Ra_{\mathrm{bottom}} \leq 10^7 \text{ (unstable)}, \\ 0.15 Ra_{\mathrm{bottom}}^{1/3}, & T_{s,\mathrm{bottom}} < T_\infty,\ Ra_{\mathrm{bottom}} > 10^7 \text{ (unstable)}. \end{cases} \tag{112}$$

Here "stable" and "unstable" refer to buoyancy stability of the adjacent air layer, not numerical stability: unstable orientation promotes overturning/plume formation and higher free-convection transport, while stable orientation suppresses overturning.

For insulated side walls, covered tops, and elevated bottoms, T_s denotes the exterior insulation-surface temperature from the wall-stack fixed-point solve; for open top/vent openings, the same correlation family is applied to the exposed top-surface temperature proxy.

Type. Empirical correlations.
Repository-native narrative. `HeatTransferStudy/VermiHeatLoss.tex:L30-L57`; `HeatTransferStudy/VermiHeatLoss.tex:L203-L244`; `HeatTransferStudy/VermiHeatLoss.tex:L862-L961`.
Implementation. `vermicomposter_air_heater_design_model.m:L1111-L1155`; `vermicomposter_air_heater_design_model.m:L2372-L2381`; `heat_transfer_study.py:L3710-L3754`.
Primary-source citations. Equation (107) is Churchill and Bernstein's overall cylinder-in-crossflow correlation, equation (9) [R1, p. 303]. Equation (109) is the Churchill–Chu isothermal vertical-plate relation, equation (9) [R4, p. 1325].
Deviation note. The same Churchill–Bernstein form is reused for wire cooling; hole-scale evaporation coupling in the current Python path uses the Frössling-Sherwood closure documented in Section 11.

8 Conduction and thermal resistance networks

The implemented wall-stack coefficient is

$$U_{\mathrm{wall}} = \left(\frac{1}{h_{\mathrm{inside,bin}}} + \frac{t_{\mathrm{sheet}}}{k_{\mathrm{sheet}}} + \frac{t_{\mathrm{ins}}}{k_{\mathrm{ins}}} + \frac{1}{h_{\mathrm{ext}}} \right)^{-1}. \tag{113}$$

The same algebraic structure is used for the heater tube and aeration tube:

$$U_{\mathrm{heater}} = \left(\frac{D_{\mathrm{heater,exp}}}{D_{i,\mathrm{heater}}\, h_{\mathrm{inside,heater}}} + \frac{D_{\mathrm{heater,exp}}}{2k_{\mathrm{wall,heater}}} \ln\frac{D_{o,\mathrm{heater}}}{D_{i,\mathrm{heater}}} + \frac{D_{\mathrm{heater,exp}}}{2k_{\mathrm{ins,heater}}} \ln\frac{D_{\mathrm{heater,exp}}}{D_{o,\mathrm{heater}}} + \frac{1}{h_{\mathrm{outside,heater}}} \right)^{-1}, \tag{114}$$

$$U_{\mathrm{aer}} = \left(\frac{D_{o,\mathrm{aer}}}{D_{i,\mathrm{aer}}\, h_{\mathrm{inside,aer}}} + \frac{D_{o,\mathrm{aer}}}{2k_{\mathrm{wall,aer}}} \ln\frac{D_{o,\mathrm{aer}}}{D_{i,\mathrm{aer}}} + \frac{1}{h_{\mathrm{outside,aer}}} \right)^{-1}. \tag{115}$$

The coefficients used in these resistance networks are defined explicitly as

$$h_{\text{inside,bin}} = h_{\text{bin,in,cfg}}, \quad h_{\text{inside,heater}} = M_{\text{wall}} \frac{Nu_{\text{heater}} k}{D_{i,\text{heater}}}, \quad h_{\text{outside,heater}} = h_{\text{heater,ext,cfg}}, \tag{116}$$

$$h_{\text{inside,aer}} = M_{\text{wall}} \frac{Nu_{\text{aer}} k}{D_{i,\text{aer}}}, \quad h_{\text{outside,aer}} = M_{\text{bed}} h_{\text{bed,cfg}}, \quad h_{\text{wire}} = M_{\text{wire}} \frac{Nu_{\text{wire}} k}{D_{\text{wire}}}. \tag{117}$$

Here $M_{\text{wire}} \equiv$ `calibration.hWireMultiplier`, $M_{\text{wall}} \equiv$ `calibration.hWallMultiplier`, and $M_{\text{bed}} \equiv$ `calibration.bedHTMultiplier`. The configured scalar coefficients are $h_{\text{bin,in,cfg}} \equiv$ `binWall.internalH_W_m2K`, $h_{\text{heater,ext,cfg}} \equiv$ `heaterTube.externalH_W_m2K`, and $h_{\text{bed,cfg}} \equiv$ `aeration.bedH_W_m2K`. In the current default JSON, $h_{\text{bin,in,cfg}} = 6.0\ \text{W}\,\text{m}^{-2}\,\text{K}^{-1}$, $h_{\text{heater,ext,cfg}} = 10.0\ \text{W}\,\text{m}^{-2}\,\text{K}^{-1}$, $h_{\text{bed,cfg}} = 8.0\ \text{W}\,\text{m}^{-2}\,\text{K}^{-1}$, and $M_{\text{wire}} = M_{\text{wall}} = M_{\text{bed}} = 1.0$. **Type.** Simplified thermal resistance networks.
Repository-native narrative. `HeatTransferStudy/VermiHeatLoss.tex:L103-L200`; `HeatTransferStudy/VermiHeatLoss.tex:L203-L244`.
Implementation. `vermicomposter_air_heater_design_model.m:L1102-L1110`; `vermicomposter_air_heater_design_model.m:L1884-L1894`; `heat_transfer_study.py:L3737-L3778`.
Deviation note. Tube-wall radial terms now use exact cylindrical $\ln(r_o/r_i)$ resistances on an outer-area basis, consistent with the current Python implementation (`cylindrical_tube_overall_u_on_outer_basis`). The exponential factor Θ_i is not a fitted correlation; it is the exact segmentwise solution of (32) under the repository's piecewise-constant marching assumptions, consistent with the generic thermal-resistance and differential energy-balance background cited in [R5].

9 Heater-side overall heat transfer treatment

The wire-side convective coefficient is

$$h_{\text{wire}} = \frac{Nu_{\text{wire}} k}{D_{\text{wire}}}, \tag{118}$$

with Nu_{wire} given by (107). The heater external perimeter used in the shell-loss term is

$$P_{\text{heater,ext}} = \pi D_{\text{heater,exp}}, \tag{119}$$

so the heater-shell loss per segment is

$$q_{\text{loss},i} = U_{\text{heater}} P_{\text{heater,ext}} \Delta x \left(T_{\text{air},i} - T_\infty\right), \tag{120}$$

and the local net air heating is

$$q_{\text{to air},i} = q_{\text{gen},i} - q_{\text{loss},i}. \tag{121}$$

Type. Derived reduced-order heater balance.
Repository-native narrative. `HeatTransferStudy/VermiHeatLoss.tex:L38-L121`; `HeatTransferStudy/VermiHeatLoss.tex:L1738-L1898`.
Implementation. `vermicomposter_air_heater_design_model.m:L1803-L1894`; `heat_transfer_study.py:L3775-L3895`.
Deviation note. The Python physics re-solver mirrors the MATLAB heater equations for annual and diagnostic use, but the winter operating grid itself still originates from MATLAB.

10 Aeration-tube heat transfer model

The wall-mediated sensible contribution is evaluated from the exact segment exchange after applying the wall factor (36)–(38),

$$q_{\mathrm{wall},i} = \dot{m}_{\mathrm{mean},i} c_p \left(T_{\mathrm{air},i} - T_{\mathrm{air,after\,wall},i}\right). \tag{122}$$

Here $T_{\mathrm{air},i}$ denotes the air temperature entering aeration segment i, not a constant heater-exit temperature carried unchanged along the tube. In the winter branch, $T_{\mathrm{air},1}$ is the heater-tube outlet temperature (19); in the summer branch, $T_{\mathrm{air},1}$ is reset to the actual spot-cooler supply temperature before the aeration march begins. In the implemented reduced-order closure, the released-air enthalpy term and evaporation term modify the bed-side energy balance and the remaining mass flow, but the retained in-tube air temperature state itself advances through (39) after the wall exchange. The direct released-air enthalpy term is

$$q_{\mathrm{jet},i} = \dot{m}_{\mathrm{rel},i} c_p \left(T_{\mathrm{rel},i} - T_{\mathrm{bed,ref}}\right). \tag{123}$$

Both current rooted branches use the same closed-end discharge law (9) and wall-factor exchange (36)–(40). The winter branch supplies $T_{\mathrm{bed,ref}} = T_{\mathrm{bed}}$, whereas the summer branch supplies the self-consistent branch reference temperature from the two-node cooling solve. In both cases, the net bed contribution still follows (29), so the sign of $q_{\mathrm{jet},i}$ and of the total segment contribution naturally reverses between heating and cooling.

Type. Derived reduced-order segment model.
Repository-native narrative. `HeatTransferStudy/VermiHeatLoss.tex:L151-L205`; `HeatTransferStudy/VermiHeatLoss.tex:L2168-L2238`.
Implementation. `vermicomposter_air_heater_design_model.m:L1895-L2076`; `heat_transfer_study.py:L2987-L3277`; `heat_transfer_study.py:L4182-L4323`.
Deviation note. The local released-air term remains a lumped enthalpy source or sink; the model still does not resolve the discharged jet field in the surrounding porous bed.

11 Warm-air release / jet-heating treatment

The released-air term (123) is the model's surrogate for direct pore-space heating or cooling by discharged air. It should be interpreted as an enthalpy source or sink attached to the released fraction $\dot{m}_{\mathrm{rel},i}$, not as a resolved jet correlation. In the summer branch, the same equation is reused with $T_{\mathrm{air}} < T_{\mathrm{bed,ref}}$, so $q_{\mathrm{jet},i}$ becomes negative and contributes to net cooling. The associated radial plume cooling, worm-safe distance, and habitat-exclusion mathematics are documented in full in the later plume/habitat section beginning at (138).

Type. Reduced-order enthalpy model plus repository-specific plume post-processing.
Repository-native narrative. `HeatTransferStudy/VermiHeatLoss.tex:L1342-L1585`.
Implementation. `heat_transfer_study.py:L1062-L1306`; `heat_transfer_study.py:L1686-L1736`.
Deviation note. The plume model is not used to modify the main tube-marching solution; it is a separate post-processing estimate.

12 Moisture, evaporation, and latent-heat treatment

The current wetted-area basis can be the top plan area, a manual area, or a particle-surface estimate derived from the Table 3 size fractions. For the particle-surface path, the Sauter mean diameter is

$$\frac{1}{d_{32}} = \sum_i \frac{w_i}{d_i}, \tag{124}$$

so the external surface area per occupied bed volume is

$$a_{\text{ext}} = \frac{6}{d_{32}}. \tag{125}$$

If V_{fill} is the filled bed volume and f_{access} the accessible wet-surface fraction,

$$A_{\text{wet}} = f_{\text{access}} a_{\text{ext}} V_{\text{fill}}. \tag{126}$$

The Python water-property package imports `IAPWS_95.py` and uses it as the backend for all pure-water saturation properties used by the summer psychrometric and evaporation calculations. In that module, the saturation pressure is evaluated from the IAPWS ancillary relation

$$\ln\left(\frac{p_{\text{sat}}}{p_c}\right) = \frac{T_c}{T} \sum_{i=1}^{6} a_i \tau^{b_i}, \qquad \tau = 1 - \frac{T}{T_c}, \tag{127}$$

and the saturated liquid and vapor densities are taken directly from the Wagner–Pruss auxiliary saturation relations

$$\frac{\rho_{\ell,\text{sat}}}{\rho_c} = 1 + \sum_{i=1}^{6} b_i \tau^{n_i}, \qquad \ln\left(\frac{\rho_{v,\text{sat}}}{\rho_c}\right) = \sum_{i=1}^{6} c_i \tau^{m_i}, \tag{128}$$

with $(n_i) = (1/3,\ 2/3,\ 5/3,\ 16/3,\ 43/3,\ 110/3)$ and $(m_i) = (2/6,\ 4/6,\ 8/6,\ 18/6,\ 37/6,\ 71/6)$. In the current implementation, these auxiliary densities define the saturation states, and the corresponding saturated liquid and vapor enthalpies are then evaluated from the IAPWS-95 Helmholtz enthalpy relation at $(\rho_{\ell,\text{sat}}, T)$ and $(\rho_{v,\text{sat}}, T)$. For humid-air calculations, the vapor-component saturation density still enters on the standard partial-pressure basis $\rho_{v,\text{sat}}^{(\text{air})} = p_{\text{sat}}/(R_v T)$, because the air-side evaporation model tracks water vapor as a constituent of a gas mixture rather than as a pure saturated steam phase. The latent heat is obtained from that Helmholtz-based saturated enthalpy difference,

$$h_{fg}(T) = h_{v,\text{sat}}(T) - h_{\ell,\text{sat}}(T). \tag{129}$$

The model does not resolve the full three-dimensional jet/interfacial field around each perforation. In the current Python implementation, hole-scale evaporation coupling uses a Frössling-Sherwood reduced-order closure. The local perforation discharge velocity u_{hole} defines

$$Re_{\text{hole}} = \frac{\rho u_{\text{hole}} D_{\text{hole}}}{\mu}, \tag{130}$$

and with $Sc \approx 0.60$, the Sherwood number is

$$Sh_{\text{hole}} = M_{\text{Sh}} \left(2 + 0.552\, Re_{\text{hole}}^{1/2} Sc^{1/3}\right), \tag{131}$$

where $M_{\mathrm{Sh}} \equiv$ `calibration.evaporationSherwoodMultiplier` with backward-compatible fallback to `evaporationHMultiplier`. The effective vapor diffusivity and mass-transfer coefficient are then

$$D_{va} = \frac{\mu}{\rho Sc}, \qquad h_m = \frac{Sh_{\mathrm{hole}} D_{va}}{D_{\mathrm{hole}}}. \tag{132}$$

The vapor-density driving force is

$$\Delta\rho_v = \max[\rho_{v,\mathrm{sat}}(T_s) - \rho_{v,\infty},\, 0]\,. \tag{133}$$

In winter/heating and summer/cooling branches, the inlet relative humidity baseline is taken from `evaporation.relativeHumidity`; in summer, spot-cooler condensate removal can further update the effective inlet vapor state before these limits are evaluated. The diffusion-limited, carrying-capacity-limited, and heat-limited evaporation rates are

$$\dot{m}_{\mathrm{diff}} = h_m A_{\mathrm{seg}} \Delta\rho_v, \tag{134}$$

$$\dot{m}_{\mathrm{cap}} = \dot{V}_{\mathrm{rel,seg}} \Delta\rho_v, \tag{135}$$

$$\dot{m}_{\mathrm{heat}} = \frac{q_{\mathrm{avail}}}{h_{fg}}, \qquad q_{\mathrm{avail}} = \max[\dot{m}_{\mathrm{rel,seg}} c_p (T_{\mathrm{air}} - T_{\mathrm{bed}}),\, 0]\,. \tag{136}$$

The implemented segment evaporation rate and latent sink are

$$\dot{m}_{\mathrm{evap,seg}} = \max[0,\, \min(\dot{m}_{\mathrm{diff}}, \dot{m}_{\mathrm{cap}}, \dot{m}_{\mathrm{heat}})]\,, \qquad q_{\mathrm{evap,seg}} = \dot{m}_{\mathrm{evap,seg}} h_{fg}. \tag{137}$$

Type. Combination of IAPWS-based pure-water property evaluation, empirical transport relations, and repository-specific area bookkeeping.
Repository-native narrative. `HeatTransferStudy/VermiHeatLoss.tex:L1015-L1369`; `HeatTransferStudy/VermiHeatLoss.tex:L1586-L1609`; `HeatTransferStudy/VermiHeatLoss.tex:L2238-L2245`.
Implementation. `vermicomposter_air_heater_design_model.m:L891-L937`; `vermicomposter_air_heater_design_model.m:L2000-L2122`; `IAPWS_95.py:L614-L691`; `heat_transfer_study.py:L1379-L1459`; `heat_transfer_study.py:L2794-L2823`; `heat_transfer_study.py:L3946-L4003`.
Primary-source citations. The particle-size fractions entering (124) come from Frederickson et al., Table 3 [R11, pp. S320–S326]. Equation (127) matches the Wagner–Pruss vapor-pressure auxiliary equation, equation (2.5) [R9, p. 399]. The saturated-liquid and saturated-vapor density closures in (128) are Wagner–Pruss equations (2.6)–(2.7) [R9, p. 399] and are now used directly in the Python implementation through `IAPWS_95.py`. Equation (131) uses the Frössling Sherwood form documented in Towler and Sinnott [R8], and (132) gives the implemented h_m conversion. Equation (129) is implemented from the IAPWS-95 Helmholtz enthalpy relation evaluated at those auxiliary saturation densities; Wagner and Pruss equations (2.10)–(2.11) [R9, p. 400] are the corresponding literature-side saturation-boundary enthalpy relations, but they are not the direct computational path used in the current Python backend.
Deviation note. The summer branch computes inlet humidity after spot-cooler condensate removal, so the effective $\rho_{v,\infty}$ is not always the raw ambient-RH state.

13 Radial plume, worm-safe stand-off, and habitat-exclusion post-processing

The released-air post-processing model starts from the released mass flow rate per unit aeration-tube length,

$$\dot{m}'_{\mathrm{rel}} = \frac{\dot{m}_{\mathrm{tube,in}}}{L_{\mathrm{aer}}}, \tag{138}$$

and the wetted-area density

$$a_{\mathrm{wet}} = \frac{A_{\mathrm{wet}}}{V_{\mathrm{fill}}}. \tag{139}$$

The local wet-surface sink temperature is clamped between the configured surface temperature, the bed-reference temperature, and the released-air temperature:

$$T_{\mathrm{sink}} = \min(\max(T_{s,\mathrm{set}}, T_{\mathrm{bed,ref}}), T_{\mathrm{air,out}})\,. \tag{140}$$

The plume radius is grown as

$$b(\varsigma) = \min(b_{\max},\, b_0 + \alpha_{\mathrm{spr}}\varsigma)\,, \tag{141}$$

so the contacted wetted area per radial length is

$$A'_{\mathrm{wet}}(\varsigma) = \pi a_{\mathrm{wet}} b(\varsigma)^2. \tag{142}$$

In the implemented plume post-processing, the sensible and evaporation-side coefficients are reused from configured bed-side closures rather than re-derived from a new local Nusselt solve:

$$h_{\mathrm{sens}} = M_{\mathrm{bed}} h_{\mathrm{bed,cfg}}, \qquad h_{\mathrm{evap}} = M_{\mathrm{Sh}} h_{\mathrm{sens}}, \tag{143}$$

with $M_{\mathrm{bed}} \equiv$ `calibration.bedHTMultiplier` and $M_{\mathrm{Sh}} \equiv$ `calibration.evaporationSherwoodMultiplier` (legacy fallback to `evaporationHMultiplier`). The sensible and latent radial sink terms are

$$q'_{\mathrm{sens}}(\varsigma) = h_{\mathrm{sens}} A'_{\mathrm{wet}}(\varsigma)\,(T - T_{\mathrm{sink}})\,, \tag{144}$$

$$q'_{\mathrm{lat}}(\varsigma) = \dot{m}'_{\mathrm{evap}}(\varsigma) h_{fg}(T_s), \tag{145}$$

and the plume temperature satisfies

$$\dot{m}'_{\mathrm{rel}} c_p \frac{dT}{d\varsigma} = -\left[q'_{\mathrm{sens}}(\varsigma) + q'_{\mathrm{lat}}(\varsigma)\right]. \tag{146}$$

The local mass-transfer coefficient is formed by the Lewis relation using local air properties,

$$h_m(\varsigma) = \frac{h_{\mathrm{evap}}}{\rho_{\mathrm{air}}(T) c_{p,\mathrm{air}} \mathrm{Le}^{2/3}}, \tag{147}$$

which gives the potential diffusion-driven evaporation rate per radial length,

$$\dot{m}'_{\mathrm{evap,pot}} = h_m(\varsigma) A'_{\mathrm{wet}}(\varsigma)\left[\rho_{v,\mathrm{sat}}(T_s) - \rho_{v,\mathrm{bulk}}(\varsigma)\right]. \tag{148}$$

The code advances the plume humidity explicitly and caps the bulk vapor density at saturation at the updated plume temperature `heat_transfer_study.py:L1241-L1292`. If the worm-safety threshold is T_{safe}, the safe radial offset is the first ς such that

$$T(\varsigma) = T_{\mathrm{safe}}, \tag{149}$$

located in the code by interpolation on the numerically marched profile `heat_transfer_study.py: L1307-L1332`.

For habitat accounting, the filled-bed height and volume are

$$H_{\text{fill}} = f_{\text{fill}} H, \qquad V_{\text{fill}} = LWH_{\text{fill}}, \tag{150}$$

with filled-bed cross section

$$\Omega_{\text{fill}} = \{(x, y) : 0 \le x \le W,\ 0 \le y \le H_{\text{fill}}\}. \tag{151}$$

If the safe offset is ς_{safe}, the exclusion radius around each tube is

$$R_{\text{exc}} = r_0 + \varsigma_{\text{safe}}. \tag{152}$$

For tube i, the local surface-offset coordinate of a Cartesian point (x, y) is

$$\varsigma_i(x, y) = \sqrt{(x - x_i)^2 + (y - y_i)^2} - r_0. \tag{153}$$

Hence the plume-defined unsafe condition $\varsigma_i(x, y) \le \varsigma_{\text{safe}}$ is equivalent to

$$\sqrt{(x - x_i)^2 + (y - y_i)^2} \le r_0 + \varsigma_{\text{safe}} = R_{\text{exc}}, \tag{154}$$

so the cross-sectional exclusion set is the union of the corresponding disks. The pipe and exclusion indicator functions are

$$I_{\text{pipe}}(x, y) = \begin{cases} 1, & \exists i :\ (x - x_i)^2 + (y - y_i)^2 \le r_0^2, \\ 0, & \text{otherwise}, \end{cases} \tag{155}$$

$$I_{\text{exc}}(x, y) = \begin{cases} 1, & \exists i :\ (x - x_i)^2 + (y - y_i)^2 \le R_{\text{exc}}^2, \\ 0, & \text{otherwise}. \end{cases} \tag{156}$$

The overlap-aware set measures for the pipe and plume-exclusion regions are

$$A_{\text{pipe,set}} = \iint_{\Omega_{\text{fill}}} I_{\text{pipe}}(x, y)\, dA, \tag{157}$$

$$A_{\text{exc,set}} = \iint_{\Omega_{\text{fill}}} I_{\text{exc}}(x, y)\, dA, \tag{158}$$

with volumes obtained by extrusion over aeration-tube length,

$$V_{\text{pipe}} = A_{\text{pipe,set}} L_{\text{aer}}, \qquad V_{\text{exc}} = A_{\text{exc,set}} L_{\text{aer}}. \tag{159}$$

The full unsafe set additionally includes wall-return bands and is assembled after those wall-band depths are defined. When the radial plume reaches the left, right, bottom, or top wall, the code computes an inside-wall temperature from the same wall stack used in the main bin-loss model,

$$q''_{\text{wall}} = \frac{T_{\text{plume,wall}} - T_\infty}{\frac{1}{h_{\text{in}}} + \frac{t_{\text{sheet}}}{k_{\text{sheet}}} + \frac{t_{\text{ins}}}{k_{\text{ins}}} + \frac{1}{h_{\text{ext}}}}, \tag{160}$$

$$T_{\text{wall,in}} = T_{\text{plume,wall}} - \frac{q''_{\text{wall}}}{h_{\text{in}}}, \tag{161}$$

then converts that to an added unsafe band depth with a linear wall-normal interpolation between the bed temperature and the wall-contact temperature `heat_transfer_study.py:L1531-L1555`. In the current implementation, nearest tube-center offsets from each wall are computed directly from realized tube-center coordinates,

$$\delta_\ell = \min_i x_i, \qquad \delta_r = W - \max_i x_i, \qquad \delta_b = \min_i y_i, \qquad \delta_t = H_{\text{fill}} - \max_i y_i, \tag{162}$$

so wall-depth limits are derived directly from these tube-to-wall center distances and no separate edge-index set is required for wall localization in this path. Under the default rule-based layout with $f_{w,x} = f_{w,y} = 0.10$, these reduce to $\delta_\ell = \delta_r = 0.10W$ and $\delta_b = \delta_t = 0.10H_{\text{fill}}$. The corresponding wall-to-nearest-tube-surface plume-sampling offsets are

$$g_\ell = \max(\delta_\ell - r_0, 0)\,, \qquad g_r = \max(\delta_r - r_0, 0)\,, \qquad g_b = \max(\delta_b - r_0, 0)\,, \qquad g_t = \max(\delta_t - r_0, 0)\,, \tag{163}$$

and the geometric caps used in the wall-band depth interpolation are

$$d_\ell^{\max} = \delta_\ell, \quad d_r^{\max} = \delta_r, \quad d_b^{\max} = \delta_b, \quad d_t^{\max} = \delta_t. \tag{164}$$

This is an explicit reduced-order assumption: once wall-driven warming has reached the nearest tube centerline from a given wall, additional bed-air temperature rise due only to wall convection is treated as comparatively unsubstantial because direct aeration from nearby tubes is assumed to dominate; accordingly, this wall-return contribution is not propagated as an extra modifier of the plume profile or safe stand-off beyond that geometric limit. For each wall $w \in \{\ell, r, b, t\}$, the active band depth is

$$d_w = \begin{cases} 0, & T_{\text{wall,in},w} \le \max(T_{\text{bed,ref}}, T_{\text{safe}}) \text{ or } d_w^{\max} \le 0, \\ d_w^{\max}, & T_{\text{bed,ref}} \ge T_{\text{safe}}, \\ \operatorname{clip}\left(d_w^{\max} \dfrac{T_{\text{wall,in},w} - T_{\text{safe}}}{T_{\text{wall,in},w} - T_{\text{bed,ref}}}, 0,\ d_w^{\max}\right), & \text{otherwise}, \end{cases} \tag{165}$$

with $d_\ell \equiv d_{w=\ell}$, $d_r \equiv d_{w=r}$, $d_b \equiv d_{w=b}$, and $d_t \equiv d_{w=t}$. The wall-band indicator is

$$I_{\text{wall}}(x, y) = \begin{cases} 1, & x \le d_\ell \ \vee\ x \ge W - d_r \ \vee\ y \le d_b \ \vee\ y \ge H_{\text{fill}} - d_t, \\ 0, & \text{otherwise}, \end{cases} \tag{166}$$

where I_{wall} is a Boolean membership test on Cartesian points, not a returned coordinate. The full unsafe indicator is

$$I_{\text{unsafe}}(x, y) = \max(I_{\text{exc}}(x, y), I_{\text{wall}}(x, y))\,. \tag{167}$$

Define the resulting unsafe set by

$$\Omega_{\text{unsafe}} = \Omega_{\text{exc}} \cup \Omega_{\text{wall}} = \{(x, y) \in \Omega_{\text{fill}} : I_{\text{unsafe}}(x, y) = 1\}.$$

The area integrals are over Cartesian coordinates (x, y); index i appears only inside I_{exc} as a membership test for disk inclusion. In other words, $A_{\text{unsafe,set}}$ is the union-aware planar set measure produced by the Boolean field $I_{\text{unsafe}}(x, y)$, not a sum over separately indexed boundary coordinates. This union-aware Boolean construction automatically counts overlap between wall bands and tube-exclusion disks only once, because I_{unsafe} saturates at 1.

$$A_{\text{unsafe,set}} = \iint_{\Omega_{\text{fill}}} I_{\text{unsafe}}(x, y)\, dA. \tag{168}$$

Equivalent explicit bounds on the rectangular filled domain are

$$A_{\text{unsafe,set}} = \int_0^W \int_0^{H_{\text{fill}}} I_{\text{unsafe}}(x, y)\, dy\, dx. \tag{169}$$

$$V_{\text{unsafe,set}} = A_{\text{unsafe,set}} L_{\text{aer}}. \tag{170}$$

$$V_{\text{unsafe,total}} = V_{\text{unsafe,set}}. \tag{171}$$

$$V_{\text{unsafe,shell}} = V_{\text{unsafe,total}} - V_{\text{pipe}}. \tag{172}$$

$$V_{\text{hab}} = V_{\text{fill}} - V_{\text{unsafe,total}}. \tag{173}$$

For overlap reporting, the naive no-overlap sums are

$$A_{\text{pipe,sum}} = N_{\text{tubes}} \pi r_0^2, \qquad A_{\text{exc,sum}} = N_{\text{tubes}} \pi R_{\text{exc}}^2, \tag{174}$$

with overlap corrections

$$A_{\text{pipe,ov}} = A_{\text{pipe,sum}} - A_{\text{pipe,set}}, \qquad A_{\text{exc,ov}} = A_{\text{exc,sum}} - A_{\text{exc,set}}. \tag{175}$$

Numerically, these set areas are evaluated on a midpoint grid:

$$A_{\text{set}} \approx \left(\frac{W H_{\text{fill}}}{N_g^2} \right) \sum_{j=1}^{N_g} \sum_{k=1}^{N_g} I(x_j^*, y_k^*). \tag{176}$$

In code outputs, the corresponding keys remain `pipe_area_union_m2`, `exclusion_area_union_m2`, `wall_area_union_m2`, and `unsafe_area_union_m2`; there, "union" denotes the overlap-aware Boolean union operation used to construct the set indicator before area integration.

Type. Repository-specific radial thermal/moisture post-processing and overlap-aware geometry bookkeeping.
Repository-native narrative. `HeatTransferStudy/VermiHeatLoss.tex:L1277-L1571`.
Implementation. `heat_transfer_study.py:L1119-L1367`; `heat_transfer_study.py:L1568-L1880`.
Deviation note. The plume model is not part of the core heater or cooling-march solve. It is a post-processing estimator used for the radial cooling plot and the habitat-exclusion metrics.

14 Bin-wall and ambient heat-loss model

The vessel geometry is

$$A_{\text{side}} = 2HL + 2HW, \qquad A_{\text{bottom}} = LW, \qquad A_{\text{top}} = LW. \tag{177}$$

With N_{vent} top vent holes of diameter d_{vent},

$$A_{\text{vent}} = N_{\text{vent}} \frac{\pi d_{\text{vent}}^2}{4}, \qquad A_{\text{top,cov}} = A_{\text{top}} - A_{\text{vent}}. \tag{178}$$

For open-top convection, the characteristic length is taken as

$$L_{c,\text{top}} = \frac{A_{\text{top}}}{P_{\text{top}}}, \qquad P_{\text{top}} = 2(L + W), \tag{179}$$

so the top natural-convection coefficient is

$$h_{\text{top}} = \frac{Nu_{\text{top}} k}{L_{c,\text{top}}}, \tag{180}$$

with Nu_{top} given by (110). The total vessel heat-loss forms can then be written explicitly as

$$Q_{\text{loss,closed}} = U_{\text{wall}}(A_{\text{side}} + A_{\text{bottom}} + A_{\text{top}})(T_{\text{bin}} - T_\infty), \tag{181}$$

$$Q_{\text{loss,open}} = U_{\text{wall}}(A_{\text{side}} + A_{\text{bottom}})(T_{\text{bin}} - T_\infty) + h_{\text{top}} A_{\text{top}}(T_{\text{bin}} - T_\infty). \tag{182}$$

For an open top, the top-node and total ambient conductances are

$$UA_{\text{top,open}} = U_{\text{wall}} A_{\text{top,wallshare}} + h_{\text{top}} A_{\text{top}}, \tag{183}$$

$$UA_{\text{tot,open}} = U_{\text{wall}}(A_{\text{side}} + A_{\text{bottom}}) + h_{\text{top}} A_{\text{top}}. \tag{184}$$

For a covered top with a small vent,

$$UA_{\text{top,vented}} = U_{\text{wall}}(A_{\text{top,cov}} + A_{\text{top,wallshare}}) + h_{\text{top,vent}} A_{\text{vent}}, \tag{185}$$

$$UA_{\text{tot,vented}} = U_{\text{wall}}(A_{\text{side}} + A_{\text{bottom}} + A_{\text{top,cov}}) + h_{\text{top,vent}} A_{\text{vent}}. \tag{186}$$

For a fully covered top,

$$UA_{\text{tot,closed}} = U_{\text{wall}}(A_{\text{side}} + A_{\text{bottom}} + A_{\text{top}}). \tag{187}$$

The orientation-specific exterior coefficients used by the wall stack are

$$h_{\text{side}} = \frac{Nu_{\text{side}} k}{L_{\text{side}}}, \quad h_{\text{bottom}} = \frac{Nu_{\text{bottom}} k}{L_{c,\text{bottom}}}, \quad h_{\text{top,cov}} = \frac{Nu_{\text{top,cov}} k}{L_{c,\text{top}}}, \quad h_{\text{top,vent}} = \frac{Nu_{\text{top,vent}} k}{L_{c,\text{vent}}}, \tag{188}$$

where both $Nu_{\text{top,cov}}$ and $Nu_{\text{top,vent}}$ are evaluated from the same horizontal free-convection family (110); the difference is the characteristic length and area assignment of the covered-top region versus the vent opening.

$$Nu_{\text{top,cov}} = \mathcal{N}_{\text{hor}}(Ra_{\text{top,cov}}), \qquad Nu_{\text{top,vent}} = \mathcal{N}_{\text{hor}}(Ra_{\text{top,vent}}), \tag{189}$$

with

$$L_{c,\text{bottom}} = \frac{A_{\text{bottom}}}{P_{\text{bottom}}}, \qquad L_{c,\text{vent}} = \frac{A_{\text{vent}}}{P_{\text{vent}}}. \tag{190}$$

Using the same planar wall stack form on each orientation,

$$U_{\text{side}} = \left(\frac{1}{h_{\text{inside,bin}}} + \frac{t_{\text{sheet}}}{k_{\text{sheet}}} + \frac{t_{\text{ins}}}{k_{\text{ins}}} + \frac{1}{h_{\text{side}}} \right)^{-1}, \tag{191}$$

$$U_{\text{bottom}} = \left(\frac{1}{h_{\text{inside,bin}}} + \frac{t_{\text{sheet}}}{k_{\text{sheet}}} + \frac{t_{\text{ins}}}{k_{\text{ins}}} + \frac{1}{h_{\text{bottom}}} \right)^{-1}, \quad U_{\text{top,cov}} = \left(\frac{1}{h_{\text{inside,bin}}} + \frac{t_{\text{sheet}}}{k_{\text{sheet}}} + \frac{t_{\text{ins}}}{k_{\text{ins}}} + \frac{1}{h_{\text{top,cov}}} \right)^{-1}. \tag{192}$$

The geometry-inclusive exterior effective conductance and coefficient can be written as

$$\begin{aligned} UA_{\text{tot,open}} &= U_{\text{side}} A_{\text{side}} + U_{\text{bottom}} A_{\text{bottom}} + h_{\text{top}} A_{\text{top}}, \\ UA_{\text{tot,cov}} &= U_{\text{side}} A_{\text{side}} + U_{\text{bottom}} A_{\text{bottom}} + U_{\text{top,cov}} A_{\text{top}}, \\ UA_{\text{tot,vent}} &= U_{\text{side}} A_{\text{side}} + U_{\text{bottom}} A_{\text{bottom}} + U_{\text{top,cov}} A_{\text{top,cov}} + h_{\text{top,vent}} A_{\text{vent}}, \end{aligned} \tag{193}$$

$$A_{\text{ext}} = A_{\text{side}} + A_{\text{bottom}} + A_{\text{top}}, \qquad U_{\text{ext}} = \frac{UA_{\text{tot}}}{A_{\text{ext}}}. \tag{194}$$

The lumped required heating or cooling load is

$$Q_{\text{req}} = UA_{\text{tot}} \left(T_{\text{set}} - T_{\infty}\right). \tag{195}$$

The characteristic length and Biot estimate are

$$L_c = \frac{V_{\text{fill}}}{A_{\text{side}} + A_{\text{bottom}} + A_{\text{top}}}, \qquad Bi = \frac{h_{\text{ext}} L_c}{k_{\text{bed}}}. \tag{196}$$

The lumped time constant is

$$\tau = \frac{C}{UA_{\text{tot}}}, \tag{197}$$

and the associated cooldown curve is

$$T(t) = T_{\infty} + \left(T_0 - T_{\infty}\right) e^{-t/\tau}. \tag{198}$$

Type. Derived thermal-network relations plus a repository-specific vented-cover extension.
Repository-native narrative. `HeatTransferStudy/VermiHeatLoss.tex:L603-L847`.
Implementation. `vermicomposter_air_heater_design_model.m:L971-L1171`; `heat_transfer_study.py:L4326-L4431`.
Deviation note. The rooted execution path only plots the lumped cooldown curve when $Bi < 0.1$; the equation exists regardless, but the code treats the plot as conditionally valid rather than unconditional.

14.1 Summer spot-cooler and assist-blower surrogate

In the summer branch, the requested sensible cooling duty for target supply temperature $T_{\text{supply,target}}$ is

$$\dot{Q}_{\text{req,cool}} = \max\left[\dot{m}c_p \left(T_{\text{amb}} - T_{\text{supply,target}}\right), 0\right]. \tag{199}$$

The actual sensible cooling applied by the spot cooler is capped at the configured or derived sensible capacity:

$$\dot{Q}_{\text{actual,cool}} = \min\left(\dot{Q}_{\text{req,cool}}, \dot{Q}_{\text{max,cool}}\right), \qquad T_{\text{supply}} = T_{\text{amb}} - \frac{\dot{Q}_{\text{actual,cool}}}{\dot{m}c_p}. \tag{200}$$

If a coefficient of performance is supplied,

$$P_{\text{spot}} = \frac{\dot{Q}_{\text{actual,cool}}}{\text{COP}}. \tag{201}$$

The aeration-network inlet is then reset to the actual spot-cooler supply state,

$$T_{\text{air,in}} = T_{\text{supply,actual}}, \tag{202}$$

while the electrical heater is disabled:

$$P_{\text{tot}} = 0, \qquad I_{\text{tot}} = 0. \tag{203}$$

The post-cooler condensate calculation uses the dry-air flow rate $\dot{m}_{\mathrm{dry}}$ and the saturated outlet humidity ratio $w_{\mathrm{sat,out}}$:

$$\dot{m}_{\mathrm{cond,psych}} = \max[\dot{m}_{\mathrm{dry}}\,(w_{\mathrm{in}} - w_{\mathrm{sat,out}})\,,\,0]\,. \tag{204}$$

If a rated condensate capacity $\dot{m}_{\mathrm{cond,rated}}$ is provided, the applied condensate rate is

$$\dot{m}_{\mathrm{cond}} = \min(\dot{m}_{\mathrm{cond,psych}}, \dot{m}_{\mathrm{cond,rated}})\,, \tag{205}$$

otherwise $\dot{m}_{\mathrm{cond}} = \dot{m}_{\mathrm{cond,psych}}$. The supply humidity ratio is then

$$w_{\mathrm{out}} = \max\left(w_{\mathrm{in}} - \frac{\dot{m}_{\mathrm{cond}}}{\dot{m}_{\mathrm{dry}}},\,0\right). \tag{206}$$

The assist blower is represented by a linear static curve. With shutoff pressure $\Delta p_{\mathrm{shutoff}}$, rated-duty point $(\dot{V}_{\mathrm{rated}}, \Delta p_{\mathrm{rated}})$, and implied free-air flow $\dot{V}_{\mathrm{free}}$,

$$\dot{V}_{\mathrm{free}} = \dot{V}_{\mathrm{rated}} \frac{\Delta p_{\mathrm{shutoff}}}{\Delta p_{\mathrm{shutoff}} - \Delta p_{\mathrm{rated}}}, \tag{207}$$

$$\Delta p_{\mathrm{avail}}(\dot{V}) = \max\left[\Delta p_{\mathrm{shutoff}}\left(1 - \frac{\dot{V}}{\dot{V}_{\mathrm{free}}}\right),\,0\right]. \tag{208}$$

The sampled summer flow is now treated as a requested target $\dot{V}_{\mathrm{req}}$, not as an automatically achieved operating point. The code first evaluates the full reduced-order system pressure requirement $\Delta p_{\mathrm{tot}}(\dot{V})$ from the spot-cooler state, distribution losses, closed-end aeration-tube march, evaporation model, and two-node bed closure. It then forms the scalar blower/system residual

$$R_{\dot{V}}(\dot{V}) = \Delta p_{\mathrm{tot}}(\dot{V}) - \Delta p_{\mathrm{avail}}(\dot{V}). \tag{209}$$

The achieved cooling flow is

$$\dot{V}_{\mathrm{ach}} = \begin{cases} \dot{V}_{\mathrm{req}}, & R_{\dot{V}}(\dot{V}_{\mathrm{req}}) \le 0, \\ \dot{V} \in [0, \dot{V}_{\mathrm{req}}] \text{ such that } R_{\dot{V}}(\dot{V}) = 0, & R_{\dot{V}}(\dot{V}_{\mathrm{req}}) > 0. \end{cases} \tag{210}$$

All reported summer quantities are then recomputed at $\dot{V}_{\mathrm{ach}}$, so the returned aeration inlet gauge pressure, branch pressure profile, segment releases, and cooling duty are mechanically consistent with the assist-blower curve rather than merely post-checked against it. The total summer water-loss rate is

$$\dot{m}_{\mathrm{water,day}} = 86400\, N_{\mathrm{tubes}} \sum_i \dot{m}_{\mathrm{evap},i}, \tag{211}$$

the reported net cooling is

$$Q_{\mathrm{cool}} = \max(-Q_{\mathrm{bed,total}},\,0)\,, \tag{212}$$

and the corresponding lumped cooling requirement at the cooling setpoint is

$$Q_{\mathrm{req,cool}} = U A_{\mathrm{tot}}\,(T_{\infty,\mathrm{summer}} - T_{\mathrm{set,cool}})\,. \tag{213}$$

Type. Repository-specific surrogate model. Equations (199)–(201) are simplified sensible-capacity relations; (204)–(206) are psychrometric mass-balance relations; (207)–(210) are a linearized blower curve plus repository operating-point closure.

Repository-native narrative. `HeatTransferStudy/VermiHeatLoss.tex:L2084-L2284`.

Implementation. `heat_transfer_study.py:L2760-L3378`.

Deviation note. The blower curve is still a two-point linear approximation rather than a full manufacturer fan map, and the spot-cooler model does not include a detailed refrigeration-cycle calculation. The new operating-point solve corrects the previous inconsistency where an infeasible requested flow could be marched thermally and only later flagged by a static-pressure residual.

14.2 Explicit assumptions for Sections 6–14

- Section 6 (Internal convection correlations): heater and aeration in-tube convection are treated as smooth, circular, one-dimensional internal flows with a laminar/turbulent branch switch at $Re = 2300$. The implementation does not include explicit roughness corrections, transition blending beyond that branch switch, or secondary-flow corrections.
- Section 7 (External convection correlations): external free-convection coefficients are represented by Churchill–Chu / horizontal-plate reduced-order closures evaluated at a film state. For insulated boundaries, T_s is the *exterior insulation-surface temperature*; for open/vent top openings, the same correlation family is applied to the exposed top-surface proxy.
- Section 8 (Conduction and thermal resistance networks): the vessel wall is modeled as a planar layered wall with lumped inside/outside convection resistances, while tube walls use exact cylindrical logarithmic resistances on an exposed-outer-area basis. Material properties and configured h-values are treated as piecewise constants within each evaluation state.
- Section 9 (Heater-side treatment): wire-to-air convection is correlation-derived, but the heater-shell exterior coefficient is a configured effective value representing unresolved ambient-side losses around the insulated heater tube. The helical-wire obstruction is captured by a reduced-order projected-area drag surrogate rather than by a resolved ribbed-passage CFD model.
- Section 10 (Aeration-tube model): the aeration tube is solved as a one-dimensional axial march with closed-end mass closure and segmentwise wall exchange. The released-air enthalpy and evaporation effects are coupled into segment energy bookkeeping, but the in-tube temperature state itself advances through the retained marching relation.
- Section 11 (Warm-air release / jet-heating): direct released-air heating/cooling is represented as an enthalpy source/sink term attached to released mass flow, not as a resolved jet-impingement field. This term is sufficient for branch-scale energy closure but does not capture three-dimensional local plume/recirculation structure.
- Section 12 (Moisture and latent heat): hole-scale evaporation coupling uses a Frössling-Sherwood mass-transfer closure driven by local perforation discharge velocity, with additional carrying-capacity and available-heat limits. Reusing this reduced-order closure is reasonable at the current model fidelity because it preserves the dominant Re-dependent transfer trend without adding a full interfacial transport solve.
- Section 13 (Radial plume and habitat exclusion): plume cooling and humidity evolution are post-processing estimates in a radial coordinate ς, and unsafe habitat metrics are set-based geometric measures on the Cartesian bed cross-section. Wall-return heating is capped at nearest tube-center offsets, assuming direct aeration dominates additional thermal influence beyond that distance.
- Section 14 (Bin-wall and ambient heat loss): side, bottom, covered-top, and vent-opening contributions are modeled with orientation-specific h and U values then collapsed to mode-dependent total UA. Covered-top and vent-opening natural convection use the same horizontal Nusselt family with different characteristic lengths/areas.

15 Constraints and feasibility conditions

The current heating hard-constraint residual vector is

$$\mathbf{g}_{\text{heat}} = \begin{bmatrix} \max(0, I - I_{\max}) \\ \max(0, T_{\text{wire}} - T_{\text{wire,max}}) \\ \max(0, T_{\text{air,out}} - T_{\text{air,out,max}}) \\ \max(0, \Delta p - \Delta p_{\max}) \\ \max(0, \dot{m}_{\text{water}} - \dot{m}_{\text{water,max}}) \\ \max(0, u_h - u_{h,\max}) \\ \max(0, T_{b,\min} - T_b) \\ \max(0, T_{t,\min} - T_t) \\ \max(0, T_b - T_{b,\max}) \\ \max(0, T_t - T_{t,\max}) \\ \max(0, |T_b - T_t| - \Delta T_{\max}) \\ \max(0, \text{dutyNeeded} - 1) \\ \max(0, -Q_{\text{bed,total}}) \end{bmatrix}. \tag{214}$$

The current cooling hard-constraint residual vector is

$$\mathbf{g}_{\text{cool}} = \begin{bmatrix} \max(0, \Delta p - \Delta p_{\text{avail}}) \\ \max(0, \Delta p - \Delta p_{\max}) \\ \max(0, \dot{m}_{\text{water}} - \dot{m}_{\text{water,max}}) \\ \max(0, u_h - u_{h,\max}) \\ \max(0, T_{b,\min} - T_b) \\ \max(0, T_{t,\min} - T_t) \\ \max(0, T_b - T_{b,\max}) \\ \max(0, T_t - T_{t,\max}) \\ \max(0, |T_b - T_t| - \Delta T_{\max}) \\ \max(0, -Q_{\text{cool}}) \end{bmatrix}, \tag{215}$$

where $Q_{\text{cool}} = \max(-Q_{\text{bed,total}}, 0)$.

Type. Repository-defined feasibility residuals.
Repository-native narrative. Heating constraints appear in `HeatTransferStudy/VermiHeatLoss.tex:L1614-L1738`. The historical summer lexicographic block at `HeatTransferStudy/VermiHeatLoss.tex:L2251-L2279` is intentionally *not* treated as governing because the live Python code now uses constrained NSGA-II for cooling.
Implementation. `heat_transfer_study.py:L2220-L2451`; `heat_transfer_study.py:L2963-L3147`.
Deviation note. The cooling NSGA-II constraints require positive cooling capacity, but they do *not* currently enforce $Q_{\text{cool}} \geq Q_{\text{req,cool}}$. Cooling shortfall is computed for diagnostics, not as a hard feasibility constraint.

16 Objective functions and optimization quantities

Heating now uses constrained NSGA-II on the discrete candidate grid with objective vector

$$\mathbf{f}_{\text{heat}} = \begin{bmatrix} P_{\text{tot}} \\ -\min(T_b, T_t) \\ |T_b - T_t| \end{bmatrix}. \tag{216}$$

The reported heating design is then selected from the feasible Pareto set by the report-priority tuple

$$\pi_{\text{heat,report}} = \begin{bmatrix} -\min(T_b, T_t) \\ P_{\text{tot}} \\ |T_b - T_t| \\ -\frac{1}{2}(T_b + T_t) \\ \max(0, \text{dutyNeeded} - 1) \end{bmatrix}. \tag{217}$$

Cooling now uses constrained NSGA-II as well. The default cooling objective vector is

$$\mathbf{f}_{\text{cool}} = \begin{bmatrix} \max(T_b, T_t) \\ \frac{1}{2}(T_b + T_t) \\ P_{\text{spot}} + P_{\text{blower}} \end{bmatrix}, \tag{218}$$

and the reported cooling design is selected from the feasible cooling Pareto set by

$$\pi_{\text{cool,report}} = \begin{bmatrix} \max(T_b, T_t) \\ \frac{1}{2}(T_b + T_t) \\ P_{\text{spot}} + P_{\text{blower}} \\ \dot{V}_{\text{tot}} \\ |T_b - T_t| \\ -Q_{\text{cool}} \end{bmatrix}. \tag{219}$$

If no feasible cooling Pareto point exists, the code falls back to a best-available diagnostic ranking using the configured ordered tuple

$$\pi_{\text{cool,fallback}} = \left[g_{\sigma(1)}, g_{\sigma(2)}, \ldots, g_{\sigma(m)}\right], \tag{220}$$

where σ is the configured `priority_order` permutation over the implemented cooling metrics and residuals `heat_transfer_study.py:L3133-L3149`; `heat_transfer_study.py:L3173-L3177`; `heat_transfer_study.py:L3244-L3246`. This is a live fallback/reporting rule, not the stale historical summer primary-selection rule described in `HeatTransferStudy/VermiHeatLoss.tex:L2251-L2279`.

Type. Repository-defined optimization quantities.
Repository-native narrative. Heating NSGA-II is described in `HeatTransferStudy/VermiHeatLoss.tex:L1614-L1738`. The stale summer lexicographic primary-selection narrative in `HeatTransferStudy/VermiHeatLoss.tex:L2251-L2279` is intentionally excluded from the governing-equation set here because it is not implemented.
Implementation. `heat_transfer_study.py:L51-L121`; `heat_transfer_study.py:L2181-L2499`; `heat_transfer_study.py:L3148-L3208`; `heat_transfer_study.py:L6220-L6490`.
Deviation note. NSGA-II is used on an already evaluated discrete set. It is not replacing the physics solve with a continuous surrogate optimizer.

17 Numerical approximations, interpolation, and discretization

- The heating design space is generated by exhaustive discrete sweep in MATLAB, not by direct continuous optimization `vermicomposter_air_heater_design_model.m:L1172-L1476`.
- Electrical resistance depends on average wire temperature, so each operating point uses a short fixed-point iteration on $T_{\text{wire,avg}}$ `vermicomposter_air_heater_design_model.m:L1497-L1536`.

- Heater and aeration transport are solved by explicit axial marching over configured segment counts `vermicomposter_air_heater_design_model.m:L1803-L1999`; `heat_transfer_study.py:L2707-L2959`; `heat_transfer_study.py:L3775-L3959`.
- When the assist blower is enabled, the summer branch treats the sampled flow as a request and applies an outer scalar bisection on $\dot{V}$ so that $\Delta p_{\mathrm{tot}}(\dot{V})$ matches the available blower static if the requested flow is not directly achievable `heat_transfer_study.py:L2793-L2858`; `heat_transfer_study.py:L3061-L3378`.
- The summer branch solves for a bed-reference temperature by bracketing and bisection over the scalar residual relating branch cooling to the lumped required cooling target `heat_transfer_study.py:L2774-L2899`.
- Display interpolation in the plotting layer is presentation-only and does not alter the underlying evaluated candidate set `heat_transfer_study.py:L555-L562`.
- The annual climate layer uses periodic Gaussian-kernel regression over day-of-year, overlays freeze-day events, and applies heat-wave spell anomalies through seeded bounded-normal sampling within NOAA-derived monthly spell min/max bounds `heat_transfer_study.py:L4633-L4858`.
- Radial plume and habitat calculations use midpoint-grid union-area estimates rather than closed-form geometric intersection formulas `heat_transfer_study.py:L1383-L1595`.
- NSGA-II is applied to integer-valued indices or discrete point indices through `pymoo`; the optimizer explores the discrete repository candidate set rather than an interpolated continuous space `heat_transfer_study.py:L2410-L2499`; `heat_transfer_study.py:L3123-L3208`.

Detailed marching and fixed-point closures

The current heating solver uses the temperature-dependent wire resistivity

$$\rho_{\mathrm{wire}}(T) = \rho_{\mathrm{wire},20}\left[1 + \alpha_{\mathrm{wire}}(T - 20^{\circ}\mathrm{C})\right], \tag{221}$$

which yields a tube or branch resistance of the form already introduced in (14). The fixed-point iteration updates the average wire temperature by under-relaxation:

$$\overline{T}_{\mathrm{wire}}^{(k+1)} = 0.65\,\overline{T}_{\mathrm{wire}}^{(k)} + 0.35\,\overline{T}_{\mathrm{wire,new}}^{(k)}, \tag{222}$$

and stops when

$$\left|\overline{T}_{\mathrm{wire,new}}^{(k)} - \overline{T}_{\mathrm{wire}}^{(k)}\right| < 0.05^{\circ}\mathrm{C}, \tag{223}$$

or when the configured iteration cap is reached `vermicomposter_air_heater_design_model.m:L1497-L1536`. The winter aeration pressure-drop march uses the segment contribution

$$\Delta p_{\mathrm{seg}} = f\frac{\Delta x}{D}\frac{\rho u^2}{2}, \tag{224}$$

with the area-change and distribution-network losses added separately through the full Section 5 assembly `vermicomposter_air_heater_design_model.m:L1953-L1956`; `vermicomposter_air_heater_design_model.m:L2123-L2232`.

For the closed-end aeration inlet-pressure shooting solve, the scalar residual is the terminal remaining flow above the mass-balance tolerance,

$$F(p_{\mathrm{in}}) = \dot{m}_{\mathrm{rem},N_{\mathrm{aer}}+1}(p_{\mathrm{in}}) - \max\left(10^{-9},\, 10^{-4}\dot{m}_{\mathrm{tube,in}}\right). \tag{225}$$

Within each marched segment, the local released mass flow is itself found from an inner scalar root because the Ergun backpressure depends on the unknown release rate. With

$$G_i(\dot{m}_{\mathrm{rel},i}) = C_d A_{h,\mathrm{seg}} \sqrt{2\rho_i \max(p_{\mathrm{tube},i} - p_{\mathrm{bed},i}(\dot{m}_{\mathrm{rel},i}), 0)} - \dot{m}_{\mathrm{rel},i},$$

the code brackets $\dot{m}_{\mathrm{rel},i} \in [0, \dot{m}_{\mathrm{rem},i}]$ and applies bounded bisection on that interval until the local discharge and the local porous-bed backpressure closure are mutually consistent. If the release implied by the local pressure difference exceeds the remaining segment inlet flow, the upper bound $\dot{m}_{\mathrm{rem},i}$ is accepted directly. The code starts from $p_{\mathrm{lo}} = 0$ and an initial high-side estimate based on the whole-hole-area discharge approximation described earlier. A pressure bracket is considered found when the marched state at the current upper bound satisfies

$$F(p_{\mathrm{hi}}) \leq 0,$$

because the low side still leaves too much flow at the cap while the high side releases enough mass that the terminal residual is within tolerance. The code therefore doubles p_{hi} until this condition is first met, then bisects on p_{in} within $[p_{\mathrm{lo}}, p_{\mathrm{hi}}]$ `heat_transfer_study.py:L2914-L2955`; `vermicomposter_air_heater_design_model.m:L1895-L1946`. If no such upper bound is found within the configured doubling limit, the highest tried pressure state is returned as the best available unresolved case. Here $\dot{m}_{\mathrm{rem},N_{\mathrm{aer}}+1}$ should be interpreted as the numerical residual of a trial pressure march, not as a physical steady leakage through the sealed cap. A trial inlet pressure that is too low leaves positive residual flow after the last perforated segment because the guessed pressure field does not drive enough side release. A trial inlet pressure that is too high drives release too aggressively upstream and can deplete the branch early; because each segment release is clamped so it cannot exceed the remaining flow, the residual never becomes meaningfully negative. The accepted solution is therefore the *smallest* bracketed inlet pressure that satisfies the closed-end mass-balance tolerance, not an arbitrary overpressured state. For optimizer hygiene, the live Python summer workflow now records whether this inner pressure bracket was successfully found. Any summer point with an unresolved closed-end pressure solve is retained only for diagnostics and is automatically excluded from Pareto eligibility before NSGA-II ranking `heat_transfer_study.py:L2914-L2955`; `heat_transfer_study.py:L3531-L3588`. This prevents a bracket-failure edge case from contaminating the reported cooling Pareto set even though the unresolved point is still available for debugging output.

The summer branch closes the bed-reference temperature by solving the scalar consistency condition

$$R(T_{\mathrm{bed,ref}}) = \frac{T_b(T_{\mathrm{bed,ref}}) + T_t(T_{\mathrm{bed,ref}})}{2} - T_{\mathrm{bed,ref}} = 0, \tag{226}$$

where T_b and T_t are obtained from the two-node bed solve using the branch heat rate predicted at that trial reference temperature `heat_transfer_study.py:L2910-L2917`. If the lower and upper trial bounds bracket a sign change, the code applies bisection:

$$T_{\mathrm{mid}} = \frac{T_{\mathrm{lo}} + T_{\mathrm{hi}}}{2}, \tag{227}$$

updating the bracket by the sign of $R(T_{\mathrm{mid}})$ until either

$$|R(T_{\mathrm{mid}})| < 0.02 \qquad \text{or} \qquad |T_{\mathrm{hi}} - T_{\mathrm{lo}}| < 0.02, \tag{228}$$

otherwise the closer endpoint is retained `heat_transfer_study.py:L2918-L2947`. These are repository numerical constructs rather than textbook closed forms.

When the assist blower is enabled, the summer branch also closes the achieved airflow through the scalar residual (209). If the requested airflow already satisfies $R_{\dot{V}}(\dot{V}_{\text{req}}) \leq 0$, that request is retained. Otherwise the code bisects over $[0, \dot{V}_{\text{req}}]$, keeping the highest bracketed flow that still satisfies $R_{\dot{V}} \leq 0$, until the flow bracket is smaller than $10^{-6} \max(\dot{V}_{\text{req}}, 1)$ in code units `heat_transfer_study.py: L2793-L2858`. The branch thermodynamics are then recomputed at that achieved flow before the cooling metrics are reported.

18 Annual climate and year-round operating layer

18.1 Detailed Methodology for Section 18: Annual Climate and Year-Round Operating Layer

This section documents, in report form, how I implemented the annual climate and daily duty simulation described by Section 18 (229–251). I wrote this layer to convert monthly climate statistics into a physically consistent daily operating profile, then compute daily heating or cooling duty, bed temperature response, electricity use, cost, and water replacement.

Important modeling clarification (Gaussian terminology). Section 18 uses periodic Gaussian-kernel regression (weighted averaging), not Gaussian Process Regression (GPR). In this implementation, no posterior covariance model is solved and no Bayesian kernel matrix inversion is performed. Instead, each day is computed as a distance-weighted average of monthly reference values using a Gaussian weighting function, with circular (wraparound) day distance.

Notation identity clarifications used in Section 18

To avoid symbol ambiguity, I use the following identities throughout this section:

- m: month index ($m = 1, \ldots, 12$).
- d: day-of-year evaluation index ($d = 1, \ldots, P$).
- P: annual period length in days ($P = 365$ in this implementation).
- d_i: day-of-year reference day associated with reference point i (month midpoint in this implementation).
- d_m: month-midpoint day-of-year for month m. With one reference point per month, d_m and d_i refer to the same reference-day map.
- L_m: number of calendar days in month m.
- i in Eqs. (229)–(230): reference-point index over the monthly climate reference values.
- i in Eqs. (232), (236): day-of-month candidate index inside one month for event placement. This is a different role than i in Eqs. (229)–(230), despite sharing the same symbol.
- y_i: reference value at reference point i. In Section 18, this is used in two separate passes:
  - pass A: $y_i = \bar{T}_i$ (monthly mean temperature),
  - pass B: $y_i = \sigma_i$ (monthly standard deviation).
- $\hat{y}(d)$: kernel-regressed daily estimate generated from one selected y_i pass at day d.
- $w_i(d)$: Gaussian kernel weight applied to reference point i when evaluating day d.
- h: Gaussian kernel bandwidth in days.
- integer allocation rule: convert fractional monthly expectations into integer monthly counts (freeze and heat-wave both use month-wise stochastic rounding in this walkthrough).

- $\beta_{f,m}$, $\rho_{f,m}$, $U^{\text{cnt}}_{f,m}$: freeze-count floor, freeze-count fractional remainder, and freeze-count uniform draw for month m.
- $\beta_{s,m}$, $\rho_{s,m}$, $U^{\text{cnt}}_{s,m}$: spell-start count floor, fractional remainder, and uniform draw for month m.
- $\beta_{h,m}$, $\rho_{h,m}$, $U^{\text{cnt}}_{h,m}$: heat-wave-day count floor, fractional remainder, and uniform draw for month m.
- N_f, N_s, N_h: realized annual totals after month-wise stochastic rounding (freeze days, spell starts, and heat-wave days).
- $s_{f,m}$, $r_{f,m}$: freeze-day Gaussian spread and repulsion length scales for month m.
- $\mathcal{S}_{f,m}$: set of freeze-day indices already selected in month m.
- $\mathcal{S}^{(k)}_{f,m}$: freeze-day selected set after k recursive selections ($\mathcal{S}^{(0)}_{f,m} = \varnothing$).
- $\mathcal{I}^{(k)}_{f,m}$: freeze-day candidates remaining at step k, defined by $\mathcal{I}^{(k)}_{f,m} = \{1, \ldots, L_m\} \setminus \mathcal{S}^{(k-1)}_{f,m}$.
- $\mathcal{I}$ is a calligraphic capital I (candidate-index set), not L.
- r: generic day-of-month index, $r \in \{1, \ldots, L_m\}$.
- $d_{f,m,k}$: freeze day selected at recursion step k.
- $R^{(k)}_{f,m}(r)$: deterministic ranking value for freeze-day candidate day r at step k; larger R means higher selection priority.
- $M_{f,m}$: freeze-anomaly monthly center value for month m (mean freeze anomaly below $\bar{T}_m$).
- $\Delta T^{\text{low}}_{f,m}, \Delta T^{\text{high}}_{f,m}$: lower/upper freeze-anomaly bounds for month m.
- $\sigma_{\Delta T,f,m}$: freeze-anomaly spread parameter for month m.
- $\Delta T_{f,m,q}$: sampled anomaly for selected freeze event q in month m (bounded to $[\Delta T^{\text{low}}_{f,m}, \Delta T^{\text{high}}_{f,m}]$).
- $A_{f,m}(r)$: day-of-month freeze-anomaly field; $A_{f,m}(r) = \Delta T_{f,m,q}$ when day r is the q-th selected freeze day, and $A_{f,m}(r) = 0$ otherwise.
- $s_{h,m}$, $r_{h,m}$: heat-wave start Gaussian spread and repulsion length scales for month m.
- $\mathcal{S}_{h,m}$: set of heat-wave start-day indices selected in month m.
- $\mathcal{S}^{(k)}_{h,m}$: heat-wave-start selected set after k recursive selections ($\mathcal{S}^{(0)}_{h,m} = \varnothing$).
- $\mathcal{I}^{(k)}_{h,m}$: heat-wave-start candidates remaining at step k, defined by $\mathcal{I}^{(k)}_{h,m} = \{1, \ldots, L_m\} \setminus \mathcal{S}^{(k-1)}_{h,m}$.
- $d_{h,m,k}$: heat-wave start day selected at recursion step k.
- $R^{(k)}_{h,m}(r)$: deterministic ranking value for heat-wave-start candidate day r at step k; larger R means higher selection priority.
- q: heat-wave spell index within month m, ordered by projected start day.
- $r_m(d)$: day-of-month index corresponding to day-of-year d in month m.
- $\Delta T^{\text{low}}_{h,m}, \Delta T^{\text{high}}_{h,m}$: lower/upper spell-anomaly bounds (above $\bar{T}_m$) for month m, loaded from NOAA-derived monthly spell statistics.
- $\sigma_{\Delta T,h,m}$: configured or fallback standard deviation used for spell-anomaly sampling in month m.
- $\Delta T_{h,m,q}$: sampled anomaly assigned to spell q in month m, bounded to $[\Delta T^{\text{low}}_{h,m}, \Delta T^{\text{high}}_{h,m}]$.
- $M_{h,m,q}$: spell-magnitude variable for spell q in month m; in code-equivalent notation $M_{h,m,q} \equiv \Delta T_{h,m,q}$.
- $A_{h,m}(r)$: day-of-month spell-anomaly field for month m, obtained by allocating each spell magnitude $M_{h,m,q}$ to all days in spell block $\mathcal{B}_{h,m,q}$.
- $g_{\text{std}}(z)$, $G_{\text{std}}(z)$: standard-normal density and CDF used in the explicit truncated-normal equations.
- $\varnothing$: empty set; $A \setminus B$: set difference; $A \cup \{x\}$: add element x to set A; $|A|$: set size.
- $D_h(d)$, $D_c(d)$: mode-specific clipped duty fractions.
- $Q_{\text{to bed},h}(T_\infty(d))$: signed net heat-to-bed returned by the heating reference-point re-solve at day ambient $T_\infty(d)$ (positive when the heater adds heat to the bed).
- $Q_{\text{to bed},c}(T_\infty(d))$: signed net heat-to-bed returned by the cooling reference-point re-solve at day

ambient $T_\infty(d)$ (typically negative when cooling removes heat from the bed).

- $Q_{h,\text{cap}}(d)$, $Q_{c,\text{cap}}(d)$: positive heating and cooling capacities used in duty equations, with $Q_{h,\text{cap}}(d) = Q_{\text{to bed},h}(T_\infty(d))$ and $Q_{c,\text{cap}}(d) = -Q_{\text{to bed},c}(T_\infty(d))$.
- $P_h(T_\infty(d))$: heating-mode electrical power from the day-ambient heating re-solve.
- $P_{\text{spot cooler}}(d)$: spot-cooler electric power on day d from the cooling re-solve.
- $P_{\text{assist blower}}(d)$: assist-blower electric power on day d from the cooling re-solve.
- $P_{h,\text{elec}}(d)$, $P_{c,\text{elec}}(d)$: mode electric powers used in daily energy equations, where $P_{h,\text{elec}}(d) = P_h(T_\infty(d))$ and $P_{c,\text{elec}}(d) = P_{\text{spot cooler}}(d) + P_{\text{assist blower}}(d)$.
- $D(d)$ in Eq. (251): active duty fraction for the current mode/day (equal to $D_h(d)$ on heating days or $D_c(d)$ on cooling days).
- f_{day} in Eq. (244): generic clipped-duty expression; in code this is implemented as `dutyApplied`.
- $\dot{Q}_{\text{load}}$, $\dot{Q}_{\text{nom}}$ in Eq. (244): generic load and nominal-capacity placeholders corresponding to the active mode (heating or cooling).

Temperature-variable identity and derivation map

The following temperature symbols are all explicit in this implementation:

- $\bar{T}_m$: monthly Connecticut mean ambient from `climate_data.monthlyMean_C`.
- σ_m: monthly ambient spread from `climate_data.monthlyStd_C`.
- $T_{\infty,\text{base}}(d)$: regressed daily ambient baseline from periodic Gaussian kernel regression on $\bar{T}_m$.
- $\sigma(d)$: regressed daily spread from the same kernel pass on σ_m.
- $T_{\infty,\text{preH}}(d)$: intermediate daily ambient after freeze-day cold overrides (input to 18.3 heat-wave overlay).
- $T_{\infty,\text{heatRaw}}(d)$: optional diagnostic track exported by code; mathematically, Eq. 18.3-G is written directly in combined form for $T_\infty(d)$.
- $T_m(d)$: month-local daily ambient profile used during event overlays inside month m.
- $T_\infty(d)$: final day ambient passed to thermal load and capacity re-solves.
- $T_{h,\text{set}}, T_{c,\text{set}}$: resolved heating/cooling setpoints from the annual config (or inherited model defaults).
- $T_b(d), T_t(d)$: bottom/top bed-node temperatures from the two-node steady solve under $Q_{\text{bed}}(d)$ and $T_\infty(d)$.
- For each heat-wave spell q: threshold temperature is $\bar{T}_m + \Delta T_{h,m,q}$, where $\Delta T_{h,m,q}$ is a bounded normal draw.
- NOAA spell-bound files used by config generation:
  - raw daily station rows: `Heat Transfer Study/data/climate/noaa_daily_summaries_USW00014740_2016_20`
  - derived monthly freeze bounds: `Heat Transfer Study/data/climate/noaa_freeze_day_anomaly_bounds_US`
  - derived monthly spell bounds: `Heat Transfer Study/data/climate/noaa_heatwave_spell_anomaly_bounds`
  - extraction utility: `Heat Transfer Study/tools/derive_noaa_heatwave_spell_bounds.py`.

Direct NOAA-derived inputs vs model-computed variables (explicit provenance)

To make data provenance explicit, the Section 18 symbols split into two groups.

A. Direct NOAA-derived reference inputs (loaded from config/file). These are treated as fixed monthly inputs by the year-round model:

- Statewide monthly climate (NOAA CAG monthly series, 2016–2025):

$$\bar{T}_m = \texttt{climate_data.monthlyMean_C}[m], \qquad \sigma_m = \texttt{climate_data.monthlyStd_C}[m].$$

where $T_{y,m}^{\mathrm{CAG}}$ is the NOAA CAG statewide monthly mean temperature (in °C) for year y, month m, and Y is the number of years in the NOAA window:

$$\bar{T}_m = \frac{1}{Y}\sum_{y=1}^{Y} T_{y,m}^{\mathrm{CAG}},$$

$$\sigma_m = \sqrt{\frac{1}{Y}\sum_{y=1}^{Y}\left(T_{y,m}^{\mathrm{CAG}} - \bar{T}_m\right)^2}.$$

- Station-based freeze statistics (NOAA daily-summaries station USW00014740, 2016–2025):

$$F_m = \texttt{climate_data.freeze.averageDaysPerMonth}[m],$$

where F_m is the historical mean freeze-day count in month m:

$$F_m = \frac{1}{Y}\sum_{y=1}^{Y} C_{y,m},$$

with $C_{y,m}$ = number of freeze days observed in year y, month m, and Y = number of years in the NOAA window.

$$\mu_{f,m} = \texttt{climate_data.freeze.meanDayOfMonth}[m],$$

$$M_{f,m} = \texttt{climate_data.freeze.meanAnomalyBelowMonthlyMean_C}[m],$$

$$\Delta T_{f,m}^{\mathrm{low}} = \texttt{climate_data.freeze.minAnomalyBelowMonthlyMean_C}[m], \quad \Delta T_{f,m}^{\mathrm{high}} = \texttt{climate_data.freeze.maxA}$$

$$\sigma_{\Delta T,f,m} = \texttt{climate_data.freeze.anomalyStd_C}[m].$$

- Station-based heat-wave timing/count statistics (same station/time window):

$$S_m = \texttt{climate_data.heat_wave.averageSpellStartsPerMonth}[m],$$

$$H_m = \texttt{climate_data.heat_wave.averageDaysPerMonth}[m],$$

$$\mu_{h,m} = \texttt{climate_data.heat_wave.meanStartDayOfMonth}[m].$$

- Station-based heat-wave spell-anomaly bounds (from `noaa_heatwave_spell_anomaly_bounds_USW00014740_201` when `useDerivedSpellAnomalyBounds=true`):

$$\Delta T_{h,m} = \texttt{meanAnomalyAboveMonthlyMean_C}[m],$$

$$\Delta T_{h,m}^{\mathrm{low}} = \texttt{minSpellAnomalyAboveMonthlyMean_C}[m], \quad \Delta T_{h,m}^{\mathrm{high}} = \texttt{maxSpellAnomalyAboveMonthlyMean_C}[m],$$

$$\sigma_{\Delta T,h,m} = \texttt{spellAnomalyStd_C}[m].$$

B. Model-computed variables (not direct NOAA inputs). These are produced by equations in Section 18 from the NOAA-derived inputs above:

- kernel-regressed daily fields $T_{\infty,\text{base}}(d)$, $\sigma(d)$,
- allocated integer counts $n_{f,m}$, $n_{s,m}$, $n_{h,m}$,
- selected index sets/lists $\mathcal{S}_{f,m}$, $\mathcal{S}_{h,m}$, and spell blocks $\mathcal{B}_{h,m,q}$,
- sampled spell anomalies $\Delta T_{h,m,q}$,
- final ambient $T_\infty(d)$, duties $D_h(d), D_c(d)$, loads, energy, cost, and water outputs.

So, NOAA provides monthly/statistical inputs; Section 18 equations transform those into daily operational trajectories.

18.1 Climate reference values and regression baseline (229, 230)

I began with monthly Connecticut reference values:

- monthly mean temperature $\bar{T}_m$,
- monthly standard deviation σ_m,
- month lengths from the annual configuration.

I mapped each month to a day-of-year reference day at the month midpoint and evaluated all 365 representative days. For each day d, I computed a Gaussian-weighted average:

$$\hat{y}(d) = \frac{\sum_i w_i(d) y_i}{\sum_i w_i(d)}, \qquad w_i(d) = \exp\left[-\frac{1}{2}\left(\frac{\delta_i(d)}{h}\right)^2\right],$$

with circular distance

$$\delta_i(d) = \min(|d - d_i|,\ P - |d - d_i|)\,,$$

where $P = 365$ days and h is the configured bandwidth (default 18 days). I applied this same regression twice:

1. once to monthly means to get daily baseline ambient $T_{\infty,\text{base}}(d)$,
2. once to monthly standard deviations to get daily $\sigma(d)$ for climate-context reporting.

Explicitly, the baseline ambient used by Sections 18.2–18.5 is

$$T_{\infty,\text{base}}(d) = \frac{\sum_{i=1}^{12} w_i(d)\,\bar{T}_i}{\sum_{i=1}^{12} w_i(d)},$$

with

$$w_i(d) = \exp\left[-\frac{1}{2}\left(\frac{\delta_i(d)}{h}\right)^2\right], \qquad \delta_i(d) = \min(|d - d_i|,\ 365 - |d - d_i|)\,.$$

Here d_i is month-i midpoint day-of-year and $\bar{T}_i$ is NOAA monthly mean for month i. In other words, y_i is not a third intermediary variable family; it is whichever reference-value array is active in the current regression pass.

The circular distance is essential because it treats late December and early January as neighboring points, preventing an artificial year-boundary discontinuity.

18.2 Freeze-day allocation and anomaly override (231–234)

This subsection introduces the explicit set-recursion logic used for event-day placement in both 18.2 and 18.3.

The NOAA-derived freeze expectation in each month is fractional (for example, 26.3 days). I convert it to an integer with month-wise stochastic rounding:

$$\beta_{f,m} = \lfloor F_m \rfloor, \qquad \rho_{f,m} = F_m - \lfloor F_m \rfloor.$$

$$U_{f,m}^{\text{cnt}} \sim \text{Uniform}(0,1).$$

$$n_{f,m} = \begin{cases} \beta_{f,m} + 1, & U_{f,m}^{\text{cnt}} < \rho_{f,m}, \\ \beta_{f,m}, & \text{otherwise.} \end{cases}$$

For this rule,

$$\mathbb{E}[n_{f,m}] = F_m$$

and each month is rounded independently. The annual freeze total

$$N_f = \sum_{m=1}^{12} n_{f,m}$$

is therefore random (close to $\sum_m F_m$ in expectation, but not forced to equal round($\sum_m F_m$)). So month m receives exactly $n_{f,m}$ freeze days for that realization.

Step 18.2-A: Define the candidate universe and initialization.

$$\mathcal{U}_m = \{1, 2, \ldots, L_m\}, \qquad \mathcal{S}_{f,m}^{(0)} = \varnothing.$$

Here:

- $\mathcal{U}_m$ is the full set of day-of-month candidates,
- L_m is the number of days in month m,
- $\varnothing$ is the empty set.

Step 18.2-B: Define the remaining candidates at selection step k.

$$\mathcal{I}_{f,m}^{(k)} = \mathcal{U}_m \setminus \mathcal{S}_{f,m}^{(k-1)}, \qquad k = 1, \ldots, n_{f,m}.$$

$\setminus$ means set difference: remove already-selected days. In the current implementation, the freeze-selection spread and repulsion scales are

$$s_{f,m} = \max\left(2, \frac{L_m}{6}\right), \qquad r_{f,m} = \max\left(1.5, \frac{L_m}{14}\right),$$

with L_m the number of days in month m.

Step 18.2-C: Compute deterministic rank value for each remaining candidate.

$$W_{f,m}(r) = \exp\left[-\frac{(r - \mu_{f,m})^2}{2s_{f,m}^2}\right],$$

$$R_{f,m}^{(k)}(r) = W_{f,m}(r) \prod_{p \in \mathcal{S}_{f,m}^{(k-1)}} \left[1 - 0.65 \exp\left(-\frac{(r-p)^2}{2r_{f,m}^2}\right)\right], \qquad r \in \mathcal{I}_{f,m}^{(k)}.$$

The factor $W_{f,m}(r)$ prefers days near mean freeze day $\mu_{f,m}$. The product term repels new selections from previously selected days. So $R^{(k)}_{f,m}(r)$ is a ranking value, not a probability. Its direct effect on the selected set is:

$$M^{(k)}_{f,m} = \max_{r \in \mathcal{I}^{(k)}_{f,m}} R^{(k)}_{f,m}(r),$$

$$d_{f,m,k} \in \mathcal{I}^{(k)}_{f,m}, \qquad R^{(k)}_{f,m}(d_{f,m,k}) = M^{(k)}_{f,m},$$

$$\mathcal{S}^{(k)}_{f,m} = \mathcal{S}^{(k-1)}_{f,m} \cup \{d_{f,m,k}\}.$$

So at step k, $R^{(k)}_{f,m}$ determines exactly which new index enters $\mathcal{S}^{(k)}_{f,m}$.

Step 18.2-D: Select one day and update by union.

$$d_{f,m,k} \in \mathcal{I}^{(k)}_{f,m}, \qquad R^{(k)}_{f,m}(d_{f,m,k}) \geq R^{(k)}_{f,m}(r) \text{ for every } r \in \mathcal{I}^{(k)}_{f,m},$$

$$\mathcal{S}^{(k)}_{f,m} = \mathcal{S}^{(k-1)}_{f,m} \cup \{d_{f,m,k}\}.$$

$\cup$ means set union: add the newly selected day index. Because $d_{f,m,k} \in \mathcal{I}^{(k)}_{f,m}$, it is not already in $\mathcal{S}^{(k-1)}_{f,m}$. Therefore each update adds exactly one new day, so $|\mathcal{S}^{(k)}_{f,m}| = k$.
After $n_{f,m}$ steps:

$$\mathcal{S}_{f,m} = \mathcal{S}^{(n_{f,m})}_{f,m},$$

so $\mathcal{S}_{f,m}$ contains exactly $n_{f,m}$ distinct freeze days.

Step 18.2-E: Same logic without set notation (code-equivalent view). If set notation is unfamiliar, the same selection can be written as an index-list loop:

```
selected = []                                # freeze day indices already chosen
for k in 1..n_f,m:
    best_r = None
    best_score = -infinity
    for r in 1..L_m:
        if r in selected:
            continue
        score = exp(-((r-mu_f,m)^2)/(2*s_f,m^2))
        for p in selected:
            score *= (1 - 0.65*exp(-((r-p)^2)/(2*r_f,m^2)))
        if score > best_score:
            best_score = score
            best_r = r
    selected.append(best_r)
```

This is mathematically identical to the set form:

- "$r \in \mathcal{I}^{(k)}_{f,m}$" means "$r$ not in `selected`",
- "$\mathcal{S}^{(k)}_{f,m} = \mathcal{S}^{(k-1)}_{f,m} \cup \{d_{f,m,k}\}$" means "`selected.append(best_r)`".

Step 18.2-F: Freeze-magnitude definition and day allocation. Define freeze-anomaly monthly center:

$$M_{f,m} \equiv \Delta T_{f,m}.$$

How $M_{f,m}$, bounds, and spread are computed from NOAA data (conceptual derivation):

- Source rows: NOAA daily-summaries station `USW00014740` over 2016–2025.
- Freeze-day definition: $T_{\min} \leq 32°\mathrm{F}$.
- Daily mean proxy in °C: $T_{\mathrm{day}}^{(r)}$ from T_{AVG} if present, otherwise $0.5(T_{\max} + T_{\min})$, converted to °C.
- For each freeze day r in month m, compute cold anomaly below monthly mean:

$$a_{m,r} = \max\Big(\bar{T}_m - T_{\mathrm{day}}^{(r)},\, 0\Big)\,.$$

- Monthly center is the average freeze-day anomaly:

$$M_{f,m} = \frac{1}{N_{f,m}} \sum_{r=1}^{N_{f,m}} a_{m,r},$$

  where $N_{f,m}$ is the number of freeze days used in month m.
- Monthly bounds and spread are:

$$\Delta T_{f,m}^{\mathrm{low}} = \min_r a_{m,r}, \quad \Delta T_{f,m}^{\mathrm{high}} = \max_r a_{m,r}, \quad \sigma_{\Delta T,f,m} = \sqrt{\frac{1}{N_{f,m}} \sum_{r=1}^{N_{f,m}} (a_{m,r} - M_{f,m})^2}.$$

In this repository, these monthly arrays are precomputed and loaded from:

$$M_{f,m} = \texttt{climate_data.freeze.meanAnomalyBelowMonthlyMean_C}[m].$$

$$\Delta T_{f,m}^{\mathrm{low}} = \texttt{climate_data.freeze.minAnomalyBelowMonthlyMean_C}[m], \quad \Delta T_{f,m}^{\mathrm{high}} = \texttt{climate_data.freeze.maxAnc}$$

Define string-normalization map

$$\mathrm{norm}(s) = \mathrm{lower}\Big(\,\mathrm{replace}\,\big(\,\mathrm{replace}(s, \texttt{"-"}, \texttt{"_"}), \texttt{" "}, \texttt{"_"}\big)\Big).$$

Define freeze model key

$$\kappa_f = \mathrm{norm}(\texttt{freeze.anomalySamplingModel})\,.$$

Define explicit mode flags (no set notation):

$$\texttt{isDeterministic}_f = \begin{cases} 1, & \kappa_f = \texttt{off} \text{ or } \texttt{none} \text{ or } \texttt{disabled} \text{ or } \texttt{deterministic} \text{ or } \texttt{deterministic_mean}, \\ 0, & \text{otherwise}, \end{cases}$$

$$\texttt{isTriangular}_f = \begin{cases} 1, & \kappa_f = \texttt{triangular} \text{ or } \texttt{triangular_mean_matched} \text{ or } \texttt{triangular_between_monthly_min_m} \\ 0, & \text{otherwise}. \end{cases}$$

Conceptually:

- κ_f is not a temperature; it is a text switch read from configuration.
- $\mathrm{norm}(\cdot)$ just standardizes spelling so equivalent labels map to one key (for example, `Triangular-Mean Matched` and `triangular_mean_matched` become the same key).

- $\texttt{isDeterministic}_f = 1$ means "use one fixed freeze anomaly for all selected freeze days in month m." Then $\Delta T_{f,m,q} = \mu^\star_{f,m}$.
- $\texttt{isTriangular}_f = 1$ means "sample each selected freeze day from the triangular distribution on $[a_{f,m}, b_{f,m}]$." Then $\Delta T_{f,m,q} = X_{f,m,q}$.
- These are logic switches (yes/no), not climate variables.

Define freeze bounds and bounded monthly center:

$$a_{f,m} = \Delta T^{\text{low}}_{f,m}, \quad b_{f,m} = \Delta T^{\text{high}}_{f,m}.$$

These are anomaly-magnitude bounds (in °C), not temperatures. In code/config, they come from

`climate_data.freeze.minAnomalyBelowMonthlyMean_C`$[m]$, `climate_data.freeze.maxAnomalyBelowMonth`

Historical NOAA derivation for month m: for each historical freeze day j, define anomaly

$$a_{m,j} = \max\left(0, \bar{T}_m - T^{\text{freeze day}}_{m,j}\right),$$

then

$$\Delta T^{\text{low}}_{f,m} = \min_j a_{m,j}, \qquad \Delta T^{\text{high}}_{f,m} = \max_j a_{m,j}.$$

$$\mu^\star_{f,m} = \begin{cases} a_{f,m}, & M_{f,m} < a_{f,m}, \\ M_{f,m}, & a_{f,m} \le M_{f,m} \le b_{f,m}, \\ b_{f,m}, & M_{f,m} > b_{f,m}. \end{cases}$$

So $\mu^\star_{f,m}$ is the bounded (clipped) freeze-anomaly center for month m.

$$c^{\text{raw}}_{f,m} = 3\mu^\star_{f,m} - a_{f,m} - b_{f,m},$$

where $c^{\text{raw}}_{f,m}$ is the unconstrained mode target for the triangular draw. It is chosen so that the triangular mean $(a_{f,m} + b_{f,m} + c)/3$ equals $\mu^\star_{f,m}$ before bounds enforcement. Why: this keeps stochastic day-level draws centered on the intended monthly anomaly level instead of drifting warmer or colder on average.

$$c_{f,m} = \begin{cases} a_{f,m}, & c^{\text{raw}}_{f,m} < a_{f,m}, \\ c^{\text{raw}}_{f,m}, & a_{f,m} \le c^{\text{raw}}_{f,m} \le b_{f,m}, \\ b_{f,m}, & c^{\text{raw}}_{f,m} > b_{f,m}. \end{cases}$$

Define triangular density on $[a_{f,m}, b_{f,m}]$:

$$t_{f,m}(x) = \begin{cases} \dfrac{2(x - a_{f,m})}{(b_{f,m} - a_{f,m})(c_{f,m} - a_{f,m})}, & a_{f,m} \le x \le c_{f,m}, \\ \dfrac{2(b_{f,m} - x)}{(b_{f,m} - a_{f,m})(b_{f,m} - c_{f,m})}, & c_{f,m} \le x \le b_{f,m}, \\ 0, & \text{otherwise}. \end{cases}$$

For each selected freeze event $q = 1, \ldots, n_{f,m}$, allocate event magnitude:

$$\Delta T_{f,m,q} = \begin{cases} \mu^\star_{f,m}, & \texttt{isDeterministic}_f = 1, \\ X_{f,m,q}, & \texttt{isTriangular}_f = 1 \text{ and } b_{f,m} > a_{f,m}, \\ \mu^\star_{f,m}, & b_{f,m} \le a_{f,m}, \\ \mu^\star_{f,m} \text{ or legacy bounded-normal draw}, & \texttt{isDeterministic}_f = 0 \text{ and } \texttt{isTriangular}_f = 0, \end{cases}$$

with explicit triangular generation:

$$U_{f,m,q} \sim \text{Uniform}(0,1), \qquad p_{f,m} = \frac{c_{f,m} - a_{f,m}}{b_{f,m} - a_{f,m}}.$$

$$X_{f,m,q} = \begin{cases} a_{f,m} + \sqrt{U_{f,m,q}(b_{f,m} - a_{f,m})(c_{f,m} - a_{f,m})}, & 0 \le U_{f,m,q} \le p_{f,m}, \\ b_{f,m} - \sqrt{(1 - U_{f,m,q})(b_{f,m} - a_{f,m})(b_{f,m} - c_{f,m})}, & p_{f,m} < U_{f,m,q} \le 1. \end{cases}$$

Then enforce bounds explicitly:

$$\Delta T_{f,m,q} = \begin{cases} a_{f,m}, & \Delta T_{f,m,q} < a_{f,m}, \\ \Delta T_{f,m,q}, & a_{f,m} \le \Delta T_{f,m,q} \le b_{f,m}, \\ b_{f,m}, & \Delta T_{f,m,q} > b_{f,m}. \end{cases}$$

For any $a_{f,m} \le u \le v \le b_{f,m}$,

$$\Pr(u \le X_{f,m,q} \le v) = \int_u^v t_{f,m}(x)\, dx.$$

Define day-of-month freeze-anomaly allocation (piecewise, no indicator notation):

$$\text{Sort selected freeze days as } d_{f,m,1} < \cdots < d_{f,m,n_{f,m}}.$$

$$A_{f,m}(r) = \begin{cases} 0, & r \notin \mathcal{S}_{f,m}, \\ \Delta T_{f,m,q}, & r = d_{f,m,q} \text{ for some } q \in \{1, \ldots, n_{f,m}\}, \end{cases} \qquad r = 1, \ldots, L_m.$$

So selected freeze days can have different magnitudes (stochastic mode) or identical magnitudes (deterministic mode), depending on κ_f. Important: $A_{f,m}(r)$ itself is not a random sampler; it is a mapping from selected day index r to the already-generated value $\Delta T_{f,m,q}$. Any randomness occurs upstream when generating $\Delta T_{f,m,q}$ (when $\texttt{isTriangular}_f = 1$); then q is matched to ordered selected days $d_{f,m,1} < \cdots < d_{f,m,n_{f,m}}$ without extra random reshuffling.

Allocation matters because selection and magnitude are different tasks:

- selection ($\mathcal{S}_{f,m}$) decides *which* days are freeze days,
- allocation ($A_{f,m}(r)$) decides *what anomaly value* each selected day gets.

Step 18.2-G: Apply freeze override on selected days. Let $r_m(d) \in \{1, \ldots, L_m\}$ be the day-of-month corresponding to day-of-year d in month m. Then

$$T^{\text{th}}_{f,m}(d) = \bar{T}_m - A_{f,m}(r_m(d)).$$

Interpretation: $T^{\text{th}}_{f,m}(d)$ is the freeze-threshold ambient for day d in month m. It is the day-specific clamp value generated from the monthly mean $\bar{T}_m$ and allocated freeze anomaly $A_{f,m}(r_m(d))$.

$$T_{\infty,\text{preH}}(d) = \begin{cases} T_{\infty,\text{base}}(d), & r_m(d) \in \mathcal{S}_{f,m} \text{ and } T_{\infty,\text{base}}(d) \le T^{\text{th}}_{f,m}(d), \\ T^{\text{th}}_{f,m}(d), & r_m(d) \in \mathcal{S}_{f,m} \text{ and } T_{\infty,\text{base}}(d) > T^{\text{th}}_{f,m}(d), \\ T_{\infty,\text{base}}(d), & r_m(d) \notin \mathcal{S}_{f,m}. \end{cases}$$

So freeze days are forced colder (or equal) relative to the baseline, never warmer.

Step 18.2-H: Why allocation matters for implementation. The override reads a day-level anomaly field $A_{f,m}(r)$, not just one month scalar. This preserves the selected-day pattern and supports both deterministic and stochastic freeze magnitudes.

18.3 Heat-wave allocation and anomaly override (235–238)

Section 18.3 uses the same set-recursion logic defined in 18.2. The only change is parameter substitution from freeze-day placement to heat-wave start placement:

$$\{n_{f,m}, \mu_{f,m}, s_{f,m}, r_{f,m}, \mathcal{S}_{f,m}\} \to \{n_{s,m}, \mu_{h,m}, s_{h,m}, r_{h,m}, \mathcal{S}_{h,m}\}.$$

So the meanings of $\setminus$, $\cup$, and "already selected" are identical to 18.2. If preferred, use the exact same code-style loop from Step 18.2-E and replace $(n_{f,m}, \mu_{f,m}, s_{f,m}, r_{f,m})$ with $(n_{s,m}, \mu_{h,m}, s_{h,m}, r_{h,m})$.

Step 18.3-A: Integer monthly counts. Start from fractional monthly expectations S_m (spell starts) and H_m (heat-wave days). Use the same month-wise stochastic rounding rule as 18.2:

$$\beta_{s,m} = \lfloor S_m \rfloor, \qquad \rho_{s,m} = S_m - \lfloor S_m \rfloor,$$

$$\beta_{h,m} = \lfloor H_m \rfloor, \qquad \rho_{h,m} = H_m - \lfloor H_m \rfloor.$$

$$U^{\text{cnt}}_{s,m} \sim \text{Uniform}(0,1), \qquad U^{\text{cnt}}_{h,m} \sim \text{Uniform}(0,1).$$

$$n_{s,m} = \begin{cases} \beta_{s,m} + 1, & U^{\text{cnt}}_{s,m} < \rho_{s,m}, \\ \beta_{s,m}, & \text{otherwise}, \end{cases}$$

$$n_{h,m} = \begin{cases} \beta_{h,m} + 1, & U^{\text{cnt}}_{h,m} < \rho_{h,m}, \\ \beta_{h,m}, & \text{otherwise}. \end{cases}$$

For this rule,

$$\mathbb{E}[n_{s,m}] = S_m, \qquad \mathbb{E}[n_{h,m}] = H_m.$$

Annual totals

$$N_s = \sum_{m=1}^{12} n_{s,m}, \qquad N_h = \sum_{m=1}^{12} n_{h,m}$$

are random (unbiased in expectation), not fixed by a global remainder redistribution. $n_{s,m}$ is number of spells; $n_{h,m}$ is total heat-wave days.

Step 18.3-B: Spell lengths. Let $q = 1, \ldots, n_{s,m}$ index spells in month m. The code applies feasibility limits first:

$$n_{s,m} \leftarrow \min\left(n_{s,m}, \left\lfloor \frac{L_m + 1}{4} \right\rfloor\right).$$

This bound comes from the enforced structure "spell + at least one non-spell day gap" with minimum spell length 3. Given this $n_{s,m}$, total heat-wave days are clipped to feasible range:

$$n^{\min}_{h,m} = 3n_{s,m}, \qquad n^{\max}_{h,m} = L_m - (n_{s,m} - 1),$$

$$n_{h,m} \leftarrow \min\big(\max(n_{h,m}, n^{\min}_{h,m}),\, n^{\max}_{h,m}\big).$$

Now initialize every spell with the minimum:

$$\ell_{h,m,q} = 3, \qquad q = 1, \ldots, n_{s,m}.$$

Define extra days:

$$e_m = n_{h,m} - 3n_{s,m} \ (\geq 0).$$

The implementation distributes e_m cyclically across spells; this is equivalent to quotient–remainder form:

$$\text{for } n_{s,m} > 0: \qquad u_m = \left\lfloor \frac{e_m}{n_{s,m}} \right\rfloor, \qquad v_m = e_m - u_m n_{s,m},$$

$$\ell_{h,m,q} = \begin{cases} 3 + u_m + 1, & q \leq v_m, \\ 3 + u_m, & q > v_m, \end{cases} \qquad q = 1, \ldots, n_{s,m}.$$

So

$$\boldsymbol{\ell}_{h,m} = (\ell_{h,m,1}, \ldots, \ell_{h,m,n_{s,m}}), \qquad \sum_{q=1}^{n_{s,m}} \ell_{h,m,q} = n_{h,m}.$$

Example: if $n_{s,m} = 3$ and $n_{h,m} = 14$, then $e_m = 5$, $u_m = 1$, $v_m = 2$, and

$$(\ell_{h,m,1}, \ell_{h,m,2}, \ell_{h,m,3}) = (5, 5, 4).$$

Edge case: if $n_{s,m} = 0$, then no spells are formed and the month has no heat-wave spell-length vector.

Step 18.3-C: Start-day set construction (same recursion as 18.2).

$$\mathcal{U}_m = \{1, \ldots, L_m\}, \qquad \mathcal{S}_{h,m}^{(0)} = \varnothing.$$

For $k = 1, \ldots, n_{s,m}$:

$$\mathcal{I}_{h,m}^{(k)} = \mathcal{U}_m \setminus \mathcal{S}_{h,m}^{(k-1)},$$

$$W_{h,m}(r) = \exp\left[-\frac{(r - \mu_{h,m})^2}{2s_{h,m}^2}\right],$$

$$R_{h,m}^{(k)}(r) = W_{h,m}(r) \prod_{p \in \mathcal{S}_{h,m}^{(k-1)}} \left[1 - 0.65 \exp\left(-\frac{(r-p)^2}{2r_{h,m}^2}\right)\right], \qquad r \in \mathcal{I}_{h,m}^{(k)},$$

where $R_{h,m}^{(k)}(r)$ is a deterministic ranking value (not a probability).

$$d_{h,m,k} \in \mathcal{I}_{h,m}^{(k)}, \qquad R_{h,m}^{(k)}(d_{h,m,k}) \geq R_{h,m}^{(k)}(r) \text{ for every } r \in \mathcal{I}_{h,m}^{(k)},$$

$$\mathcal{S}_{h,m}^{(k)} = \mathcal{S}_{h,m}^{(k-1)} \cup \{d_{h,m,k}\}.$$

Final selected start set:

$$\mathcal{S}_{h,m} = \mathcal{S}_{h,m}^{(n_{s,m})}.$$

Step 18.3-D: Enforce non-overlap and month bounds. Raw starts from Step 18.3-C are projected to feasible starts $s_{h,m,q}$. Let $d^{\text{raw}}_{h,m,1} < \cdots < d^{\text{raw}}_{h,m,n_{s,m}}$ be the ordered elements of $\mathcal{S}_{h,m}$. So $d^{\text{raw}}_{h,m,q}$ means: the q-th ordered raw start day before feasibility clipping. Equivalently,

$$d^{\text{raw}}_{h,m,q} = \text{SortAsc}(\mathcal{S}_{h,m})_q, \qquad q = 1, \ldots, n_{s,m}.$$

Code-equivalent path:

$$\mathcal{S}_{h,m} \xrightarrow{\text{sort}} (d^{\text{raw}}_{h,m,1}, \ldots, d^{\text{raw}}_{h,m,n_{s,m}}) \xrightarrow{\text{clip to } [e_{h,m,q}, u_{h,m,q}]} (s_{h,m,1}, \ldots, s_{h,m,n_{s,m}}).$$

Using spell index $q = 1, \ldots, n_{s,m}$:

$$e_{h,m,q} = \begin{cases} 1, & q = 1, \\ s_{h,m,q-1} + \ell_{h,m,q-1} + 1, & q \geq 2, \end{cases}$$

$$u_{h,m,q} = L_m - \left(\sum_{p=q}^{n_{s,m}} \ell_{h,m,p} + (n_{s,m} - q) \right) + 1,$$

$$s_{h,m,q} = \begin{cases} e_{h,m,q}, & d^{\text{raw}}_{h,m,q} < e_{h,m,q}, \\ d^{\text{raw}}_{h,m,q}, & e_{h,m,q} \leq d^{\text{raw}}_{h,m,q} \leq u_{h,m,q}, \\ u_{h,m,q}, & d^{\text{raw}}_{h,m,q} > u_{h,m,q}. \end{cases}$$

Spell q occupies

$$\mathcal{B}_{h,m,q} = \{\, r : \; s_{h,m,q} \leq r \leq s_{h,m,q} + \ell_{h,m,q} - 1 \,\},$$

where r is day-of-month index. So $\mathcal{B}_{h,m,q}$ is the spell-day block (not a start-day set, and not a matrix), with

$$|\mathcal{B}_{h,m,q}| = \ell_{h,m,q}.$$

All heat-wave days in month m are

$$\mathcal{B}^{\text{all}}_{h,m} = \bigcup_{q=1}^{n_{s,m}} \mathcal{B}_{h,m,q}.$$

Because blocks are non-overlapping by construction,

$$\left| \mathcal{B}^{\text{all}}_{h,m} \right| = \sum_{q=1}^{n_{s,m}} |\mathcal{B}_{h,m,q}| = \sum_{q=1}^{n_{s,m}} \ell_{h,m,q} = n_{h,m}.$$

Step 18.3-E: Spell-magnitude definition and distribution-based allocation. Define spell magnitude for each spell q as

$$M_{h,m,q} \equiv \Delta T_{h,m,q}.$$

Also define the distribution-method key κ from `heat_wave.spellAnomalySamplingModel`. In the implementation:

$$\mathcal{D}_{\text{det}} = \{\texttt{off}, \texttt{none}, \texttt{disabled}, \texttt{deterministic}, \texttt{deterministic_mean}\}.$$

Define bounds and bounded parameters:

$$a_{h,m} = \Delta T^{\text{low}}_{h,m}, \quad b_{h,m} = \Delta T^{\text{high}}_{h,m}.$$

Here $\Delta T_{h,m}$, $\Delta T^{\text{low}}_{h,m}$, and $\Delta T^{\text{high}}_{h,m}$ are spell-anomaly-above-monthly-mean statistics (in °C), from

`heat_wave.meanAnomalyAboveMonthlyMean_C`$[m]$, `heat_wave.minSpellAnomalyAboveMonthlyMean_C`$[m]$, `hea`

Historical NOAA derivation (month m, spells indexed by q): if $\mathcal{B}_{h,m,q}$ is the set of days in historical spell q (grouped by spell start month m), define

$$a^{(h)}_{m,q} = \frac{1}{|\mathcal{B}_{h,m,q}|} \sum_{t \in \mathcal{B}_{h,m,q}} \max\bigl(0,\ T^{\text{avg}}_t - \bar{T}_{m(t)}\bigr)\,,$$

where T^{avg}_t is NOAA daily mean proxy (TAVG if present, else $(TMAX + TMIN)/2$) and $m(t)$ is the calendar month index of day t. Thus $\bar{T}_{m(t)}$ is the same monthly-mean notation $\bar{T}_m$, evaluated at day t's month. Then

$$\Delta T_{h,m} = \frac{1}{N_m} \sum_{q=1}^{N_m} a^{(h)}_{m,q}, \qquad \Delta T^{\text{low}}_{h,m} = \min_q a^{(h)}_{m,q}, \qquad \Delta T^{\text{high}}_{h,m} = \max_q a^{(h)}_{m,q}.$$

$$\mu^\star_{h,m} = \begin{cases} a_{h,m}, & \Delta T_{h,m} < a_{h,m}, \\ \Delta T_{h,m}, & a_{h,m} \le \Delta T_{h,m} \le b_{h,m}, \\ b_{h,m}, & \Delta T_{h,m} > b_{h,m}, \end{cases}$$

$$\sigma^\star_{h,m} = \begin{cases} \sigma_{\Delta T,h,m}, & \sigma_{\Delta T,h,m} \ge \varepsilon, \\ \varepsilon, & \sigma_{\Delta T,h,m} < \varepsilon. \end{cases}$$

$\sigma_{\Delta T,h,m}$ is the month-m spell-anomaly spread from `heat_wave.spellAnomalyStd_C`$[m]$. If that value is missing or nonpositive, the implementation fallback is

$$\sigma_{\Delta T,h,m} = \begin{cases} (\Delta T^{\text{high}}_{h,m} - \Delta T^{\text{low}}_{h,m})/4, & \Delta T^{\text{high}}_{h,m} > \Delta T^{\text{low}}_{h,m}, \\ 0, & \Delta T^{\text{high}}_{h,m} \le \Delta T^{\text{low}}_{h,m}. \end{cases}$$

The numerical floor is $\varepsilon = 10^{-3}\,°\text{C}$ (used in the stochastic sampling branch to avoid zero spread).

For each spell $q = 1, \ldots, n_{s,m}$, magnitude allocation follows:

$$M_{h,m,q} = \begin{cases} \mu^\star_{h,m}, & \kappa \in \mathcal{D}_{\text{det}}, \\ X_{h,m,q}, & \kappa \notin \mathcal{D}_{\text{det}} \text{ and } b_{h,m} > a_{h,m}, \\ \mu^\star_{h,m}, & \kappa \notin \mathcal{D}_{\text{det}} \text{ and } b_{h,m} \le a_{h,m}, \end{cases}$$

$$M_{h,m,q} = \begin{cases} a_{h,m}, & M_{h,m,q} < a_{h,m}, \\ M_{h,m,q}, & a_{h,m} \le M_{h,m,q} \le b_{h,m}, \\ b_{h,m}, & M_{h,m,q} > b_{h,m}. \end{cases}$$

So each spell receives one scalar magnitude $M_{h,m,q}$; all days in that spell share that same value.

Define day-of-month anomaly allocation:

$$A_{h,m}(r) = \begin{cases} 0, & r \notin \mathcal{B}_{h,m,1} \cup \cdots \cup \mathcal{B}_{h,m,n_{s,m}}, \\ M_{h,m,q}, & r \in \mathcal{B}_{h,m,q} \text{ for the unique spell } q, \end{cases} \qquad r = 1, \ldots, L_m.$$

Because $\mathcal{B}_{h,m,q}$ are non-overlapping by construction, there is at most one active spell for each day r.

Step 18.3-F: Explicit stochastic draw used when $\kappa \notin \mathcal{D}_{\text{det}}$. For each spell q, generate $X_{h,m,q}$ by rejection sampling:

$$g_{\text{std}}(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right), \qquad Z_{h,m,q,t} \sim g_{\text{std}}, \quad t = 1, \ldots, 64$$

Here z is a dimensionless standard-normal coordinate (z-score), and $Z_{h,m,q,t}$ is one random draw from that $N(0,1)$ distribution. So z itself is unbounded:

$$z \in (-\infty, \infty).$$

Practical bounds are imposed only after affine transformation to anomaly space, because accepted draws must satisfy

$$a_{h,m} \leq Y_{h,m,q,t} \leq b_{h,m}.$$

Equivalently, accepted z-draws lie in

$$\frac{a_{h,m} - \mu^{\star}_{h,m}}{\sigma^{\star}_{h,m}} \leq z \leq \frac{b_{h,m} - \mu^{\star}_{h,m}}{\sigma^{\star}_{h,m}}.$$

$$Y_{h,m,q,t} = \mu^{\star}_{h,m} + \sigma^{\star}_{h,m} Z_{h,m,q,t}.$$

$$\tau_{h,m,q} = \min\left\{t \in \{1, \ldots, 64\} : \; a_{h,m} \leq Y_{h,m,q,t} \leq b_{h,m}\right\}.$$

If $\tau_{h,m,q}$ exists, set

$$X_{h,m,q} = Y_{h,m,q,\tau_{h,m,q}}.$$

If no accepted draw appears in 64 attempts, the implementation fallback is

$$X_{h,m,q} = \min\bigl(\max\bigl(\mu^{\star}_{h,m}, a_{h,m}\bigr), \, b_{h,m}\bigr).$$

So every accepted draw lies inside the physical anomaly range $[a_{h,m}, b_{h,m}]$.

Step 18.3-G: Apply heat-wave override on selected spell days. Let $r_m(d) \in \{1, \ldots, L_m\}$ be the day-of-month corresponding to day-of-year d in month m. Then

$$T^{\text{th}}_{h,m}(d) = \bar{T}_m + A_{h,m}(r_m(d)).$$

Interpretation: $T^{\text{th}}_{h,m}(d)$ is the heat-wave-threshold ambient for day d in month m. It is the day-specific lift value generated from the monthly mean $\bar{T}_m$ and allocated spell anomaly $A_{h,m}(r_m(d))$.

$$T_{\infty}(d) = \begin{cases} \max\Bigl(T_{\infty,\text{base}}(d), \, T^{\text{th}}_{h,m}(d), \, T_{\infty,\text{preH}}(d)\Bigr), & r_m(d) \in \mathcal{B}^{\text{all}}_{h,m}, \\ T_{\infty,\text{preH}}(d), & r_m(d) \notin \mathcal{B}^{\text{all}}_{h,m}. \end{cases}$$

So 18.3 applies a single spell-day max among baseline, heat-wave threshold, and the pre-heat-wave profile. After this month-wise heat-wave override, $T_{\infty}(d)$ is passed into Section 18.4.

In implementation, random sampling uses a fixed seed (`spellAnomalyRngSeed`) for reproducibility. In this repository state, $\Delta T_{h,m}$, $\Delta T^{\text{low}}_{h,m}$, $\Delta T^{\text{high}}_{h,m}$, and $\sigma_{\Delta T,h,m}$ are NOAA-derived from 2016–2025 daily station data (`USW00014740`) with spell-level anomalies grouped by spell start month.

18.4 Daily heating and cooling loads (239, 240)

For each representative day d, after final ambient assignment $T_\infty(d)$, I rebuilt the bin thermal-loss model and computed required loads:

$$Q_{\text{req},h}(d) = UA_{\text{bin}} \max(T_{h,\text{set}} - T_\infty(d),\, 0)\,,$$

$$Q_{\text{req},c}(d) = UA_{\text{bin}} \max(T_\infty(d) - T_{c,\text{set}},\, 0)\,.$$

The first equation activates only when ambient is below heating setpoint; the second activates only when ambient is above cooling setpoint.

18.5 Capacity re-solve and duty application (241–246)

For each day, I did not reuse fixed design capacities. Instead, I re-solved the selected heating or cooling operating point at that day's ambient:

$$Q_{h,\text{cap}}(d) = Q_{\text{to bed},h}(T_\infty(d)), \quad P_{h,\text{elec}}(d) = P_h(T_\infty(d)),$$

$$Q_{c,\text{cap}}(d) = -Q_{\text{to bed},c}(T_\infty(d)), \quad P_{c,\text{elec}}(d) = P_{\text{spot cooler}}(d) + P_{\text{assist blower}}(d).$$

Identity clarification for these symbols: $Q_{\text{to bed},h}(T_\infty(d))$ and $Q_{\text{to bed},c}(T_\infty(d))$ are the signed net bed heat-transfer rates returned by the heating and cooling day-ambient reference-point re-solves, respectively. $P_h(T_\infty(d))$ is the heating electrical input from the heating re-solve. $P_{\text{spot cooler}}(d)$ and $P_{\text{assist blower}}(d)$ are the two cooling electrical components returned by the cooling re-solve. Therefore $Q_{h,\text{cap}}(d)$ and $Q_{c,\text{cap}}(d)$ are positive capacity magnitudes used for duty calculation, and $P_{h,\text{elec}}(d)$, $P_{c,\text{elec}}(d)$ are the corresponding mode electric powers used for $E_h(d)$, $E_c(d)$.

What "re-solve" means mathematically (explicit). For each day d, the model re-runs the selected heating/cooling reference physics at ambient $T_\infty(d)$, not at a fixed design ambient.

Heating electrical power $P_h(T_\infty(d))$. Wire resistance is temperature-corrected:

$$R_{\text{tube},g}(d) = \rho_{20}\big[1 + \alpha\big(\bar{T}_{w,g}(d) - 20\big)\big]\,\frac{L_{w,g}}{A_{w,g}},$$

then branch/tube current and power are

$$I_{\text{tube},g}(d) = \frac{V}{N_{s,g} R_{\text{tube},g}(d)}, \qquad P_{\text{tube},g}(d) = I^2_{\text{tube},g}(d) R_{\text{tube},g}(d),$$

and total heating electrical input is

$$P_h(T_\infty(d)) = \sum_g n_{\text{tube},g}\, P_{\text{tube},g}(d).$$

Signed heating $Q_{\text{to bed},h}(T_\infty(d))$. For each active branch b and axial segment s:

$$Q_{\text{wall},b,s} = \dot{m}_{\text{mean},b,s} c_{p,b,s} (T_{\text{air,in},b,s} - T_{\text{air,out},b,s})\,,$$

$$Q_{\text{jet},b,s} = \dot{m}_{\text{rel},b,s} c_{p,b,s} (T_{\text{rel},b,s} - T_{\text{bed,ref}})\,,$$

$$Q_{\text{evap},b,s} = \dot{m}_{\text{evap},b,s}\, h_{fg,b,s},$$

and the branch/segment balance is accumulated as

$$Q_{\text{to bed},h}(T_\infty(d)) = \sum_{b=1}^{N_b} \sum_{s=1}^{N_{\text{seg}}} \left(Q_{\text{wall},b,s} + Q_{\text{jet},b,s} - Q_{\text{evap},b,s}\right).$$

Spot-cooler and assist-blower powers.

$$\dot{Q}_{\text{spot,req}}(d) = \dot{m}(d) c_p(d) \max(T_\infty(d) - T_{\text{target}}, 0),$$

$$\dot{Q}_{\text{spot,act}}(d) = \min\Big(\dot{Q}_{\text{spot,req}}(d), \dot{Q}_{\text{spot,max}}(d)\Big),$$

$$P_{\text{spot cooler}}(d) = \frac{\dot{Q}_{\text{spot,act}}(d)}{\text{COP}},$$

$$\Delta p_{\text{avail}}(\dot{V}) = \max\left(\Delta p_{\text{shutoff}}\left(1 - \frac{\dot{V}}{\dot{V}_{\text{free}}}\right), 0\right),$$

$$P_{\text{assist blower}}(d) = \begin{cases} P_{\text{motor}}, & \dot{V}_{\text{ach}}(d) > 0 \text{ and blower enabled,} \\ 0, & \text{otherwise.} \end{cases}$$

Cooling operating flow is solved from the scalar residual

$$R_{\dot{V}}(\dot{V}) = \Delta p_{\text{sys}}(\dot{V}) - \Delta p_{\text{avail}}(\dot{V}),$$

with bisection over $[0, \dot{V}_{\text{req}}]$ to return

$$\dot{V}_{\text{ach}}(d) = \max\Big\{\dot{V} \in [0, \dot{V}_{\text{req}}(d)] : R_{\dot{V}}(\dot{V}) \le 0\Big\}.$$

At this solved $\dot{V}_{\text{ach}}(d)$, the same segment energy balance gives

$$Q_{\text{to bed},c}(T_\infty(d)) = \sum_{s=1}^{N_{\text{seg}}} \left(Q_{\text{wall},s} + Q_{\text{jet},s} - Q_{\text{evap},s}\right),$$

which is usually negative in cooling mode (heat removed from bed).

Matrix solve vs residual solve (clarification). The two-node temperature step is a direct linear solve, not an iterative matrix residual solve:

$$\mathbf{A}(d)\mathbf{T}(d) = \mathbf{b}(d), \qquad \mathbf{T}(d) = \mathbf{A}(d)^{-1}\mathbf{b}(d).$$

The iterative residual solves in this layer are scalar bisections: $R_{\dot{V}}(\dot{V})$ (blower/system match), $R_{\text{inlet}}(p_{\text{in}}) = \dot{m}_{\text{remaining}}(p_{\text{in}})$ (closed-end inlet pressure), and $R_{\text{bed ref}}(T_{\text{ref}}) = \frac{1}{2}(T_b + T_t) - T_{\text{ref}}$ (cooling branch bed-reference closure).

Variables used in the equations above (explicit definitions).

- d: representative day index $(1, \ldots, 365)$.
- g: heater-tube group index; $n_{\text{tube},g}$ is the number of identical tubes in group g.
- b: aeration branch index $(1, \ldots, N_b)$.
- s: axial segment index $(1, \ldots, N_{\text{seg}})$.
- $R_{\text{tube},g}(d)$: electrical resistance of one tube in group g at day-d operating temperature.
- ρ_{20}: wire resistivity at $20\,^\circ\text{C}$.

- α: wire temperature coefficient of resistivity.
- $\bar{T}_{w,g}(d)$: representative wire temperature for group g, day d.
- $L_{w,g}$, $A_{w,g}$: wire length and wire cross-sectional area for group g.
- V: applied electrical voltage across each series string.
- $N_{s,g}$: number of tube elements in series in group g.
- $I_{\text{tube},g}(d)$: current through one tube/string in group g.
- $P_{\text{tube},g}(d)$: electrical power of one tube/string in group g.
- $P_h(T_\infty(d))$: total heating electrical input at ambient $T_\infty(d)$.
- $Q_{\text{wall},b,s}$: sensible heat transfer from branch airflow to bed-side walls in branch b, segment s.
- $\dot{m}_{\text{mean},b,s}$: mean airflow through segment s of branch b.
- $c_{p,b,s}$: specific heat of segment airflow in branch b, segment s.
- $T_{\text{air,in},b,s}$, $T_{\text{air,out},b,s}$: segment inlet/outlet air temperatures.
- $Q_{\text{jet},b,s}$: sensible heat carried by released jet flow relative to bed reference temperature.
- $\dot{m}_{\text{rel},b,s}$: mass flow released through perforations into bed in branch b, segment s.
- $T_{\text{rel},b,s}$: release/jet air temperature to the bed in branch b, segment s.
- $T_{\text{bed,ref}}$: branch-level bed reference temperature used for local segment balances.
- $Q_{\text{evap},b,s}$: latent heat sink due to evaporation in branch b, segment s.
- $\dot{m}_{\text{evap},b,s}$: evaporation mass flow rate in branch b, segment s.
- $h_{fg,b,s}$: latent heat of vaporization used in branch b, segment s.
- $Q_{\text{to bed},h}(T_\infty(d))$: net signed heating-mode bed heat-transfer rate; positive means heat delivered to bed.
- $\dot{Q}_{\text{spot,req}}(d)$: requested sensible spot-cooling load at day d.
- $\dot{m}(d)$, $c_p(d)$: day-d cooling-branch mass flow and effective specific heat.
- T_{target}: spot-cooler outlet/control target temperature.
- $\dot{Q}_{\text{spot,max}}(d)$: available spot-cooler sensible capacity at day d.
- $\dot{Q}_{\text{spot,act}}(d)$: actual spot-cooler sensible load after capacity cap.
- COP: coefficient of performance used to convert sensible load to electric power.
- $P_{\text{spot cooler}}(d)$: spot-cooler electrical power.
- $\Delta p_{\text{shutoff}}$, $\dot{V}_{\text{free}}$: blower shutoff pressure and free-delivery volumetric flow (fan-curve endpoints).
- $\Delta p_{\text{avail}}(\dot{V})$: blower available pressure at trial flow $\dot{V}$.
- $\Delta p_{\text{sys}}(\dot{V})$: system-required pressure at trial flow $\dot{V}$.
- $R_{\dot{V}}(\dot{V})$: scalar residual for blower/system matching; root/feasible sign gives operating flow.
- $\dot{V}_{\text{req}}(d)$: requested volumetric flow for day d.
- $\dot{V}_{\text{ach}}(d)$: achieved volumetric flow from bisection on $R_{\dot{V}}$.
- P_{motor}: assist-blower motor power rating used when blower is active.
- $P_{\text{assist blower}}(d)$: assist-blower electric power for day d (rated value or zero).
- $Q_{\text{to bed},c}(T_\infty(d))$: net signed cooling-mode bed heat-transfer rate; typically negative (heat removed from bed).
- $\mathbf{A}(d)$, $\mathbf{T}(d)$, $\mathbf{b}(d)$: day-d two-node linear-system matrix, unknown temperature vector, and right-hand-side vector.
- $R_{\text{inlet}}(p_{\text{in}})$: inlet-pressure scalar residual in the closed-end pressure closure.
- $\dot{m}_{\text{remaining}}(p_{\text{in}})$: remaining mass-flow mismatch used in R_{inlet}.
- $R_{\text{bed ref}}(T_{\text{ref}})$: bed-reference closure residual in cooling branch.
- T_b, T_t: bottom and top bed-node temperatures used by the closure.
- T_{ref}: trial bed reference temperature in the cooling closure.

Derivation notes (why these forms are used).

- $R_{\text{tube},g}$, $I_{\text{tube},g}$, and $P_{\text{tube},g}$ follow temperature-corrected resistivity plus Ohm's law and Joule

heating.

- $Q_{\text{to bed},h}$ and $Q_{\text{to bed},c}$ are first-law balances over all active segments: sensible-to-wall + sensible-by-jet − latent evaporation.
- $\dot{Q}_{\text{spot,req}}$ is a sensible-load relation $\dot{m}c_p\Delta T$; $\dot{Q}_{\text{spot,act}}$ applies equipment capacity capping.
- $P_{\text{spot cooler}} = \dot{Q}_{\text{spot,act}}/\text{COP}$ is the standard load-to-electric relation under the model COP assumption.
- $\Delta p_{\text{avail}}(\dot{V})$ is the linear fan-curve model between shutoff and free-delivery points.
- $R_{\dot{V}}(\dot{V}) = 0$ enforces operating-point consistency: system pressure requirement equals blower available pressure.
- $\mathbf{AT} = \mathbf{b}$ is solved directly for the two thermal nodes; only the closure constraints use scalar residual bisection.

I then computed required duty:

$$D_h^*(d) = \frac{Q_{\text{req},h}(d)}{Q_{h,\text{cap}}(d)}, \qquad D_c^*(d) = \frac{Q_{\text{req},c}(d)}{Q_{c,\text{cap}}(d)},$$

and clipped applied duty to physical bounds:

$$D_h(d) = \min(\max(D_h^*(d), 0), 1), \qquad D_c(d) = \min(\max(D_c^*(d), 0), 1).$$

The generic clipped-duty form f_{day} in Eq. (244) maps to these mode-specific quantities through the active heating/cooling branch.

The applied bed load is then:

$$Q_{\text{bed}}(d) = \begin{cases} D_h(d)\, Q_{h,\text{cap}}(d), & \text{heating day,} \\ -D_c(d)\, Q_{c,\text{cap}}(d), & \text{cooling day,} \\ 0, & \text{idle day.} \end{cases}$$

If required duty exceeds unity, I preserve $D = 1$ as the applied maximum and record unmet load separately; this is how capacity shortfall is represented without violating equipment limits.

18.6 Daily bed temperatures, electricity, cost, and water (247–251)

Given applied $Q_{\text{bed}}(d)$, I solved the same two-node steady bed model used elsewhere:

$$\begin{bmatrix} UA_{b,\infty} + UA_{bt} & -UA_{bt} \\ -UA_{bt} & UA_{t,\infty} + UA_{bt} \end{bmatrix} \begin{bmatrix} T_b(d) \\ T_t(d) \end{bmatrix} = \begin{bmatrix} f_b Q_{\text{bed}}(d) + UA_{b,\infty} T_\infty(d) \\ (1 - f_b) Q_{\text{bed}}(d) + UA_{t,\infty} T_\infty(d) \end{bmatrix}.$$

I computed daily electrical energy by mode:

$$E_h(d) = \frac{P_{h,\text{elec}}(d)\, D_h(d)\, 24}{1000}, \qquad E_c(d) = \frac{P_{c,\text{elec}}(d)\, D_c(d)\, 24}{1000},$$

then

$$E_{\text{day}} = E_h(d) + E_c(d).$$

Daily cost is:

$$C(d) = c_{\text{elec}} E_{\text{day}},$$

and daily water replacement estimate is:

$$m_{\text{water}}(d) = \dot{m}_{\text{water},24h}\, D(d),$$

where $D(d)$ is the active heating or cooling duty on that day.

Section 18 feed-forward map (explicit variable dependency order)

18.1 Climate reference values and regression baseline (229, 230). **Inputs:**

- month lengths $\{L_m\}$,
- month-midpoint reference days $\{d_i\}$ (equivalently $\{d_m\}$),
- monthly means $\{\bar{T}_m\}$,
- monthly standard deviations $\{\sigma_m\}$,
- kernel bandwidth h.

Feed-forward chain:

$$\{\bar{T}_m, d_i, h\} \to \delta_i(d),\ w_i(d) \to T_{\infty,\text{base}}(d),$$
$$\{\sigma_m, d_i, h\} \to \delta_i(d),\ w_i(d) \to \sigma(d).$$

Outputs passed forward:

- baseline ambient profile $T_{\infty,\text{base}}(d)$,
- daily spread profile $\sigma(d)$.

18.2 Freeze-day allocation and anomaly override (231–234). **Inputs:**

- expected freeze days F_m,
- mean freeze day $\mu_{f,m}$,
- freeze monthly center $M_{f,m}$,
- freeze anomaly bounds $\Delta T^{\text{low}}_{f,m}, \Delta T^{\text{high}}_{f,m}$,
- freeze anomaly spread $\sigma_{\Delta T,f,m}$,
- baseline ambient $T_{\infty,\text{base}}(d)$ from 18.1.

Feed-forward chain:

$$F_m \to n_{f,m} \text{ from } \beta_{f,m} = \lfloor F_m \rfloor,\ \rho_{f,m} = F_m - \beta_{f,m},\ U^{\text{cnt}}_{f,m} \sim \text{Uniform}(0,1) \to \mathcal{S}_{f,m},$$

$$\{T_{\infty,\text{base}}(d), \bar{T}_m, M_{f,m}, \Delta T^{\text{low}}_{f,m}, \Delta T^{\text{high}}_{f,m}, \sigma_{\Delta T,f,m}, \mathcal{S}_{f,m}\} \to M_{f,m},\ A_{f,m}(r) \to T_{\infty,\text{preH}}(d) \text{ via the 18.2-G conditional ov}$$

Output passed forward:

- pre-heat-wave ambient profile $T_{\infty,\text{preH}}(d)$.

18.3 Heat-wave allocation and anomaly override (235–238). **Inputs:**

- expected spell starts S_m,
- expected heat-wave days H_m,
- mean start day $\mu_{h,m}$,
- heat-wave mean anomaly $\Delta T_{h,m}$,
- heat-wave anomaly bounds $\Delta T^{\text{low}}_{h,m}, \Delta T^{\text{high}}_{h,m}$,
- heat-wave anomaly spread $\sigma_{\Delta T,h,m}$,
- pre-heat-wave profile from 18.2.

Feed-forward chain:

$$\{S_m, H_m\} \to \{n_{s,m}, n_{h,m}\} \to \mathcal{S}_{h,m} \text{ (using the same recursive set logic as 18.2)},$$

$$\{\Delta T_{h,m}, \Delta T^{\text{low}}_{h,m}, \Delta T^{\text{high}}_{h,m}, \sigma_{\Delta T,h,m}, \mathcal{S}_{h,m}\} \to \{\Delta T_{h,m,q}\}_{q=1}^{n_{s,m}},$$

$$\{T_{\infty,\text{base}}(d), \bar{T}_m, \Delta T_{h,m,q}, T_{\infty,\text{preH}}(d)\} \to T_\infty(d)$$
$$= \max\big(T_{\infty,\text{base}}(d),\, \bar{T}_m + \Delta T_{h,m,q(d)},\, T_{\infty,\text{preH}}(d)\big) \text{ on spell days } r_m(d) \in \mathcal{B}^{\text{all}}_{h,m}.$$

Output passed forward:

- final ambient profile $T_\infty(d)$ used in day-level thermal solves.

18.4 Daily heating and cooling loads (239, 240). Inputs:

- final ambient $T_\infty(d)$ from 18.3,
- setpoints $T_{h,\text{set}}$, $T_{c,\text{set}}$,
- day-resolved effective bin conductance terms (UA_{bin}).

Feed-forward chain:

$$\{T_\infty(d), T_{h,\text{set}}, UA_{\text{bin}}\} \to Q_{\text{req},h}(d),$$
$$\{T_\infty(d), T_{c,\text{set}}, UA_{\text{bin}}\} \to Q_{\text{req},c}(d).$$

Outputs passed forward:

- required day loads $Q_{\text{req},h}(d)$, $Q_{\text{req},c}(d)$.

18.5 Capacity re-solve and duty application (241–246). Inputs:

- required day loads from 18.4,
- final ambient $T_\infty(d)$ from 18.3,
- selected heating and cooling reference points.

Feed-forward chain:

$$\{T_\infty(d), \text{heating reference}\} \to \{Q_{h,\text{cap}}(d), P_{h,\text{elec}}(d)\},$$
$$\{T_\infty(d), \text{cooling reference}\} \to \{Q_{c,\text{cap}}(d), P_{c,\text{elec}}(d)\},$$
$$\{Q_{\text{req},h}(d), Q_{h,\text{cap}}(d)\} \to D^*_h(d) \to D_h(d),$$
$$\{Q_{\text{req},c}(d), Q_{c,\text{cap}}(d)\} \to D^*_c(d) \to D_c(d),$$
$$\{D_h(d), D_c(d), Q_{h,\text{cap}}(d), Q_{c,\text{cap}}(d)\} \to Q_{\text{bed}}(d).$$

Outputs passed forward:

- applied bed load $Q_{\text{bed}}(d)$,
- mode duty $D(d)$,
- mode electrical power ($P_{h,\text{elec}}(d)$ on heating days or $P_{c,\text{elec}}(d)$ on cooling days).

18.6 Daily temperatures, electricity, cost, and water (247–251). Inputs:

- $Q_{\text{bed}}(d)$, duty, and power from 18.5,
- ambient $T_\infty(d)$ from 18.3,
- two-node conductance parameters and split fraction f_b,
- electricity price c_{elec},
- nominal daily water rate $\dot{m}_{\text{water},24h}$.

Feed-forward chain:

$$\{Q_{\text{bed}}(d), T_\infty(d), UA_{b,\infty}, UA_{t,\infty}, UA_{bt}, f_b\} \to \{T_b(d), T_t(d)\},$$
$$\{P_{h,\text{elec}}(d), D_h(d)\} \to E_h(d), \quad \{P_{c,\text{elec}}(d), D_c(d)\} \to E_c(d) \to E_{\text{day}},$$
$$\{E_{\text{day}}, c_{\text{elec}}\} \to C(d),$$
$$\{\dot{m}_{\text{water},24h}, D(d)\} \to m_{\text{water}}(d).$$

Annual aggregation closure:

$$\sum_{d=1}^{365} \{E_h(d), E_c(d), C(d), m_{\text{water}}(d)\} \to \{\text{annual energy, annual cost, annual water}\}.$$

18.7 Literature-backed microbial heat source q_{bio} for winter heating

To model partial winter reliance on in-bed microbial activity, I added an explicit volumetric source term q_{bio} in the bottom-mat reduced heating solver. The source is parameterized from oxygen uptake rate (OUR) and oxygen-to-heat conversion enthalpy, then converted to volumetric units for direct insertion into the bed energy equation.

Primary source references used in this implementation.

- Sonowal et al. report dewatered-sludge vermicomposting OUR values of 18.9 ± 0.5 mg O_2/g VS/day initially, and late-stage values as low as 2.0–3.1 mg O_2/g VS/day [R20].
- de Guardia et al. use the biodegradation heat model $H_{\text{BIO}} = r_{O_2} h_{\text{BIO}}$, and report literature h_{BIO} spans roughly 304–448 kJ/mol O_2, with 440 kJ/mol O_2 used in their calibrated runs [R19].
- Harper et al. report an empirical oxygen-to-heat slope of about 9.8 MJ/kg O_2, equivalent to 313.6 kJ/mol O_2, providing a lower empirical conversion reference [R18].
- Gea et al. show OUR is the most reliable process activity indicator under dynamic aeration monitoring, and provide broad OUR ranges across wastes that support dynamic activity modeling rather than static maturity-only values [R21].

Step 1: OUR time profile at the selected winter operating stage. I represent process-stage decay from an initial value to a mature value with:

$$\text{OUR}(t_d) = \text{OUR}_{\text{mature}} + \left(\text{OUR}_{\text{initial}} - \text{OUR}_{\text{mature}}\right) \exp(-k_d t_d),$$

where t_d is operating day in days and k_d is in day^{-1}. The default $k_d = 0.0353$ day^{-1} is the mean implied by reported 45-day fold reductions from Sonowal et al. [R20].

Step 2: Convert OUR to heat per unit VS mass. Using the selected scenario enthalpy h_{O_2} (kJ/mol O_2):

$$q_{\text{bio,VS}}\ [\text{W/kgVS}] = \text{OUR}\ [\text{mg } O_2\text{/gVS/day}] \cdot \frac{h_{O_2}}{32} \cdot \frac{1000}{86400}.$$

The script exposes scenario bounds:

- conservative: $h_{O_2} = 313.6$ kJ/mol O_2 (Harper reference),
- base: $h_{O_2} = 376.8$ kJ/mol O_2 (midpoint working value),
- aggressive: $h_{O_2} = 440$ kJ/mol O_2 (de Guardia working value).

Step 3: Convert to volumetric source term for the bed field solve.

$$q_{\text{bio,vol}}\ [\text{W/m}^3] = q_{\text{bio,VS}} \cdot \rho_{\text{bulk}} \cdot f_{\text{VS,wet}},$$

where ρ_{bulk} is wet bulk density and $f_{\text{VS,wet}}$ is wet-mass VS fraction.

Step 4: Apply local temperature response and clipping. I use a bounded Q_{10} scaling:

$$f_T(T) = \text{clip}\Big(Q_{10}^{(T-T_{\text{ref}})/10}, f_{T,\text{min}}, f_{T,\text{max}}\Big), \qquad q_{\text{bio,local}} = q_{\text{bio,vol}}\, f_T(T).$$

Defaults are $Q_{10} = 2.0$, $T_{\text{ref}} = 22°\text{C}$, and clip 0.35–1.65, preventing runaway positive feedback in the explicit relaxation loop.

Step 5: Insert into the bed PDE update. For the 2D reduced field, the interior update adds:

$$\frac{\partial T}{\partial t} \supset \frac{q_{\text{bio,local}}}{\rho c_p},$$

so the implemented source competes directly with evaporation and wall-loss sinks in the same units.

Step 6: Net winter external-heating requirement. For duty and feasibility calculations, the winter external requirement is decomposed as:

$$Q_{\text{loss,gross}} = UA\,(T_{\text{set}} - T_\infty), \qquad Q_{\text{ext,req}} = \max(Q_{\text{loss,gross}} - Q_{\text{bio,ref}},\, 0)\,,$$

where $Q_{\text{bio,ref}}$ is the temperature-adjusted reference bioheat at T_{set}. This is now printed explicitly in the heating summary as gross loss, microbial credit, and residual external requirement.

18.8 Practical interpretation for report writing

The implemented q_{bio} model is intentionally transparent and auditable:

- every conversion step is explicit (OUR $\rightarrow$ W/kgVS $\rightarrow$ W/m^3),
- all literature-sensitive terms are configuration parameters (OUR bounds, h_{O_2} scenario, decay rate, VS fraction),
- solver outputs now separate mat heat and microbial heat contributions,
- the net-load decomposition is reported explicitly, so winter feasibility claims can be traced to assumptions instead of hidden fitting.

Implementation trace (how I actually executed it in code)

I executed this workflow through the following function chain:

- configuration and climate defaults: `year_round_analysis_config(...)`,
- integer allocation and event placement: month-wise stochastic-rounding equations from 18.2/18.3, plus `gaussian_weighted_day_selection(...)` and `heatwave_day_mask(...)`,
- spell anomaly sampling and overlays: `sample_triangular_mean_matched(...)` (freeze default), `sample_truncated_normal(...)` (heat-wave and legacy freeze mode), `circular_gaussian_kernel_regression(...)`, `build_representative_climate_year(...)`,

- microbial source assembly for winter heating: `_bioheat_base_terms(...)`, `_bioheat_temperature_factor(...)`, `_bioheat_reference_power_w(...)`,
- daily simulation task: `simulate_year_round_day_task(...)`,
- ambient-specific re-solve of operating points: `simulate_heating_reference_point(...)` and `simulate_summer_cooling_point(...)`,
- bottom-mat field solve with explicit q_{bio} insertion: `simulate_bottom_mats_state(...)`,
- daily and annual aggregation: `build_year_round_payload(...)`,
- report export: `write_year_round_summary(...)`.

Reproducibility notes

To ensure reproducibility, I preserved deterministic timing-selection equations (fixed ranking rules over remaining start-day candidates), then used seeded stochastic spell sampling (`spellAnomalyRngSeed`) with explicit bound-enforcement equations, plus explicit annual summary totals for freeze days, heat-wave starts, heat-wave days, energy, cost, water, and unmet-load equivalent.

In summary, the Section 18 implementation is an equation-driven annual operations layer with deterministic timing and seeded stochastic spell intensities: smooth baseline climate from monthly reference values, event overlays with bounded spell anomalies, ambient-specific capacity re-solving, physically bounded duty application, and day-level thermo-economic output accumulation.

18.2 Climate reference values and regression baseline

The annual layer stores for each month m a Connecticut monthly mean temperature $\bar{T}_m$ and monthly standard deviation σ_m derived from the NOAA series described in `HeatTransferStudy/VermiHeatLoss.tex:L2576-L2616`. The representative-year ambient baseline uses circular Gaussian-kernel regression:

$$\hat{y}(d) = \frac{\sum_i w_i(d) y_i}{\sum_i w_i(d)}, \qquad w_i(d) = \exp\left[-\frac{1}{2}\left(\frac{\delta_i(d)}{h}\right)^2\right], \tag{229}$$

with wraparound day distance

$$\delta_i(d) = \min(|d - d_i|,\ P - |d - d_i|)\,. \tag{230}$$

Identity clarification for Eqs. (229)–(230): d is the evaluated day-of-year; P is the annual period length (365 days in this implementation); d_i is the day-of-year reference day for reference point i (month midpoint in this implementation); i indexes monthly reference points; h is the Gaussian bandwidth in days; and y_i is the active reference-value family at i, used in two separate passes ($y_i = \bar{T}_i$ for baseline ambient and $y_i = \sigma_i$ for daily spread). When month-midpoint notation d_m appears, it is the same reference-day map as d_i under the one-reference-point-per-month construction. This same periodic Gaussian regression is applied both to the monthly mean reference values and to the monthly standard-deviation reference values `heat_transfer_study.py:L4794-L4808`; `heat_transfer_study.py:L4866-L4879`. Therefore, for the mean-temperature pass used by the annual layer,

$$T_{\infty,\text{base}}(d) = \frac{\sum_{i=1}^{12} w_i(d)\,\bar{T}_i}{\sum_{i=1}^{12} w_i(d)}, \tag{231}$$

with $w_i(d)$ and $\delta_i(d)$ from Eqs. (229)–(230).

Temperature-variable identity and derivation map

- $\bar{T}_m$: configured monthly mean ambient (from `climate_data.monthlyMean_C`).
- σ_m: configured monthly ambient spread (from `climate_data.monthlyStd_C`).
- $T_{\infty,\text{base}}(d)$: daily baseline ambient from periodic Gaussian regression of $\bar{T}_m$.
- $\sigma(d)$: daily spread profile from periodic Gaussian regression of σ_m.
- $T_m(d)$: month-local ambient profile while freeze/heat-wave overrides are being applied.
- $T_\infty(d)$: final day ambient after all event overlays, passed to daily load and capacity solves.
- $T_{h,\text{set}}, T_{c,\text{set}}$: resolved annual heating/cooling setpoints from the year-round config (or inherited model defaults).
- $T_b(d), T_t(d)$: daily bottom/top bed-node temperatures from Eq. (291), solved using $Q_{\text{bed}}(d)$ and $T_\infty(d)$.
- Heat-wave spell threshold temperature: $\bar{T}_m + \Delta T_{h,m,q}$, where $\Delta T_{h,m,q}$ is the spell-specific bounded-normal draw.
- NOAA spell-bound artifacts: raw rows in `Heat Transfer Study/data/climate/noaa_daily_summaries_USW000` derived bounds in `Heat Transfer Study/data/climate/noaa_heatwave_spell_anomaly_bounds_USW00014740` extraction via `Heat Transfer Study/tools/derive_noaa_heatwave_spell_bounds.py`.

Capacity/power identity map (used in Eqs. (275)–(276)).

- $Q_{\text{to bed},h}(T_\infty(d))$: signed net bed heat-transfer rate from the heating reference-point re-solve at day ambient $T_\infty(d)$ (positive for heating-to-bed).
- $Q_{\text{to bed},c}(T_\infty(d))$: signed net bed heat-transfer rate from the cooling reference-point re-solve at day ambient $T_\infty(d)$ (typically negative for cooling/removal from bed).
- $Q_{h,\text{cap}}(d)$, $Q_{c,\text{cap}}(d)$: positive capacity magnitudes used in duty equations, where $Q_{h,\text{cap}}(d) = Q_{\text{to bed},h}(T_\infty(d))$ and $Q_{c,\text{cap}}(d) = -Q_{\text{to bed},c}(T_\infty(d))$.
- $P_h(T_\infty(d))$: heating electrical power returned by the heating re-solve at day ambient $T_\infty(d)$.
- $P_{\text{spot cooler}}(d)$: spot-cooler electrical power returned by the cooling re-solve on day d.
- $P_{\text{assist blower}}(d)$: assist-blower electrical power returned by the cooling re-solve on day d.
- $P_{h,\text{elec}}(d)$, $P_{c,\text{elec}}(d)$: mode electric powers used in Eqs. (292)–(293), with $P_{h,\text{elec}}(d) = P_h(T_\infty(d))$ and $P_{c,\text{elec}}(d) = P_{\text{spot cooler}}(d) + P_{\text{assist blower}}(d)$.

18.3 Freeze-day allocation and anomaly override

Expected monthly freeze-day counts are converted to representative integer counts by largest-remainder allocation:

$$n_{f,m} = \text{LR}(F_m). \tag{232}$$

Within each month, day-of-month weights are centered on the observed mean freeze day $\mu_{f,m}$:

$$w_{f,m}(r) = \exp\left[-\frac{(r-\mu_{f,m})^2}{2s_{f,m}^2}\right], \tag{233}$$

and sequential selection applies a repulsion modifier

$$\tilde{w}_{f,m}(r) = w_{f,m}(r) \prod_{p \in \mathcal{S}_{f,m}} \left[1 - 0.65 \exp\left(-\frac{(r-p)^2}{2r_{f,m}^2}\right)\right]. \tag{234}$$

Here r and p are day-of-month indices within month m (not kernel indices from Eq. (229)); $s_{f,m}$ is the freeze-timing spread parameter; $r_{f,m}$ is the repulsion length scale; and $\mathcal{S}_{f,m}$ is the set of freeze days already selected in that month. In the current implementation:

$$s_{f,m} = \max\left(2, \frac{L_m}{6}\right), \qquad r_{f,m} = \max\left(1.5, \frac{L_m}{14}\right),$$

where L_m is the number of days in month m.

Freeze-anomaly magnitude derivation (explicit). Define monthly freeze-anomaly center and bounds:

$$M_{f,m} \equiv \Delta T_{f,m}, \quad a_{f,m} = \Delta T^{\text{low}}_{f,m}, \quad b_{f,m} = \Delta T^{\text{high}}_{f,m}, \quad \mu^{\star}_{f,m} = \text{clip}(M_{f,m}, a_{f,m}, b_{f,m}). \tag{235}$$

Here $\Delta T^{\text{low}}_{f,m}, \Delta T^{\text{high}}_{f,m}$ are anomaly-magnitude bounds (not temperatures), loaded from `climate_data.freeze.minAno` and `climate_data.freeze.maxAnomalyBelowMonthlyMean_C`$[m]$. Historical NOAA derivation in month m: for each freeze day j,

$$a_{m,j} = \max\left(0, \bar{T}_m - T^{\text{freeze day}}_{m,j}\right),$$

$$\Delta T^{\text{low}}_{f,m} = \min_j a_{m,j}, \qquad \Delta T^{\text{high}}_{f,m} = \max_j a_{m,j}.$$

So $\mu^{\star}_{f,m}$ is the bounded (clipped) freeze-anomaly center for month m. Define mean-matched triangular mode:

$$c^{\text{raw}}_{f,m} = 3\mu^{\star}_{f,m} - a_{f,m} - b_{f,m}, \qquad c_{f,m} = \text{clip}(c^{\text{raw}}_{f,m}, a_{f,m}, b_{f,m}). \tag{236}$$

Here $c^{\text{raw}}_{f,m}$ is the unconstrained triangular-mode target, chosen so that

$$\frac{a_{f,m} + b_{f,m} + c^{\text{raw}}_{f,m}}{3} = \mu^{\star}_{f,m}$$

before clipping to $[a_{f,m}, b_{f,m}]$. Why: this mean-matching preserves the intended monthly anomaly center while still allowing bounded day-to-day stochastic variation. Define triangular density on $[a_{f,m}, b_{f,m}]$:

$$t_{f,m}(x) = \begin{cases} \dfrac{2(x - a_{f,m})}{(b_{f,m} - a_{f,m})(c_{f,m} - a_{f,m})}, & a_{f,m} \le x \le c_{f,m}, \\ \dfrac{2(b_{f,m} - x)}{(b_{f,m} - a_{f,m})(b_{f,m} - c_{f,m})}, & c_{f,m} \le x \le b_{f,m}, \\ 0, & \text{otherwise}. \end{cases} \tag{237}$$

For any $a_{f,m} \le u \le v \le b_{f,m}$,

$$\Pr(u \le X_{f,m,q} \le v) = \int_u^v t_{f,m}(x)\, dx. \tag{238}$$

Code-equivalent freeze-event draw for $X_{f,m,q}$. With seed s_f = `freeze.anomalyRngSeed`, default freeze draw is

$$X_{f,m,q} = \text{Triangular}(a_{f,m}, c_{f,m}, b_{f,m}). \tag{239}$$

Then clip:

$$X_{f,m,q} \leftarrow \text{clip}(X_{f,m,q}, a_{f,m}, b_{f,m}). \tag{240}$$

Define string normalization

$$\mathrm{norm}(s) = \mathrm{lower}\Big(\mathrm{replace}\left(\mathrm{replace}(s, \texttt{"-"}, \texttt{"_"}), \texttt{" "}, \texttt{"_"}\right)\Big).$$

Define freeze model key

$$\kappa_f = \mathrm{norm}(\texttt{freeze.anomalySamplingModel}).$$

Define explicit mode flags (equivalent to code logic):

$$\texttt{isDeterministic}_f = \begin{cases} 1, & \kappa_f = \texttt{off} \text{ or } \texttt{none} \text{ or } \texttt{disabled} \text{ or } \texttt{deterministic} \text{ or } \texttt{deterministic_mean}, \\ 0, & \text{otherwise}, \end{cases}$$

$$\texttt{isTriangular}_f = \begin{cases} 1, & \kappa_f = \texttt{triangular} \text{ or } \texttt{triangular_mean_matched} \text{ or } \texttt{triangular_between_monthly_min_m} \\ 0, & \text{otherwise}. \end{cases}$$

Conceptual meaning:

- κ_f is a configuration key (text), not a measured climate variable.
- $\mathrm{norm}(\cdot)$ converts formatting variants to one canonical key so branch selection is robust.
- $\texttt{isDeterministic}_f = 1$ selects deterministic freeze magnitude assignment:

$$\Delta T_{f,m,q} = \mu^{\star}_{f,m} \text{ for all selected } q.$$

- $\texttt{isTriangular}_f = 1$ selects triangular stochastic assignment:

$$\Delta T_{f,m,q} = X_{f,m,q}, \quad X_{f,m,q} \in [a_{f,m}, b_{f,m}].$$

- These are logic flags (yes/no), not climate variables.

Then event magnitude assignment is

$$\Delta T_{f,m,q} = \begin{cases} \mu^{\star}_{f,m}, & \texttt{isDeterministic}_f = 1, \\ X_{f,m,q}, \quad X_{f,m,q} = \mathrm{Triangular}(a_{f,m}, c_{f,m}, b_{f,m}), & \texttt{isTriangular}_f = 1 \text{ and } b_{f,m} > a_{f,m}, \\ \mu^{\star}_{f,m}, & b_{f,m} \le a_{f,m}, \\ \mu^{\star}_{f,m} \text{ or legacy bounded-normal draw}, & \texttt{isDeterministic}_f = 0 \text{ and } \texttt{isTriangular}_f = 0, \end{cases} \tag{241}$$

followed by $\Delta T_{f,m,q} \leftarrow \mathrm{clip}(\Delta T_{f,m,q}, a_{f,m}, b_{f,m})$. Define day allocation:

$$A_{f,m}(r) = \begin{cases} 0, & r \notin \mathcal{S}_{f,m}, \\ \Delta T_{f,m,q}, & r = d_{f,m,q} \text{ for some } q \in \{1, \dots, n_{f,m}\}, \end{cases} \qquad r = 1, \dots, L_m. \tag{242}$$

where $d_{f,m,1} < \cdots < d_{f,m,n_{f,m}}$ are selected freeze days. Here $A_{f,m}(r)$ is a deterministic day-to-magnitude map; it does not itself generate random values. Stochasticity (when $\texttt{isTriangular}_f = 1$) enters only when drawing $\Delta T_{f,m,q}$, after which q is paired with ordered selected days $d_{f,m,1} < \cdots < d_{f,m,n_{f,m}}$ with no additional random reshuffle.

On selected freeze days, the ambient profile is overridden by

$$T^{\mathrm{th}}_{f,m}(d) = \bar{T}_m - A_{f,m}(r_m(d)). \tag{243}$$

$T^{\mathrm{th}}_{f,m}(d)$ is the day-level freeze-threshold ambient (the clamp value used by the freeze override).

$$T_m(d) = \min\Big(T_{\mathrm{base},m}(d),\, T^{\mathrm{th}}_{f,m}(d)\Big). \tag{244}$$

Here $r_m(d)$ is the day-of-month index for day-of-year d, and $T_m(d)$ denotes the month-local daily ambient profile after applying freeze overrides. These relations correspond to `heat_transfer_study.py:L4683-L4732`; `heat_transfer_study.py:L4893-L4906`.

18.4 Heat-wave allocation and anomaly override

This subsection is organized as a derivation pipeline from monthly statistics to daily ambient temperature.

Step 1: Integer event allocation. Monthly spell starts and total heat-wave days are allocated by largest-remainder rounding:

$$n_{s,m} = \mathrm{LR}(S_m), \qquad n_{h,m} = \mathrm{LR}(H_m). \tag{245}$$

Step 2: Construct spell lengths from total days. The implementation enforces a minimum spell length of 3 days and at least one non-spell day between consecutive spells. Therefore, the spell-start count is feasibility-limited by

$$n_{s,m} \leftarrow \min\left(n_{s,m}, \left\lfloor \frac{L_m + 1}{4} \right\rfloor\right). \tag{246}$$

Given this $n_{s,m}$, feasible total spell days satisfy

$$n_{h,m}^{\min} = 3n_{s,m}, \qquad n_{h,m}^{\max} = L_m - (n_{s,m} - 1), \tag{247}$$

$$n_{h,m} \leftarrow \min\big(\max(n_{h,m}, n_{h,m}^{\min}),\, n_{h,m}^{\max}\big). \tag{248}$$

Then initialize spell lengths:

$$\ell_{h,m,q} = 3, \qquad q = 1, \ldots, n_{s,m}. \tag{249}$$

Define remaining days after minimum allocation:

$$e_m = n_{h,m} - 3n_{s,m}. \tag{250}$$

Those e_m days are distributed cyclically across spells in code. Equivalent closed form:

$$\text{for } n_{s,m} > 0: \qquad u_m = \left\lfloor \frac{e_m}{n_{s,m}} \right\rfloor, \qquad v_m = e_m - u_m n_{s,m}, \tag{251}$$

$$\ell_{h,m,q} = \begin{cases} 3 + u_m + 1, & q \le v_m, \\ 3 + u_m, & q > v_m, \end{cases} \qquad q = 1, \ldots, n_{s,m}, \tag{252}$$

so

$$\sum_{q=1}^{n_{s,m}} \ell_{h,m,q} = n_{h,m}.$$

Example: $n_{s,m} = 3$, $n_{h,m} = 14 \Rightarrow e_m = 5,\ u_m = 1,\ v_m = 2$, hence $(\ell_{h,m,1}, \ell_{h,m,2}, \ell_{h,m,3}) = (5, 5, 4)$. If $n_{s,m} = 0$, the spell-length vector is empty and no heat-wave spell blocks are created for that month.

Step 3: Compute start-day preference score. The base start-day weighting is

$$w_{h,m}(r) = \exp\left[-\frac{(r - \mu_{h,m})^2}{2s_{h,m}^2}\right], \tag{253}$$

where r is day-of-month index in month m, and $s_{h,m}$ is start-day spread.

Step 4: Build the already-selected set recursively. In explicit recursive form:

$$\mathcal{S}_{h,m}^{(0)} = \varnothing, \tag{254}$$

$$\mathcal{I}_{h,m}^{(k)} = \{1, \ldots, L_m\} \setminus \mathcal{S}_{h,m}^{(k-1)}, \tag{255}$$

$$\text{score}_{h,m}^{(k)}(r) = w_{h,m}(r) \prod_{p \in \mathcal{S}_{h,m}^{(k-1)}} \left[1 - 0.65 \exp\left(-\frac{(r-p)^2}{2r_{h,m}^2}\right)\right], \qquad r \in \mathcal{I}_{h,m}^{(k)}, \tag{256}$$

$$d_{h,m,k} \in \mathcal{I}_{h,m}^{(k)}, \qquad \text{score}_{h,m}^{(k)}(d_{h,m,k}) = \max_{r \in \mathcal{I}_{h,m}^{(k)}} \text{score}_{h,m}^{(k)}(r), \tag{257}$$

$$\mathcal{S}_{h,m}^{(k)} = \mathcal{S}_{h,m}^{(k-1)} \cup \{d_{h,m,k}\}, \qquad k = 1, \ldots, n_{s,m}, \qquad \mathcal{S}_{h,m} = \mathcal{S}_{h,m}^{(n_{s,m})}. \tag{258}$$

So at iteration k, the phrase "already selected" means $\mathcal{S}_{h,m}^{(k-1)}$. Here $r_{h,m}$ is repulsion length scale.

Step 5: Enforce feasible contiguous spell windows. Selected starts are projected to feasible non-overlapping starts $s_{h,m,q}$ within month bounds. Spell q occupies

$$d_{h,m,1}^{\text{raw}} < \cdots < d_{h,m,n_{s,m}}^{\text{raw}}, \qquad \{d_{h,m,1}^{\text{raw}}, \ldots, d_{h,m,n_{s,m}}^{\text{raw}}\} = \mathcal{S}_{h,m}.$$

So $d_{h,m,q}^{\text{raw}}$ is the q-th ordered raw start day before feasibility clipping. Define feasible bounds:

$$e_{h,m,q} = \begin{cases} 1, & q = 1, \\ s_{h,m,q-1} + \ell_{h,m,q-1} + 1, & q \geq 2, \end{cases} \qquad u_{h,m,q} = L_m - \left(\sum_{p=q}^{n_{s,m}} \ell_{h,m,p} + (n_{s,m} - q)\right) + 1.$$

Then project each raw start:

$$s_{h,m,q} = \begin{cases} e_{h,m,q}, & d_{h,m,q}^{\text{raw}} < e_{h,m,q}, \\ d_{h,m,q}^{\text{raw}}, & e_{h,m,q} \leq d_{h,m,q}^{\text{raw}} \leq u_{h,m,q}, \\ u_{h,m,q}, & d_{h,m,q}^{\text{raw}} > u_{h,m,q}. \end{cases}$$

(code equivalent: $s_{h,m,q} = \text{clip}(d_{h,m,q}^{\text{raw}}, e_{h,m,q}, u_{h,m,q})$).

Construction path: $\mathcal{S}_{h,m} \xrightarrow{\text{sort}} d^{\text{raw}} \xrightarrow{\text{clip}} s$.

$$\mathcal{B}_{h,m,q} = \{\, r : \; s_{h,m,q} \leq r \leq s_{h,m,q} + \ell_{h,m,q} - 1 \,\}, \tag{259}$$

and $q(d) = q$ when $r_m(d) \in \mathcal{B}_{h,m,q}$. Thus $\mathcal{B}_{h,m,q}$ is a spell-day block (all days in spell q), not a start-day set.

$$|\mathcal{B}_{h,m,q}| = \ell_{h,m,q}. \tag{260}$$

Define month-m all-spell day set:

$$\mathcal{B}_{h,m}^{\text{all}} = \bigcup_{q=1}^{n_{s,m}} \mathcal{B}_{h,m,q}. \tag{261}$$

Because spell blocks are non-overlapping by construction,

$$\left|\mathcal{B}_{h,m}^{\text{all}}\right| = \sum_{q=1}^{n_{s,m}} |\mathcal{B}_{h,m,q}| = \sum_{q=1}^{n_{s,m}} \ell_{h,m,q} = n_{h,m}. \tag{262}$$

Step 6: Sample spell anomaly and propagate to ambient. For each spell q, anomaly magnitude is sampled as a bounded random variable:

$$\Delta T_{h,m,q} \in [a_{h,m}, b_{h,m}], \qquad \Pr(u \le \Delta T_{h,m,q} \le v) = \int_u^v f_{h,m}(x)\,dx, \tag{263}$$

for any $a_{h,m} \le u \le v \le b_{h,m}$. This uses a truncated normal density on $[a_{h,m}, b_{h,m}]$ with

$$a_{h,m} = \Delta T^{\text{low}}_{h,m}, \quad b_{h,m} = \Delta T^{\text{high}}_{h,m}, \quad \mu^\star_{h,m} = \text{clip}(\Delta T_{h,m}, a_{h,m}, b_{h,m}), \quad \sigma^\star_{h,m} = \max(\sigma_{\Delta T,h,m}, \varepsilon), \tag{264}$$

Here $\sigma_{\Delta T,h,m}$ is the month-m spell-anomaly spread from `heat_wave.spellAnomalyStd_C`$[m]$. If missing or nonpositive, implementation fallback is

$$\sigma_{\Delta T,h,m} = \begin{cases} (\Delta T^{\text{high}}_{h,m} - \Delta T^{\text{low}}_{h,m})/4, & \Delta T^{\text{high}}_{h,m} > \Delta T^{\text{low}}_{h,m}, \\ 0, & \Delta T^{\text{high}}_{h,m} \le \Delta T^{\text{low}}_{h,m}. \end{cases}$$

The stochastic sampling floor is

$$\varepsilon = 10^{-3}\,{}^\circ\text{C}.$$

Here $\Delta T_{h,m}$, $\Delta T^{\text{low}}_{h,m}$, and $\Delta T^{\text{high}}_{h,m}$ are spell-anomaly-above-monthly-mean statistics from `heat_wave.meanAnomalyAb`
`heat_wave.minSpellAnomalyAboveMonthlyMean_C`$[m]$, and `heat_wave.maxSpellAnomalyAboveMonthlyMean_C`$[m]$
Historical NOAA derivation in month m: if $\mathcal{B}_{h,m,q}$ is the day set for historical spell q (grouped by spell start month m),

$$a^{(h)}_{m,q} = \frac{1}{|\mathcal{B}_{h,m,q}|} \sum_{t \in \mathcal{B}_{h,m,q}} \max\big(0,\ T^{\text{avg}}_t - \bar{T}_{m(t)}\big)\,,$$

where T^{avg}_t is NOAA daily mean proxy (TAVG if present, else $(TMAX + TMIN)/2$) and $m(t)$ is the calendar month index of day t. Hence $\bar{T}_{m(t)}$ is the same monthly-mean notation $\bar{T}_m$, evaluated at day t's month. Then

$$\Delta T_{h,m} = \frac{1}{N_m} \sum_{q=1}^{N_m} a^{(h)}_{m,q}, \qquad \Delta T^{\text{low}}_{h,m} = \min_q a^{(h)}_{m,q}, \qquad \Delta T^{\text{high}}_{h,m} = \max_q a^{(h)}_{m,q}.$$

$$g_{\text{std}}(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right), \qquad G_{\text{std}}(z) = \int_{-\infty}^{z} g_{\text{std}}(t)\,dt, \tag{265}$$

$$f_{h,m}(x) = \frac{g_{\text{std}}\left(\frac{x-\mu^\star_{h,m}}{\sigma^\star_{h,m}}\right)}{\sigma^\star_{h,m}\left[G_{\text{std}}\left(\frac{b_{h,m}-\mu^\star_{h,m}}{\sigma^\star_{h,m}}\right) - G_{\text{std}}\left(\frac{a_{h,m}-\mu^\star_{h,m}}{\sigma^\star_{h,m}}\right)\right]}, \qquad a_{h,m} \le x \le b_{h,m}, \tag{266}$$

and $f_{h,m}(x) = 0$ outside $[a_{h,m}, b_{h,m}]$. Here z is a dimensionless standard-normal coordinate (z-score), and $Z_{h,m,q,t}$ is one random draw from that $N(0,1)$ distribution. So z itself is unbounded:

$$z \in (-\infty, \infty).$$

Bounds are enforced only through accepted anomaly values:

$$a_{h,m} \le Y_{h,m,q,t} \le b_{h,m},$$

equivalently,

$$\frac{a_{h,m} - \mu^\star_{h,m}}{\sigma^\star_{h,m}} \le z \le \frac{b_{h,m} - \mu^\star_{h,m}}{\sigma^\star_{h,m}}.$$

Implementation-equivalent draw equations for spell q are

$$Z_{h,m,q,t} \sim g_{\text{std}}, \qquad Y_{h,m,q,t} = \mu^{\star}_{h,m} + \sigma^{\star}_{h,m} Z_{h,m,q,t}, \qquad t = 1, \ldots, 64. \tag{267}$$

$$\tau_{h,m,q} = \min\left\{t \in \{1, \ldots, 64\} : a_{h,m} \le Y_{h,m,q,t} \le b_{h,m}\right\}. \tag{268}$$

If $\tau_{h,m,q}$ exists, set $\Delta T_{h,m,q} = Y_{h,m,q,\tau_{h,m,q}}$. If not, the implementation fallback is

$$\Delta T_{h,m,q} = \min\bigl(\max\bigl(\mu^{\star}_{h,m}, a_{h,m}\bigr), b_{h,m}\bigr). \tag{269}$$

with fixed seed in the implementation for reproducibility. In this repository state, $\Delta T_{h,m}$, $\Delta T^{\text{low}}_{h,m}$, $\Delta T^{\text{high}}_{h,m}$, and $\sigma_{\Delta T,h,m}$ are NOAA-derived from spell-level daily-station records (2016–2025, `USW00014740`), grouped by spell start month. The threshold implied by spell q is

$$T^{\text{th}}_{h,m,q} = \bar{T}_m + \Delta T_{h,m,q}. \tag{270}$$

The final ambient override is

$$T_m(d) = \begin{cases} \max\Bigl(T_{\text{base},m}(d),\, T^{\text{th}}_{h,m,q(d)},\, T_{\infty,\text{preH}}(d)\Bigr), & r_m(d) \in \mathcal{B}^{\text{all}}_{h,m}, \\ T_{\infty,\text{preH}}(d), & r_m(d) \notin \mathcal{B}^{\text{all}}_{h,m}. \end{cases} \tag{271}$$

Equivalent dependency chain:

$$f_{h,m}(x) \Rightarrow \Delta T_{h,m,q} \Rightarrow T^{\text{th}}_{h,m,q} \Rightarrow T_m(d) \Rightarrow T_\infty(d). \tag{272}$$

Here $q(d)$ denotes the active spell index on day d. These relations correspond to `heat_transfer_study.py:L4735-L4791`; `heat_transfer_study.py:L4907-L4911`.

18.5 Daily heating and cooling loads

For each representative day d, the required heating and cooling loads are

$$Q_{\text{req},h}(d) = UA_{\text{bin}} \max(T_{h,\text{set}} - T_\infty(d),\, 0), \tag{273}$$

$$Q_{\text{req},c}(d) = UA_{\text{bin}} \max(T_\infty(d) - T_{c,\text{set}},\, 0), \tag{274}$$

which are implemented by rebuilding the bin model at the day's ambient and calling the same lumped-loss routine used elsewhere `heat_transfer_study.py:L4977-L4983`.

18.6 Reference-point capacity re-solve and duty application

The annual layer re-solves the selected heating and cooling operating points at each day's ambient temperature to obtain

$$Q_{h,\text{cap}}(d) = Q_{\text{to bed},h}(T_\infty(d)), \qquad P_{h,\text{elec}}(d) = P_h(T_\infty(d)), \tag{275}$$

$$Q_{c,\text{cap}}(d) = -Q_{\text{to bed},c}(T_\infty(d)), \qquad P_{c,\text{elec}}(d) = P_{\text{spot cooler}}(d) + P_{\text{assist blower}}(d). \tag{276}$$

Identity clarification for Eqs. (275)–(276): $Q_{\text{to bed},h}(T_\infty(d))$ and $Q_{\text{to bed},c}(T_\infty(d))$ are the signed net bed heat-transfer outputs of the day-ambient heating/cooling re-solves. $P_h(T_\infty(d))$ is the heating electrical input at that day ambient. $P_{\text{spot cooler}}(d)$ and $P_{\text{assist blower}}(d)$ are the cooling electric-power components from the same day-ambient cooling re-solve.

The explicit derivation path for these quantities is:

$$R_{\text{tube},g}(d) = \rho_{20}\left[1 + \alpha\left(\bar{T}_{w,g}(d) - 20\right)\right] \frac{L_{w,g}}{A_{w,g}}, \tag{277}$$

$$I_{\text{tube},g}(d) = \frac{V}{N_{s,g} R_{\text{tube},g}(d)}, \qquad P_{\text{tube},g}(d) = I^2_{\text{tube},g}(d) R_{\text{tube},g}(d), \tag{278}$$

$$P_h(T_\infty(d)) = \sum_g n_{\text{tube},g} P_{\text{tube},g}(d). \tag{279}$$

For heating-mode bed transfer,

$$Q_{\text{to bed},h}(T_\infty(d)) = \sum_{b=1}^{N_b} \sum_{s=1}^{N_{\text{seg}}} \left(Q_{\text{wall},b,s} + Q_{\text{jet},b,s} - Q_{\text{evap},b,s}\right). \tag{280}$$

For cooling electrical components,

$$\dot{Q}_{\text{spot,req}}(d) = \dot{m}(d) c_p(d) \max(T_\infty(d) - T_{\text{target}}, 0), \qquad \dot{Q}_{\text{spot,act}}(d) = \min\Big(\dot{Q}_{\text{spot,req}}(d), \dot{Q}_{\text{spot,max}}(d)\Big), \tag{281}$$

$$P_{\text{spot cooler}}(d) = \frac{\dot{Q}_{\text{spot,act}}(d)}{\text{COP}}, \tag{282}$$

$$P_{\text{assist blower}}(d) = \begin{cases} P_{\text{motor}}, & \dot{V}_{\text{ach}}(d) > 0 \text{ and blower enabled}, \\ 0, & \text{otherwise}. \end{cases} \tag{283}$$

with operating flow determined from the scalar residual

$$R_{\dot{V}}(\dot{V}) = \Delta p_{\text{sys}}(\dot{V}) - \Delta p_{\text{avail}}(\dot{V}), \tag{284}$$

and bisection on $[0, \dot{V}_{\text{req}}(d)]$ returning

$$\dot{V}_{\text{ach}}(d) = \max\Big\{\dot{V} \in [0, \dot{V}_{\text{req}}(d)] : R_{\dot{V}}(\dot{V}) \le 0\Big\}. \tag{285}$$

Cooling-mode bed transfer is

$$Q_{\text{to bed},c}(T_\infty(d)) = \sum_{s=1}^{N_{\text{seg}}} \left(Q_{\text{wall},s} + Q_{\text{jet},s} - Q_{\text{evap},s}\right). \tag{286}$$

Thus "re-solve for $Q_{\text{to bed}}$" means re-running the day-ambient operating-point equations and scalar closure residuals, then evaluating the segment heat-balance sums in Eqs. (280) and (286). The 2×2 bed-node system in Eq. (291) is solved directly as $\mathbf{AT} = \mathbf{b}$, not by residual iteration.

Variable definitions for Eqs. (277)–(286).

- Indexing and counts: d day index; g heater-tube group; b aeration branch; s axial segment; $n_{\text{tube},g}$ number of tubes in group g; N_b total branches; N_{seg} total segments.
- Electrical/wire symbols: ρ_{20} wire resistivity at 20 °C; α temperature coefficient of resistivity; $\bar{T}_{w,g}(d)$ representative wire temperature; $L_{w,g}$ wire length; $A_{w,g}$ wire cross-sectional area; V applied voltage; $N_{s,g}$ number of series elements in group g; $R_{\text{tube},g}(d)$ tube resistance; $I_{\text{tube},g}(d)$ tube/string current; $P_{\text{tube},g}(d)$ tube/string electric power; $P_h(T_\infty(d))$ total heating electric power.

- Heating bed-transfer symbols: $\dot{m}_{\text{mean},b,s}$ mean segment flow; $c_{p,b,s}$ segment specific heat; $T_{\text{air,in},b,s}, T_{\text{air,out},b,s}$ segment inlet/outlet air temperatures; $\dot{m}_{\text{rel},b,s}$ released jet flow; $T_{\text{rel},b,s}$ jet/release temperature; $T_{\text{bed,ref}}$ bed reference temperature used in segment balances; $\dot{m}_{\text{evap},b,s}$ evaporation mass flow; $h_{fg,b,s}$ latent heat; $Q_{\text{wall},b,s}$, $Q_{\text{jet},b,s}$, $Q_{\text{evap},b,s}$ segment wall/jet/evaporation terms; $Q_{\text{to bed},h}(T_\infty(d))$ net signed heating-mode bed transfer.
- Cooling-load and cooler-power symbols: $\dot{m}(d)$ day cooling-branch mass flow; $c_p(d)$ day effective specific heat; $T_\infty(d)$ day ambient temperature; T_{target} spot-cooler target temperature; $\dot{Q}_{\text{spot,req}}(d)$ requested sensible cooling load; $\dot{Q}_{\text{spot,max}}(d)$ available maximum sensible capacity; $\dot{Q}_{\text{spot,act}}(d)$ actual (capped) sensible load; COP coefficient of performance; $P_{\text{spot cooler}}(d)$ spot-cooler electrical power.
- Blower and residual symbols: $\Delta p_{\text{shutoff}}$ blower shutoff pressure; $\dot{V}_{\text{free}}$ blower free-delivery volumetric flow; $\Delta p_{\text{avail}}(\dot{V})$ available blower pressure at trial flow $\dot{V}$; $\Delta p_{\text{sys}}(\dot{V})$ required system pressure at trial flow; $R_{\dot{V}}(\dot{V})$ flow residual; $\dot{V}_{\text{req}}(d)$ requested flow; $\dot{V}_{\text{ach}}(d)$ achieved flow; P_{motor} rated blower motor power; $P_{\text{assist blower}}(d)$ assist-blower electric power.
- Cooling bed-transfer symbols: $Q_{\text{wall},s}$, $Q_{\text{jet},s}$, $Q_{\text{evap},s}$ segment contributions in cooling mode; $Q_{\text{to bed},c}(T_\infty(d))$ net signed cooling-mode bed transfer (typically negative).

Derivation notes for these equations.

- Eqs. (277)–(279) follow temperature-corrected resistivity, Ohm's law, and Joule heating.
- Eqs. (280) and (286) are first-law sums of segment sensible gain/loss and latent evaporation sink terms.
- Eqs. (281) and (282) use the sensible-load relation $\dot{m}c_p\Delta T$, capacity capping, then COP conversion to electric power.
- Eqs. (284)–(285) define operating flow as the residual-consistent blower/system match solved by scalar bisection.

The required duty fractions are

$$D_h^*(d) = \frac{Q_{\text{req},h}(d)}{Q_{h,\text{cap}}(d)}, \qquad D_c^*(d) = \frac{Q_{\text{req},c}(d)}{Q_{c,\text{cap}}(d)}, \tag{287}$$

and the applied duties are clipped at unity:

$$f_{\text{day}} = \begin{cases} \min\left(\max\left(\dfrac{\dot{Q}_{\text{load}}}{\dot{Q}_{\text{nom}}}, 0\right), 1\right), & \dot{Q}_{\text{nom}} > 0, \\ 0, & \dot{Q}_{\text{nom}} = 0 \text{ and } \dot{Q}_{\text{load}} \le 0, \\ 1, & \dot{Q}_{\text{nom}} = 0 \text{ and } \dot{Q}_{\text{load}} > 0, \end{cases} \tag{288}$$

Identity clarification for Eq. (288): $\dot{Q}_{\text{load}}$ and $\dot{Q}_{\text{nom}}$ are generic placeholders for the active-mode required load and nominal capacity. In the heating branch, $\dot{Q}_{\text{load}} = Q_{\text{req},h}(d)$ and $\dot{Q}_{\text{nom}} = Q_{h,\text{cap}}(d)$, so $f_{\text{day}} = D_h(d)$. In the cooling branch, $\dot{Q}_{\text{load}} = Q_{\text{req},c}(d)$ and $\dot{Q}_{\text{nom}} = Q_{c,\text{cap}}(d)$, so $f_{\text{day}} = D_c(d)$. $D(d)$ in Eq. (295) denotes this active-branch duty fraction for day d. which, in mode-specific notation, are exactly

$$D_h(d) = \min(\max(D_h^*(d), 0), 1), \qquad D_c(d) = \min(\max(D_c^*(d), 0), 1). \tag{289}$$

The applied bed load is therefore

$$Q_{\text{bed}}(d) = \begin{cases} D_h(d)\, Q_{h,\text{cap}}(d), & \text{heating day}, \\ -D_c(d)\, Q_{c,\text{cap}}(d), & \text{cooling day}, \\ 0, & \text{idle day}. \end{cases} \tag{290}$$

18.7 Daily bed temperatures, electricity, cost, and water

The daily bed temperatures are obtained from the same two-node steady system:

$$\begin{bmatrix} UA_{b,\infty} + UA_{bt} & -UA_{bt} \\ -UA_{bt} & UA_{t,\infty} + UA_{bt} \end{bmatrix} \begin{bmatrix} T_b(d) \\ T_t(d) \end{bmatrix} = \begin{bmatrix} f_b Q_{\text{bed}}(d) + UA_{b,\infty} T_\infty(d) \\ (1 - f_b) Q_{\text{bed}}(d) + UA_{t,\infty} T_\infty(d) \end{bmatrix}. \tag{291}$$

Daily heating and cooling electricity are

$$E_h(d) = \frac{P_{h,\text{elec}}(d)\, D_h(d)\, 24}{1000}, \qquad E_c(d) = \frac{P_{c,\text{elec}}(d)\, D_c(d)\, 24}{1000}, \tag{292}$$

so the total day electrical energy is

$$E_{\text{day}} = E_h(d) + E_c(d). \tag{293}$$

With electricity price c_{elec}, the daily cost is

$$C(d) = c_{\text{elec}} E_{\text{day}}, \tag{294}$$

and the daily water-replacement estimate is

$$m_{\text{water}}(d) = \dot{m}_{\text{water},24h}\, D(d), \tag{295}$$

where $D(d)$ is the active heating or cooling duty fraction for that day. These are repository numerical constructs driven by the same daily re-solved heating/cooling reference points `heat_transfer_study.py:L4966-L5072`.

19 Implementation notes and deviations from the written model

- The authoritative reported heating recommendation is now Python-side, even though MATLAB still generates the heating candidate physics. The rooted execution path therefore treats Python as the final recommendation authority `heat_transfer_study.py:L6908-L6947`; `vermicomposter_air_heater_design_model.m:L536-L608`.
- Maximum bottom and top bed temperatures are enforced in the Python-side heating and cooling feasibility logic; the MATLAB helper `explicitlySafe(...)` still omits those maxima `vermicomposter_air_heater_design_model.m:L2647-L2659`; `heat_transfer_study.py:L2220-L2368`; `heat_transfer_study.py:L2963-L3112`.
- Cooling selection is no longer lexicographic. The current code uses constrained NSGA-II plus a report-priority ordering inside the feasible cooling Pareto set `heat_transfer_study.py:L3148-L3208`; `heat_transfer_study.py:L6220-L6490`.
- Cooling shortfall is computed and printed, but not enforced as a hard cooling feasibility condition in the current NSGA-II implementation `heat_transfer_study.py:L2915-L2959`; `heat_transfer_study.py:L2963-L3112`.
- The post-cooler relative humidity used by the summer evaporation model is recomputed after psychrometric condensate removal, limited by any configured rated condensate capacity `heat_transfer_study.py:L2662-L2673`.
- The rooted refresh/export path disables MATLAB plotting and transients in the generated export helper, so the rooted execution graph exercises steady heating physics rather than the optional MATLAB transient branch `analyze_vermicomposter_heater.py:L294-L304`.

20 Symbol glossary with units

Symbol	Meaning	Units
A_h	area of one perforation hole	m^2
$A_{h,\mathrm{seg}}$	total perforation area assigned to one aeration segment	m^2
$A_{\mathrm{bed,trib}}$	tributary porous-bed flow area assigned to one aeration segment	m^2
A_{wet}	effective wetted area assigned to evaporation	m^2
Bi	lumped-capacitance Biot number	1
C_d	perforation discharge coefficient	1
c_p	specific heat of air	$\mathrm{J\,kg^{-1}\,K^{-1}}$
d_{32}	Sauter mean diameter used in particle-area estimate	m
d_p	representative particle diameter used in the Ergun porous-bed closure; defaults to d_{32} unless overridden	m
D_i, D_o	inner and outer tube diameters	m
ε	bed porosity used in the Ergun porous-bed closure	1
f	Darcy friction factor	1
h	convective heat-transfer coefficient	$\mathrm{W\,m^{-2}\,K^{-1}}$
h_m	mass-transfer coefficient	$\mathrm{m\,s^{-1}}$
h_{fg}	latent heat of vaporization	$\mathrm{J\,kg^{-1}}$
I_{tot}	total electrical current draw	A
Nu, Re, Pr, Ra	standard dimensionless groups	1
P_{tot}	total electrical power	W
$Q_{\mathrm{bed,total}}$	net total bed heat rate; negative in cooling mode	W
Q_{cool}	positive cooling capacity, $\max(-Q_{\mathrm{bed,total}}, 0)$	W
Q_{req}	lumped required heating or cooling load	W
T_b, T_t	bottom and top bed-node temperatures	°C
T_∞	ambient air temperature	°C
UA	lumped conductance	$\mathrm{W\,K^{-1}}$
$A_{\mathrm{vent,branch}}$	total top-vent area assigned to one aeration branch in the outlet-boundary closure	m^2
$C_{d,\mathrm{vent}}$	simple-hole vent discharge coefficient used by the outlet-boundary closure	1
Δp_i	local tube-to-bed pressure difference driving perforation discharge	Pa
$L_{\mathrm{bed},i}$	representative porous escape length attached to aeration segment i	m
$p_{\mathrm{bed},i}$	local bed-side gauge pressure adjacent to aeration segment i	Pa
$p_{\mathrm{out,bnd}}$	derived outlet-boundary gauge pressure at the end of the porous escape path	Pa
$p_{\mathrm{out,offset}}$	configured outlet-boundary gauge offset or closed-top fallback pressure	Pa
$u_{h,i}$	local mean hole exit velocity	$\mathrm{m\,s^{-1}}$
$u_{\mathrm{bed},i}$	superficial porous-bed velocity attached to aeration segment i	$\mathrm{m\,s^{-1}}$
$y_{\mathrm{tube,rep}}$	representative tube-center height used by the porous-bed pressure model	m
$\phi_{\mathrm{rel},i}$	fraction of inlet tube mass released by aeration segment i	1
$\dot{m}_{\mathrm{evap}}$	evaporation mass flow rate	$\mathrm{kg\,s^{-1}}$
$\dot{V}_{\mathrm{tot}}$	total imposed volumetric flow rate	$\mathrm{m^3\,s^{-1}}$

21 Bibliography / source notes

R1 Churchill, S.W.; Bernstein, M. "A Correlating Equation for Forced Convection From Gases and Liquids to a Circular Cylinder in Crossflow." *Journal of Heat Transfer* 99(2), 1977, pp. 300–306. An accessible scan was used to verify that the implemented overall correlation appears as equation (9) on p. 303. The repository stores the citation string at `vermicomposter_air_heater_design_model.m:L4-L8`.

R2 Gnielinski, V. "New Equations for Heat and Mass Transfer in Turbulent Pipe and Channel Flow." *International Chemical Engineering* 16(2), 1976, pp. 359–368. The article page range was independently verifiable from accessible bibliographic records; the repository stores the citation string at `vermicomposter_air_heater_design_model.m:L9-L10`.

R3 Churchill, S.W. "Friction-Factor Equation Spans All Fluid-Flow Regimes." *Chemical Engineering* 84(24), 1977, pp. 91–92. The repository stores the citation string at `vermicomposter_air_heater_design_model.m:L11-L12`.

R4 Churchill, S.W.; Chu, H.H.S. "Correlating Equations for Laminar and Turbulent Free Convection from a Vertical Plate." *International Journal of Heat and Mass Transfer* 18(11), 1975, pp. 1323–1329. An accessible scan was used to verify that the implemented isothermal-wall relation appears as equation (9) on p. 1325. The repository stores the citation string at `vermicomposter_air_heater_design_model.m:L13-L15`.

R5 The repository cites Bergman, Lavine, Incropera, and DeWitt generically for thermal resistances, developing internal flow, lumped balances, and free convection `vermicomposter_air_heater_design_model.m:L16-L17`. Edition and page numbers are not stored in the repository.

R6 Idel'chik, I.E. *Handbook of Hydraulic Resistance*. The accessible English translation consulted online places the Section IV sudden-expansion discussion and the Borda–Carnot coefficient used here on p. 112. The repository source mapping for this usage is `vermicomposter_air_heater_design_model.m:L2130-L2141`; `HeatTransferStudy/VermiHeatLoss.tex:L1976-L1984`.

R7 The helix geometry relation is documented as a repository-native design relation in comments and TeX narrative `vermicomposter_air_heater_design_model.m:L1708-L1727`; `HeatTransferStudy/VermiHeatLoss.tex:L38-L121`.

R8 Towler, G.; Sinnott, R. *Chemical Engineering Design: Principles, Practice and Economics of Plant and Process Design*, 3rd ed. Butterworth-Heinemann, 2022. Chapter 15 restates the Frössling form $Sh = 2 + 0.552 Re^{1/2} Sc^{1/3}$, used here for the hole-scale Sherwood closure.

R9 Wagner, W.; Pruss, A. "The IAPWS Formulation 1995 for the Thermodynamic Properties of Ordinary Water Substance for General and Scientific Use." *Journal of Physical and Chemical Reference Data* 31(2), 2002, pp. 387–535. Section 2.3.1 reproduces the auxiliary saturation relations used for vapor pressure and the saturated liquid and vapor densities, with the vapor-pressure equation at equation (2.5) on p. 399 and the density equations at equations (2.6)–(2.7) on p. 399; Section 2.3.2 gives the saturation-boundary enthalpy relations at equations (2.10)–(2.11) on p. 400. This is the same auxiliary saturation package that the paper notes had also appeared in the earlier supplementary-release lineage. In the current Python implementation, `IAPWS_95.py` supplies the saturation-pressure call, directly evaluates equations (2.6)–(2.7) for the saturated liquid and vapor densities, and then forms latent heat from Helmholtz-relation enthalpies evaluated at those saturation densities rather than by directly coding equations (2.10)–(2.11); the repository commentary using these relations is at `vermicomposter_air_heater_design_model.m:L18-L36`; `HeatTransferStudy/VermiHeatLoss.tex:L1089-L1167`.

R10 Kloppers, J.C.; Kröger, D.G. "The Lewis factor and its influence on the performance prediction of wet-cooling towers." *International Journal of Thermal Sciences* 44(9), 2005, pp. 879–884. This corrects the author names stored in some repository comments while preserving the same title

and journal context. The repository narrative using the Lewis relation is at `HeatTransferStudy/VermiHeatLoss.tex:L1174-L1180`.

R11 Frederickson, J.; Howell, G.; Hobson, A.M. "Effect of pre-composting and vermicomposting on compost characteristics." *European Journal of Soil Biology* 43(S1), 2007, pp. S320–S326. Table 3 supplies the particle-size fractions used for the wetted-area estimate. The repository reproduces those fractions at `HeatTransferStudy/VermiHeatLoss.tex:L1015-L1088`.

R12 Idel'chik, I.E. *Handbook of Hydraulic Resistance.* The accessible English translation consulted online places the general incompressible orifice-discharge relation $Q = \mu F\sqrt{2\Delta p/\rho}$ at equation (1-48) on p. 31, the thin-wall simplification at equation (1-49) on p. 31, and representative discharge-coefficient values for orifices on p. 32; the same translation places the Section III contraction discussion on pp. 82–85 and equation (3-3), used here for the conical bellmouth contraction surrogate, on p. 83. The repository source mapping for these usages is `vermicomposter_air_heater_design_model.m:L1895-L2076`; `heat_transfer_study.py:L2825-L2849`; `vermicomposter_air_heater_design_model.m:L2148-L2232`; `HeatTransferStudy/VermiHeatLoss.tex:L1943-L1955`.

R13 Hausen, H. "Darstellung des Wärmeüberganges in Rohren durch verallgemeinerte Potenzbeziehungen." *Zeitschrift des VDI, Beiheft Verfahrenstechnik* 8(4), 1943, pp. 91–98. Later literature and thesis bibliographies preserve this citation; the repository uses the resulting Hausen-type form but does not currently store the primary citation string.

R14 Crowl, D.A.; Louvar, J.F. *Chemical Process Safety: Fundamentals with Applications*, 2nd ed. Prentice Hall, 2002. Section 4-4, "Flow of Liquids through Pipes," gives the 2-K method used here for fitting-loss surrogates: equation (4-38) on p. 124, Table 4-2 on p. 125, and the entrance/exit modification at equation (4-39) on p. 126. The same p. 126 discussion below equation (4-40) notes that a simple hole has $\sum K_f = 1.5$, giving $C_d \approx 0.63$. The current repository uses those relations as reduced-order surrogates for the unresolved air-side header and splitter-body losses, with the optional header treated as a normal entrance, one splitter branch path treated as a standard threaded tee used as an elbow followed by one full-line open gate/ball/plug valve, and the vented-top outlet boundary treated as a simple-hole discharge.

R15 Ergun, S. "Fluid Flow Through Packed Columns." *Chemical Engineering Progress* 48(2), 1952, pp. 89–94. The current repository uses the standard Ergun packed-bed pressure-drop form as a segmentwise porous-bed backpressure surrogate between each aeration-segment release zone and the nearest top opening.

R16 Fornes, F.; Mendoza-Hernández, D.; García-de-la-Fuente, R.; Abad, M.; Belda, R.M. "Composting versus vermicomposting: A comparative study of organic matter evolution through straight and combined processes." *Bioresource Technology* 118, 2012, pp. 296–305. Table 2 on p. 299 reports total porosity for the composting (COM), vermicomposting (VER), and combined composting-plus-vermicomposting (C + V) stages. The current repository uses the COM day 21 and day 63 total-porosity values (80.1 and 79.9% v/v) as the bracketing measurements around the nominal 30-day pre-composting interval, giving the default constant $\varepsilon = 0.80$, and notes the weak variation of the C + V values over the later vermicomposting interval when justifying the constant-porosity simplification.

R17 Idel'chik, I.E. *Handbook of Hydraulic Resistance.* For helical-coil obstruction context, the consulted scan was checked at the "Tubes with wire ribbing" entry in Section VIII (printed p. 337; PDF page 346) and at "Spiral-ribbed air heater,", Diagram 12-21, in Section XII (printed p. 482; PDF page 491). The current repository uses these as source-backed motivation for the added coil-obstruction surrogate rather than as a direct one-chart closed-form fit.

R18 Harper, E.; Miller, F.C.; Macauley, B.J. "Physical management and interpretation of an environmentally controlled composting ecosystem." *Australian Journal of Experimental Agriculture* 32,

1992, pp. 657–667. This source provides measured heat-output and oxygen-consumption linkage in controlled composting, including the reported linear relation equivalent to about 9.8 MJ heat per kg O_2 consumed (used here as a lower empirical oxygen-to-heat reference).

R19 de Guardia, A.; Petiot, C.; Benoist, J.C.; Druilhe, C. "Characterization and modelling of the heat transfers in a pilot-scale reactor during composting under forced aeration." *Waste Management* 32(6), 2012, pp. 1091–1105. This source formulates biodegradation heat as $H_{\mathrm{BIO}} = r_{O_2} h_{\mathrm{BIO}}$, reports literature h_{BIO} ranges around 304–448 kJ/mol O_2, and uses $h_{\mathrm{BIO}} = 440$ kJ/mol O_2 in model runs.

R20 Sonowal, P.; Khwairakpam, M.; Kalamdhad, A.S. "Stability Analysis of Dewatered Sludge of Pulp and Paper Mill During Vermicomposting." *Waste and Biomass Valorization*, 2013. This source provides vermicomposting OUR magnitudes used here for parameterization, including initial values near 18.9 mg O_2/g VS/day and mature-stage values near 2.0–3.1 mg O_2/g VS/day.

R21 Gea, T.; Barrena, R.; Artola, A.; Sánchez, A. "Monitoring the Biological Activity of the Composting Process: Oxygen Uptake Rate (OUR), Respirometric Index (RI), and Respiratory Quotient (RQ)." *Biotechnology and Bioengineering* 88(4), 2004, pp. 520–527. This source supports dynamic OUR as the preferred activity indicator and provides reported OUR ranges and static-vs-dynamic interpretation used for modeling rationale.

- NOAA and EIA URLs used by the annual layer are preserved in the configuration and narrative, not in a dedicated bibliography database `HeatTransferStudy/VermiHeatLoss.tex:L2574-L3164`; `HeatTransferStudy/study_config.json:L289-L419`.

Implementation topic map

Concept	Governing equations and source status	Implementation mapping and notes
Tube flow, perforations, electrical grouping	Equations: `eq:volflow_per_tube`–`eq:electrical_totals`. Source status: repository bookkeeping plus [R12] for the closed-end perforation discharge relation.	Implementation: `vermicomposter_air_heater_design_model.m:L763-L875`; `vermicomposter_air_heater_design_model.m:L1497-L1677`. Notes: series-parallel grouping allowed.
Internal convection and friction	Equations: `eq:internal_laminar_nu`, `eq:internal_turbulent_nu`, `eq:churchill_friction_full`. Source status: [R2], [R3], [R5].	Implementation: `vermicomposter_air_heater_design_model.m:L2382-L2407`; `heat_transfer_study.py:L3755-L3774`. Notes: used in both heater and aeration branches.
Heater axial energy balance	Equations: `eq:heater_segment_air`, `eq:wire_delta_t`. Source status: [R1], [R5] plus repository discretization.	Implementation: `vermicomposter_air_heater_design_model.m:L1803-L1894`; `heat_transfer_study.py:L3775-L3895`. Notes: explicit marching.

Concept	Governing equations and source status	Implementation mapping and notes
Aeration wall/jet/latent coupling	Equations: `eq:released_massflow`, `eq:aeration_remaining_mdot`, `eq:cooling_wall_factor`, `eq:aeration_q_jet`. Source status: [R5], [R12] plus repository reduction of sensible and latent paths.	Implementation: `vermicomposter_air_heater_design_model.m:L1895-L2076`; `heat_transfer_study.py:L2987-L3277`; `heat_transfer_study.py:L4182-L4323`. Notes: same structure in heating and cooling.
Evaporation and latent heat	Equations: `eq:sauter_diameter`–`eq:evap_selected`. Source status: [R1], [R9], [R10], [R11].	Implementation: `vermicomposter_air_heater_design_model.m:L891-L937`; `vermicomposter_air_heater_design_model.m:L2000-L2122`; `heat_transfer_study.py:L2567-L2603`; `heat_transfer_study.py:L3607-L3708`. Notes: minimum of three limiting rates.
Bin loss model and two-node bed	Equations: `eq:topmode_open`–`eq:two_node_split`. Source status: [R4], [R5] plus repository vented-cover extension.	Implementation: `vermicomposter_air_heater_design_model.m:L971-L1171`; `heat_transfer_study.py:L4326-L4438`. Notes: lumped bed closure.
Cooling spot cooler and assist blower	Equations: cooling and condensate relations in Sections 12–15. Source status: repository surrogate model.	Implementation: `heat_transfer_study.py:L2534-L3292`. Notes: includes condensate-limited RH update.
Heating and cooling NSGA-II	Equations: `eq:heating_constraint_residuals`–`eq:cooling_report_selection`. Source status: repository implementation.	Implementation: `heat_transfer_study.py:L51-L121`; `heat_transfer_study.py:L2181-L2499`; `heat_transfer_study.py:L3148-L3208`; `heat_transfer_study.py:L6220-L6490`. Notes: cooling now also NSGA-II.
Annual climate and duty layer	Equations: `eq:kernel_baseline`, `eq:daily_duty`, `eq:daily_energy`. Source status: repository numerical construct plus NOAA/EIA inputs.	Implementation: `heat_transfer_study.py:L4633-L5012`. Notes: not a literature first-principles climate model.